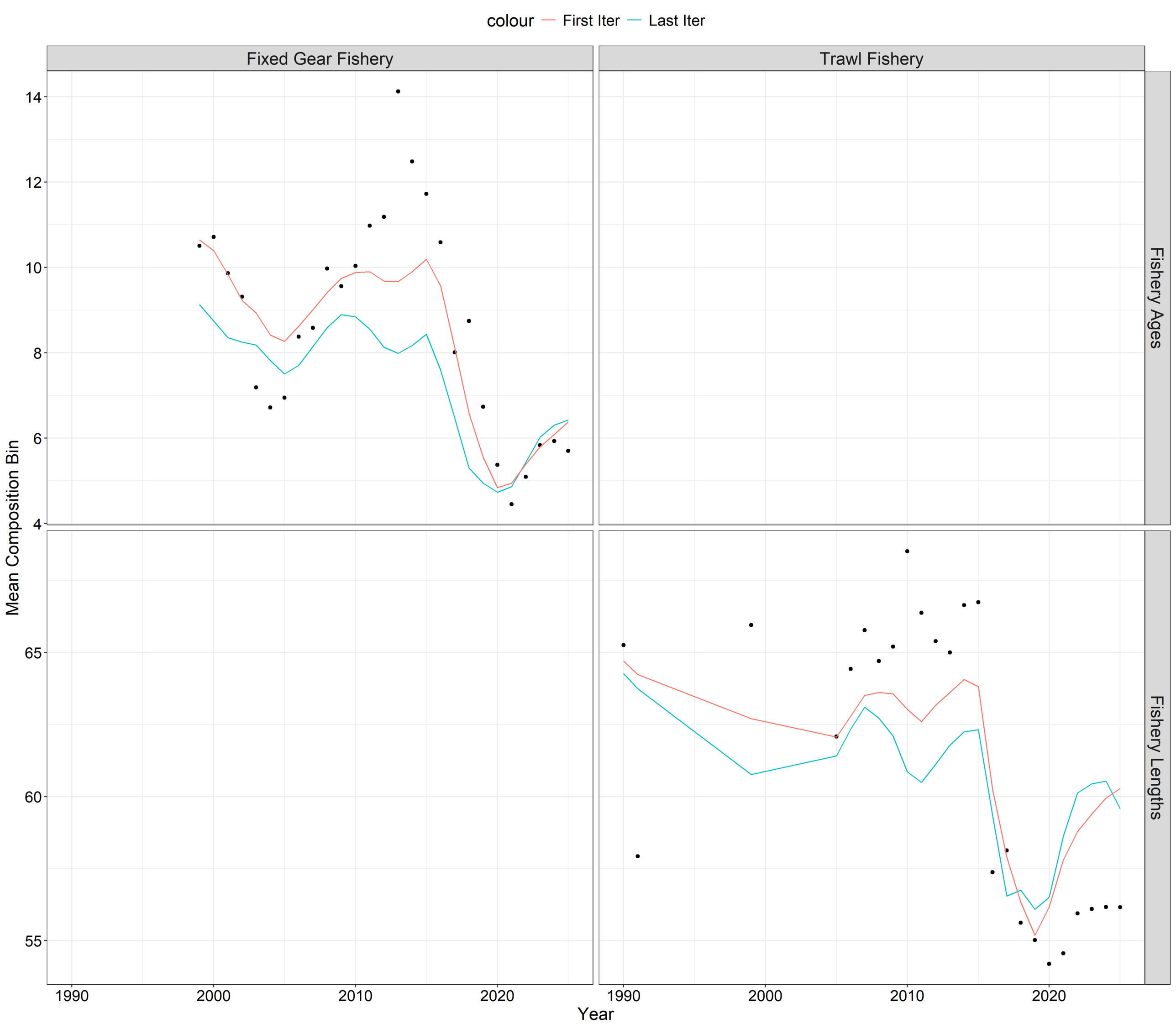
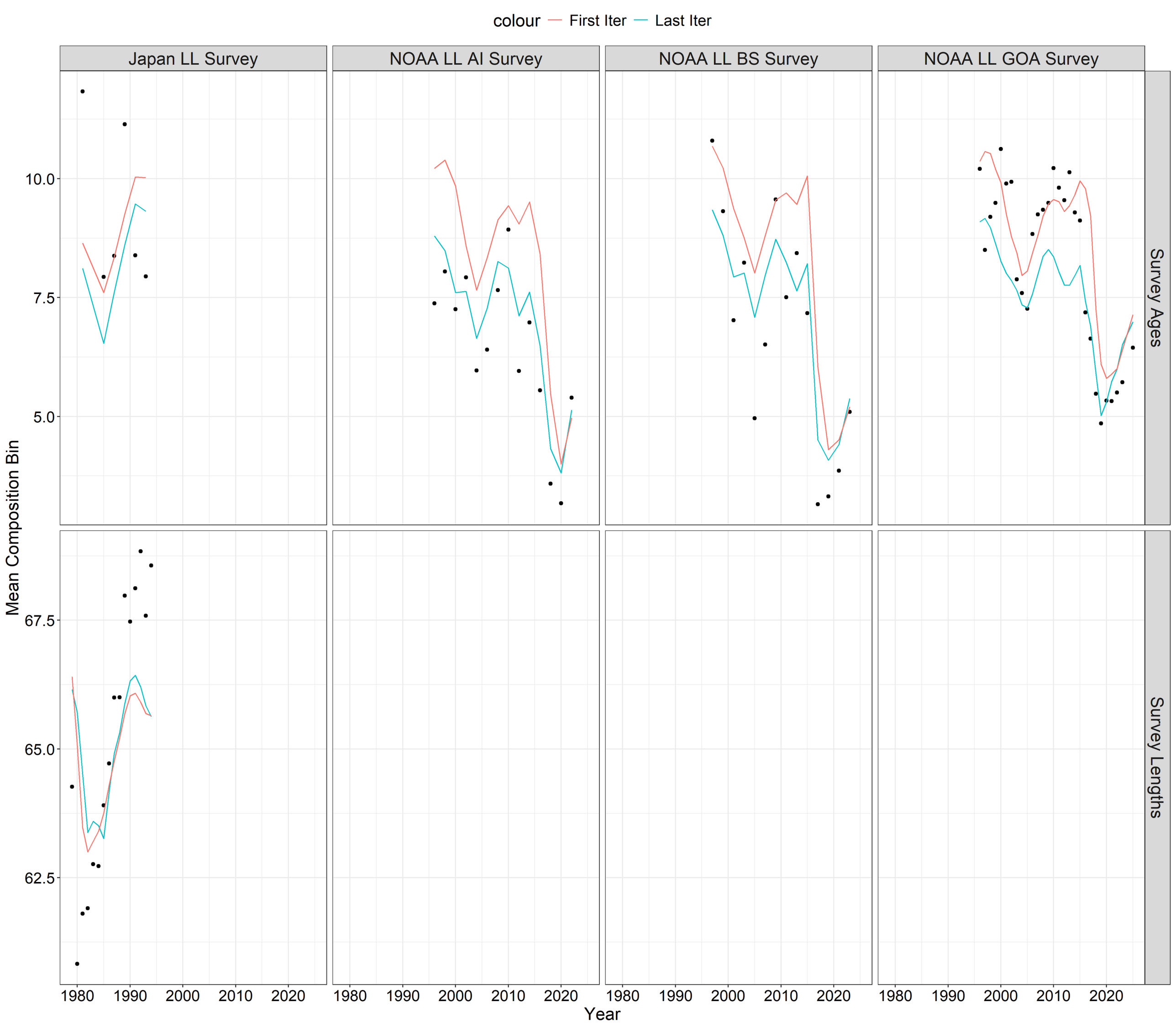
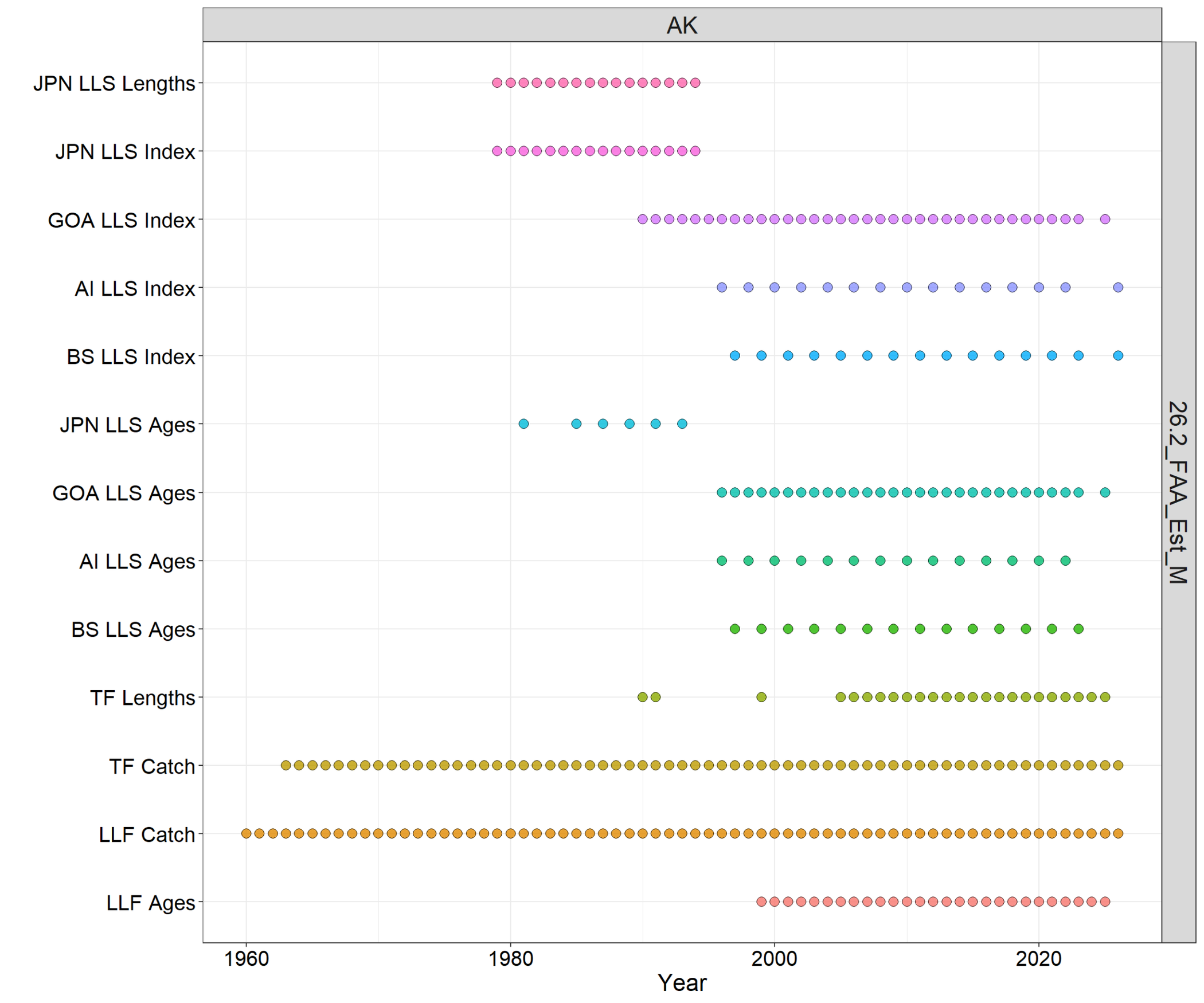
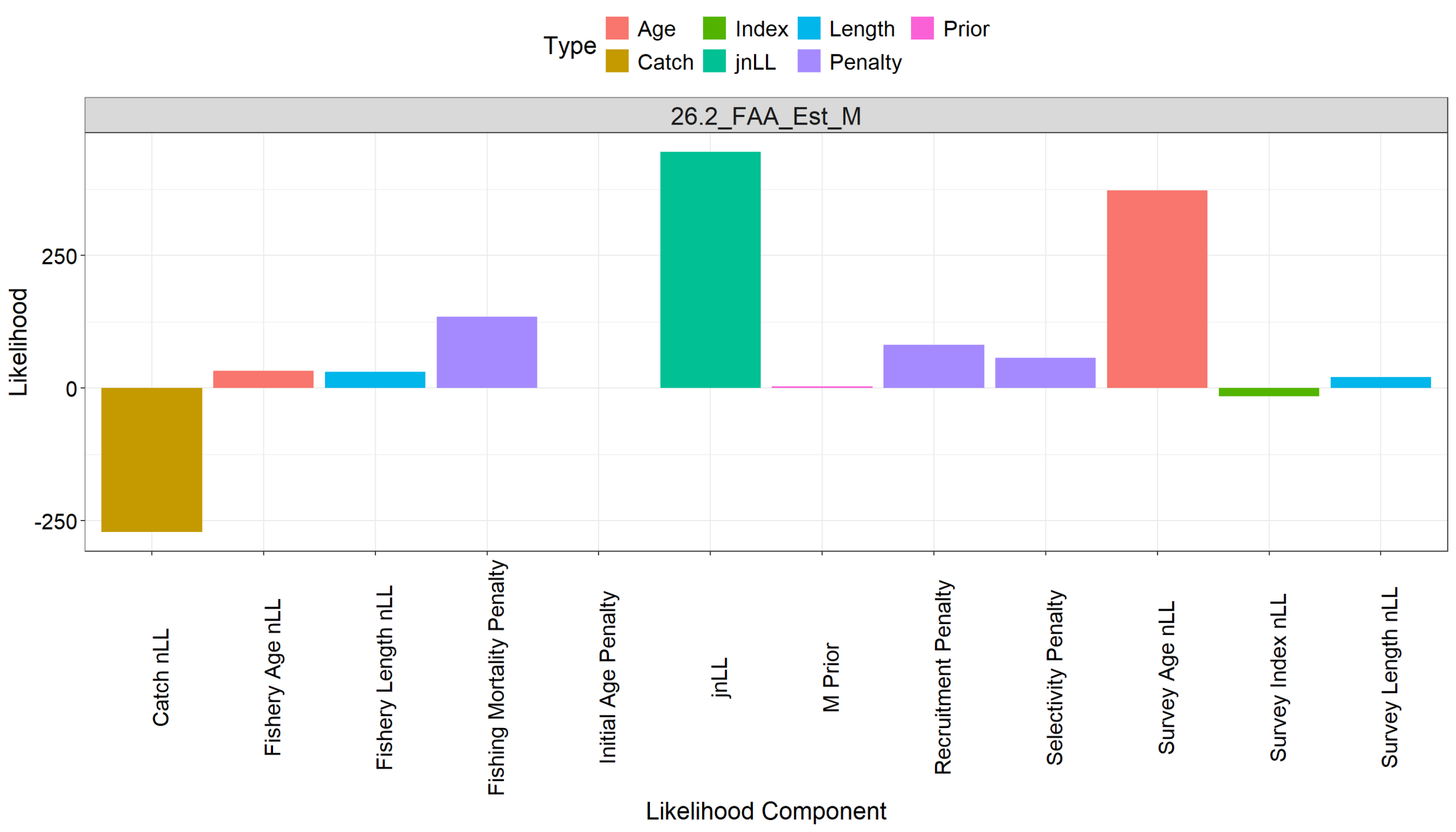

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year


colour
First Iter
Last Iter
Japan LL Survey
NOAA LL AI Survey
NOAA LL BS Survey
NOAA LL GOA Survey
Survey Ages
Survey Lengths
Mean Composition Bin
10.0
7.5
5.0
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year


AK
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.2_FAA_Est_M
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.2_FAA_Est_M
Likelihood
250
0
-250
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

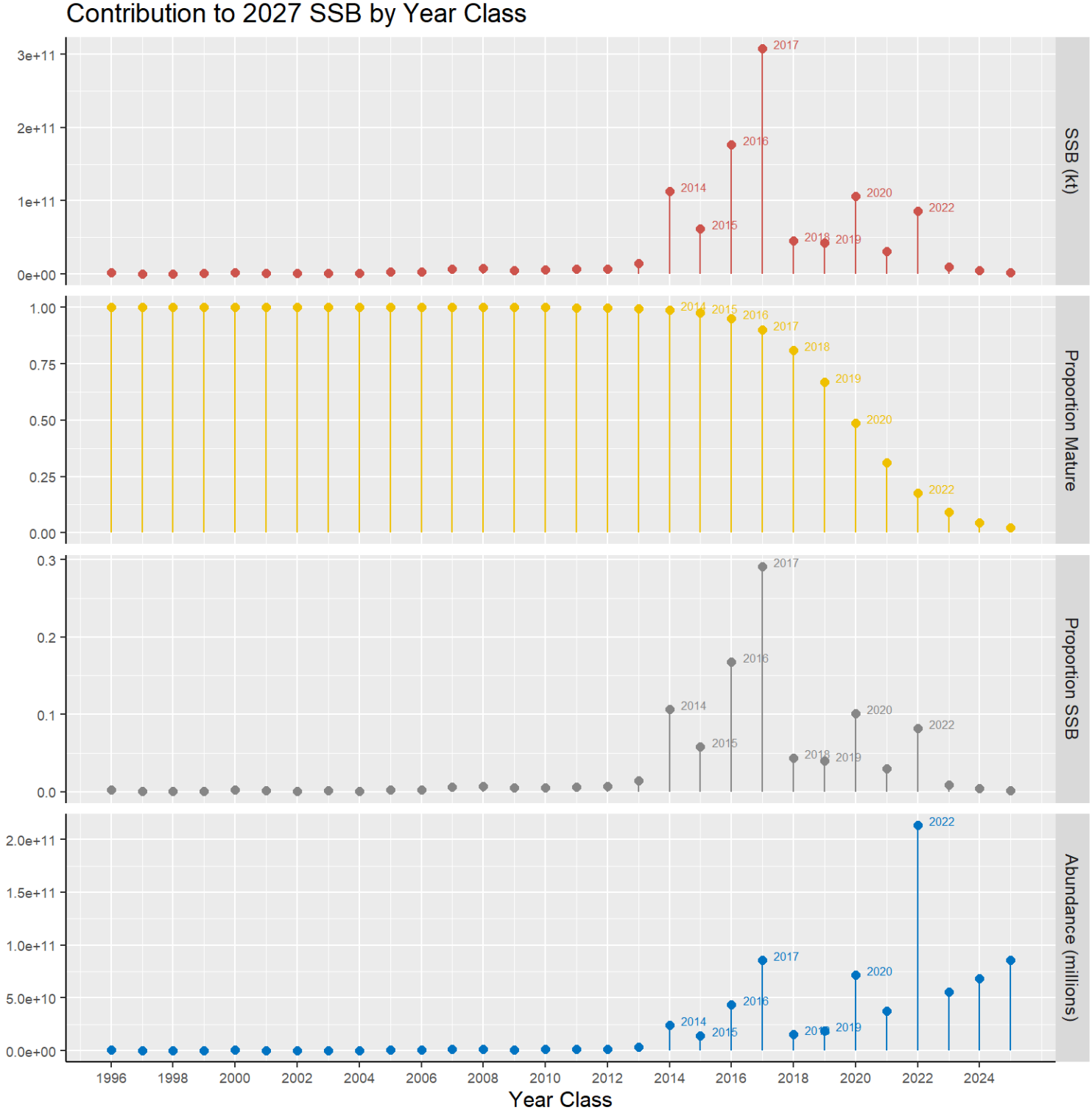
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

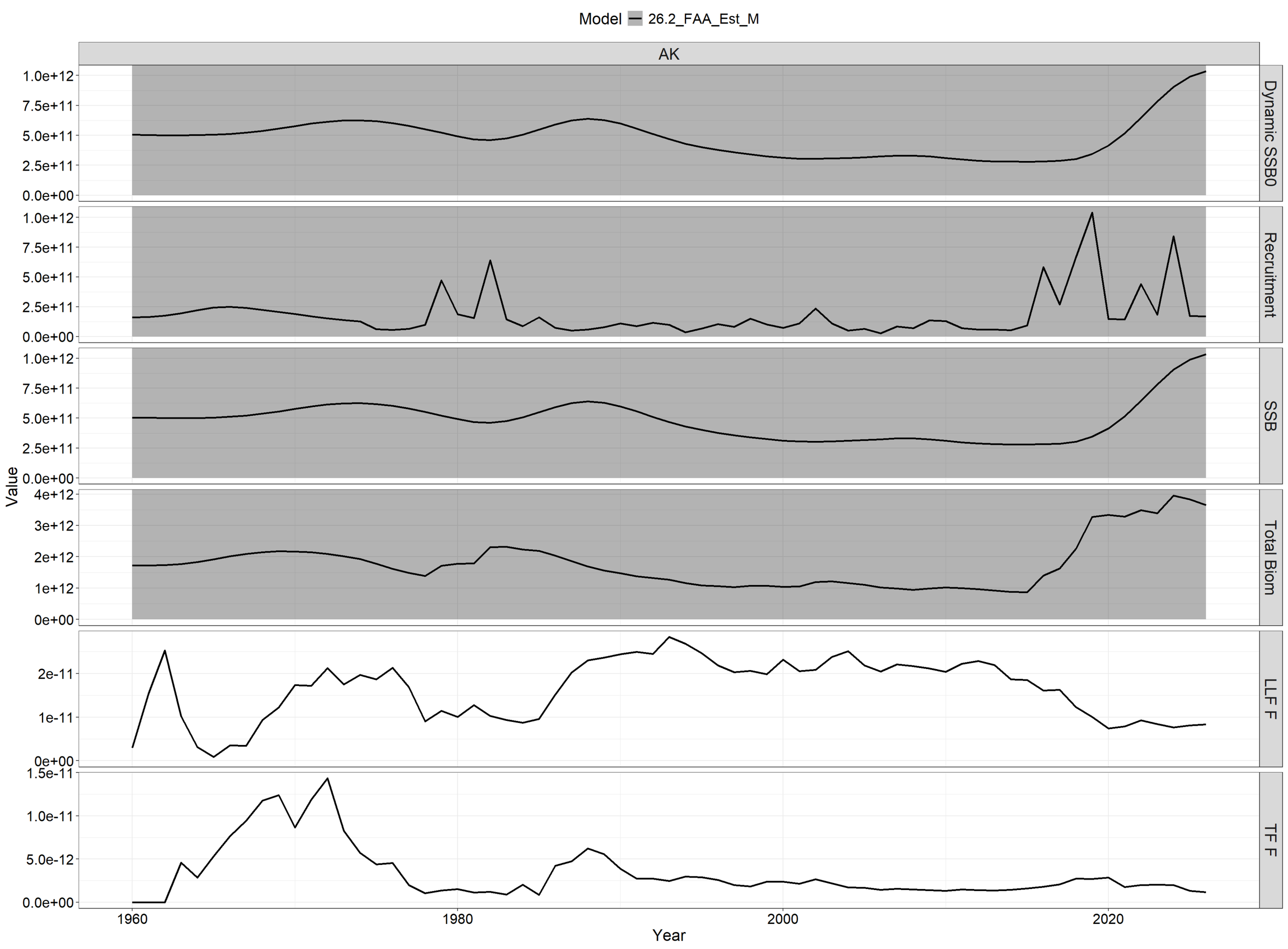
Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

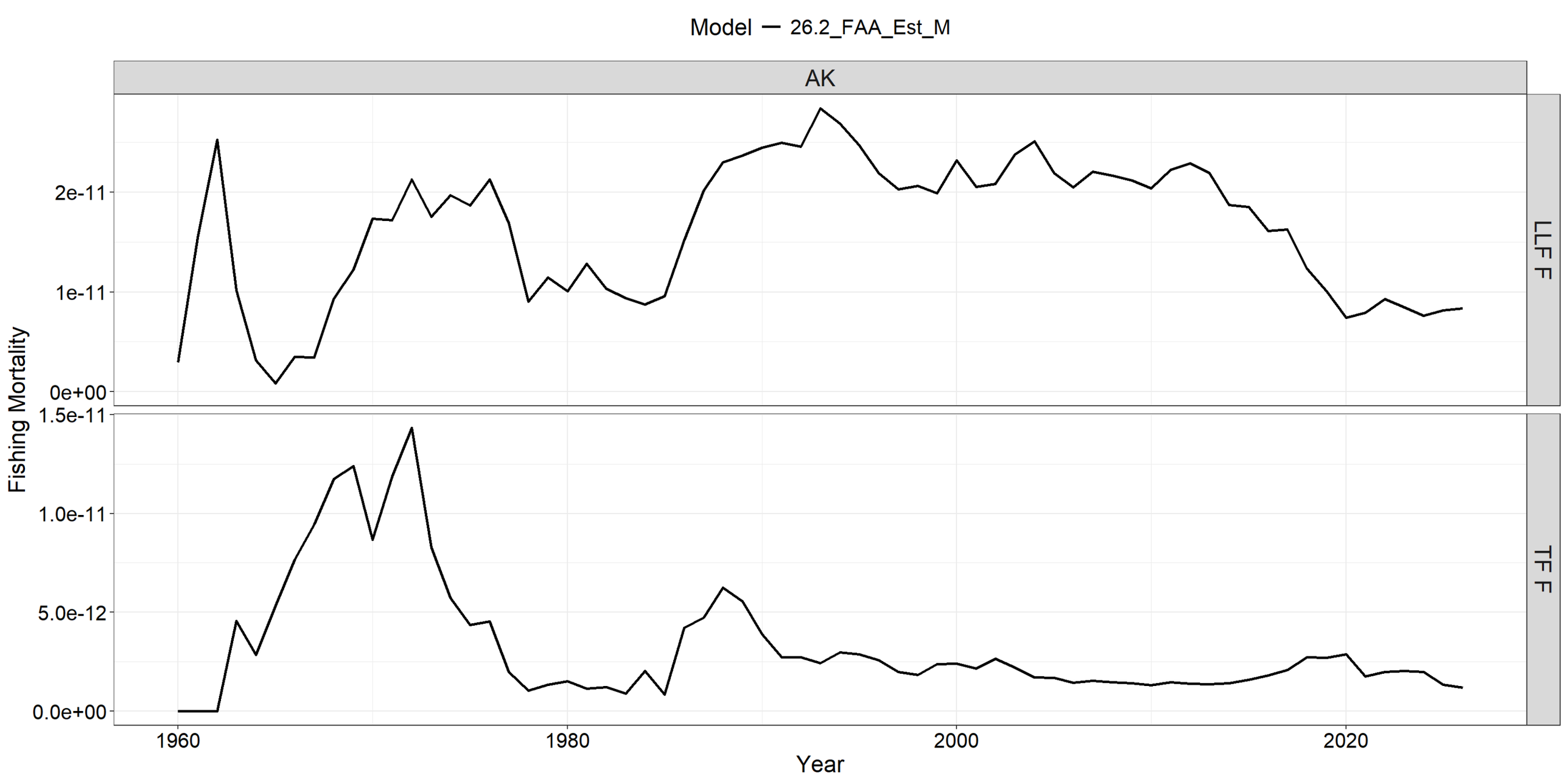
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

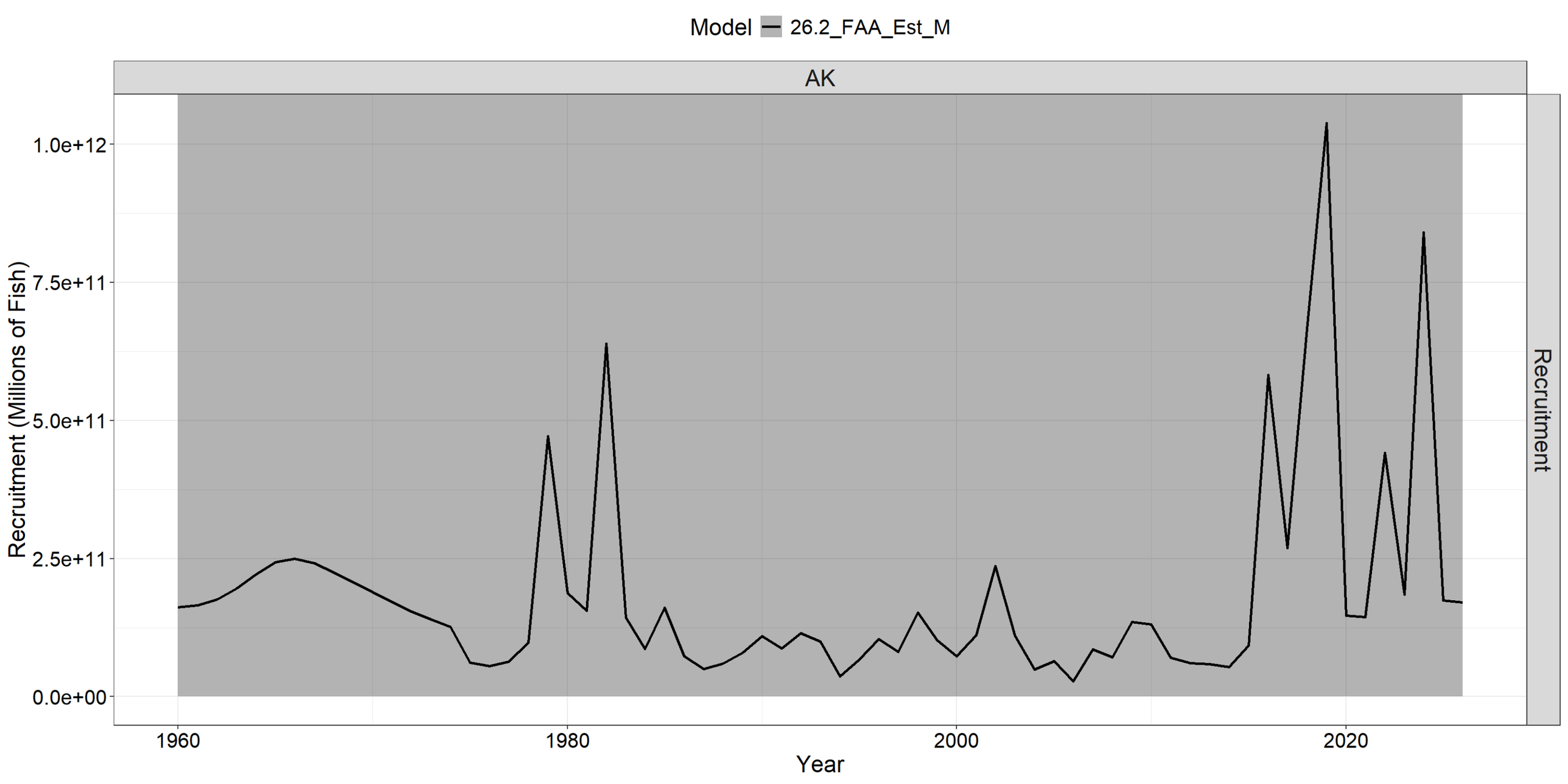
Numbers-at-Age (millions)

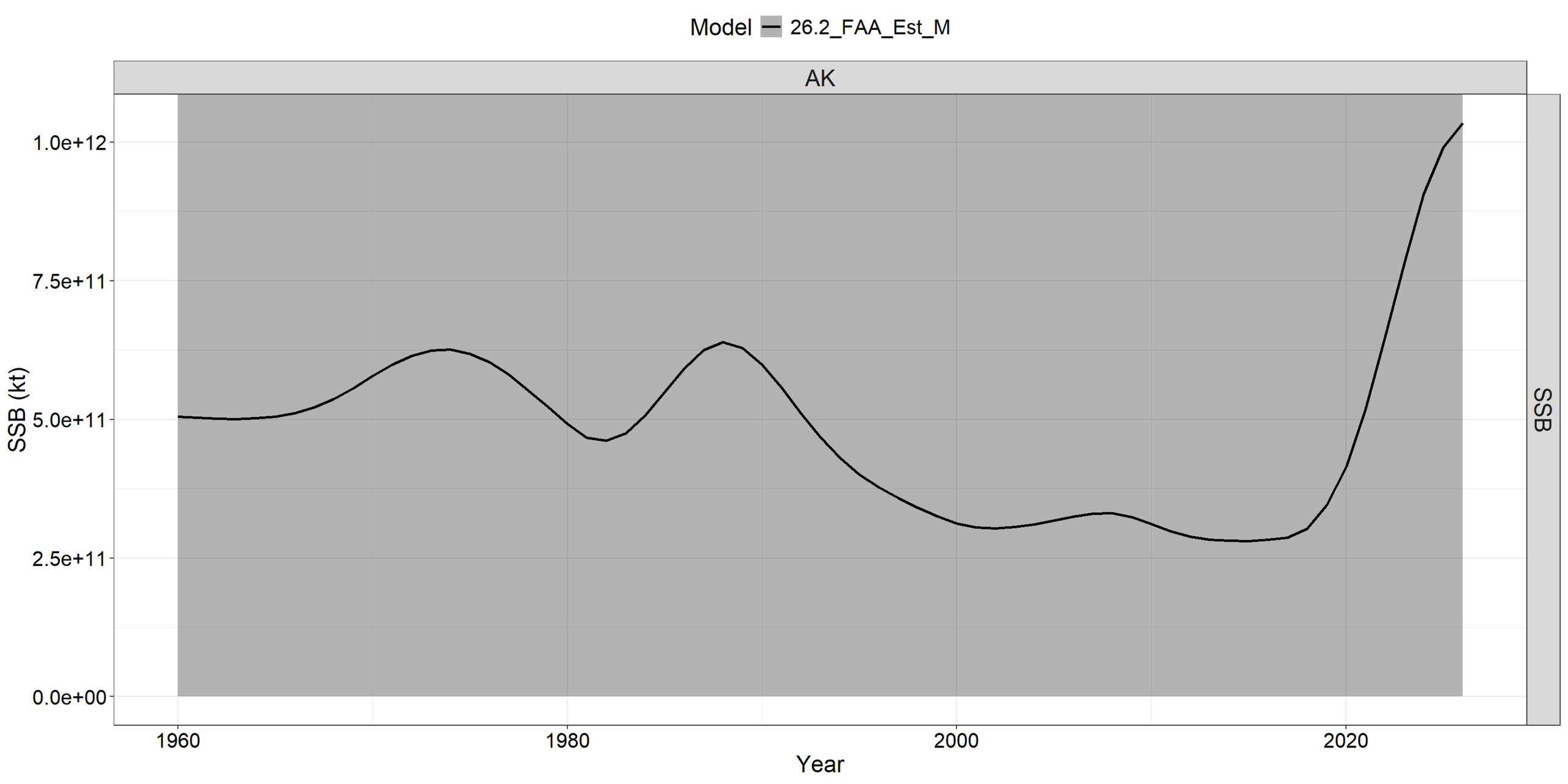
Year

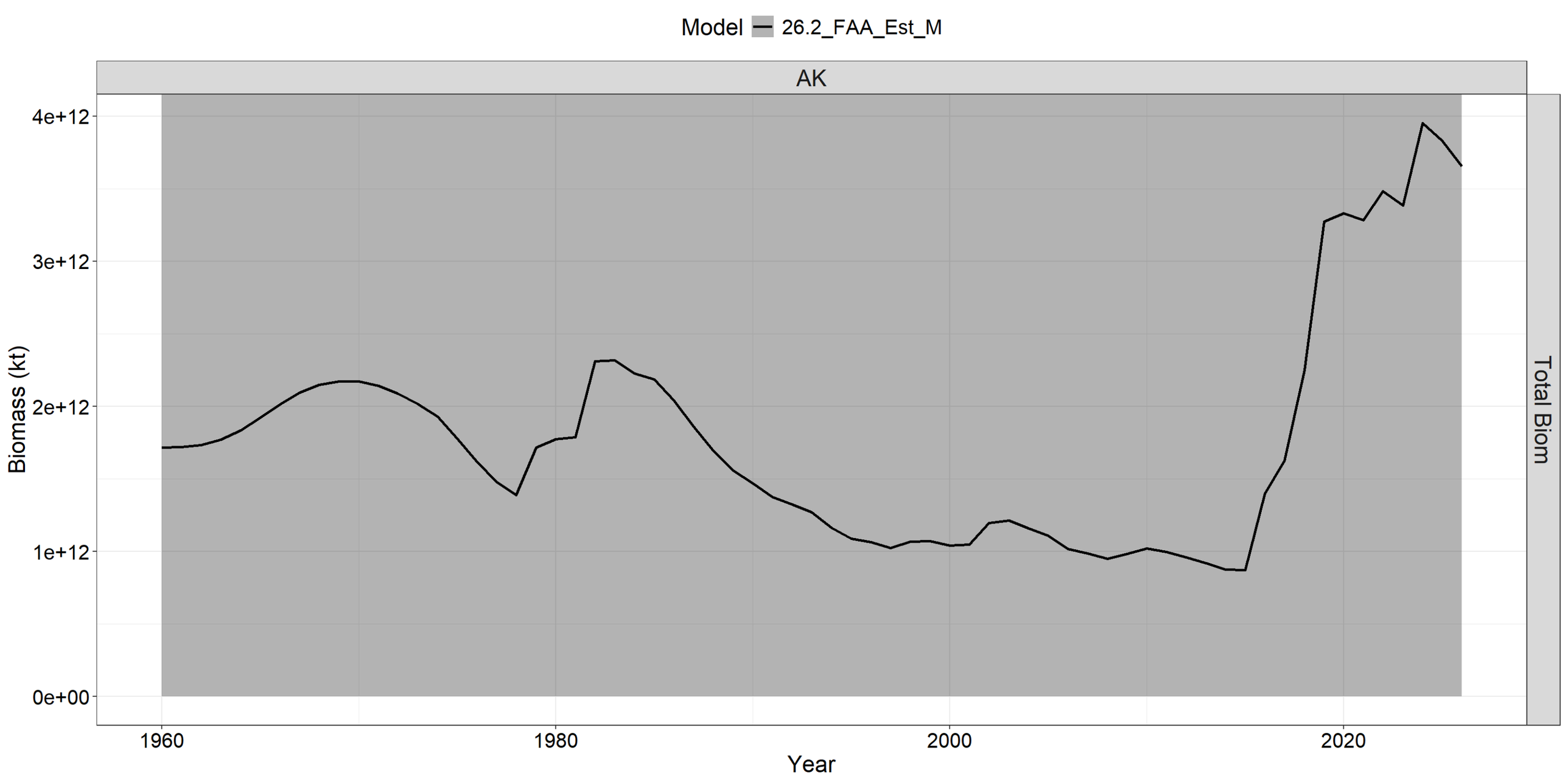

Contribution to 2027 SSB by Year Class
SSB (kt)
3e+11
2e+11
1e+11
0e+00
2017
2016
2014
2015
2020
2022
2018
2019
Proportion Mature
1.00
0.75
0.50
0.25
0.00
2014
2015
2016
2017
2018
2019
2020
2022
Proportion SSB
0.3
0.2
0.1
0.0
2017
2016
2014
2015
2020
2022
2018
2019
Abundance (millions)
2.0e+11
1.5e+11
1.0e+11
5.0e+10
0.0e+00
2022
2017
2020
2016
2014
2015
2019
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year Class

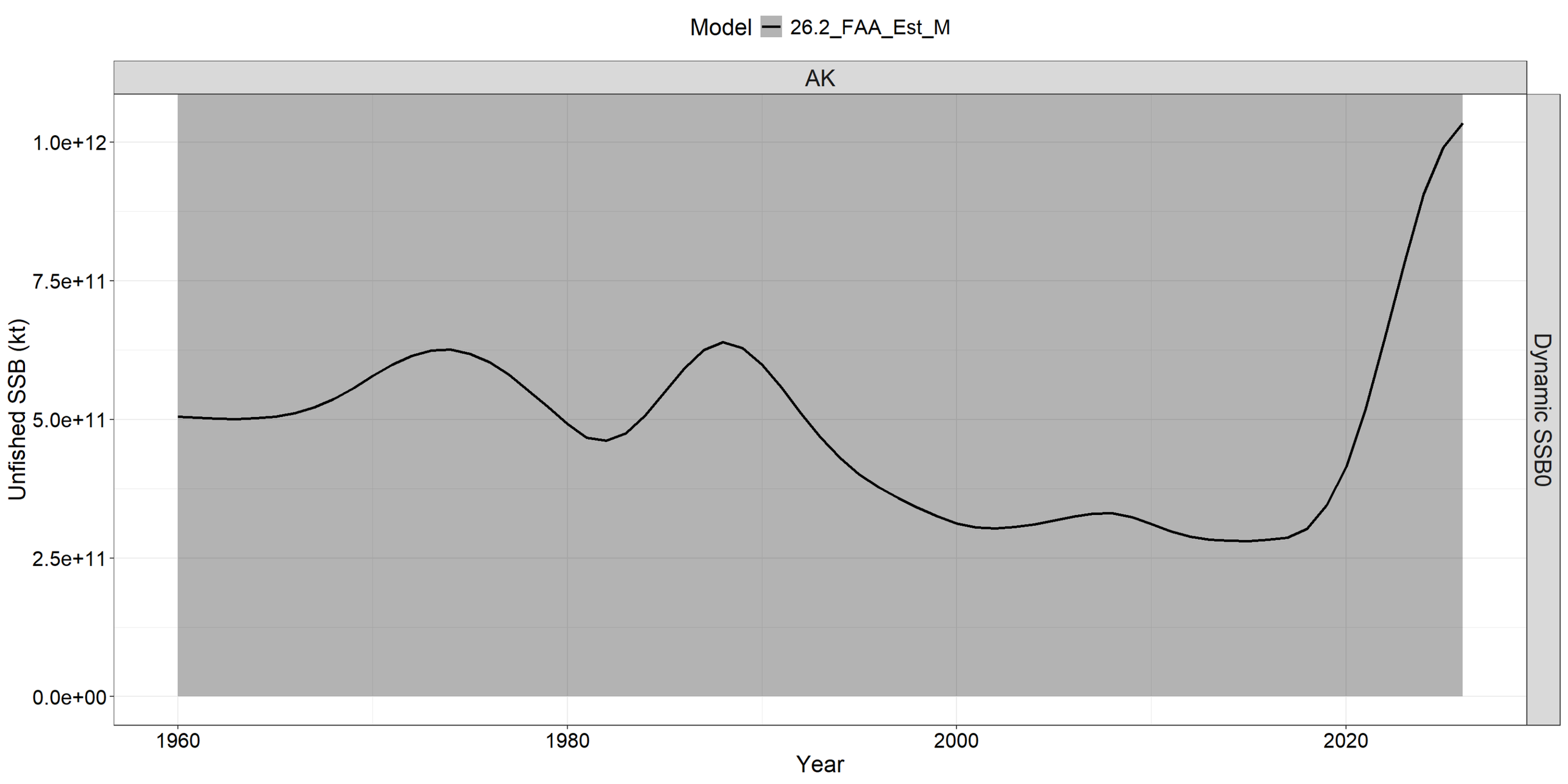

Model
26.2_FAA_Est_M
AK
Dynamic SSB0
Recruitment
SSB
Total Biom
LLF F
TF F
Value
Year
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
4e+12
3e+12
2e+12
1e+12
2e-11
1e-11
1.5e-11
1.0e-11
5.0e-12
1960
1980
2000
2020

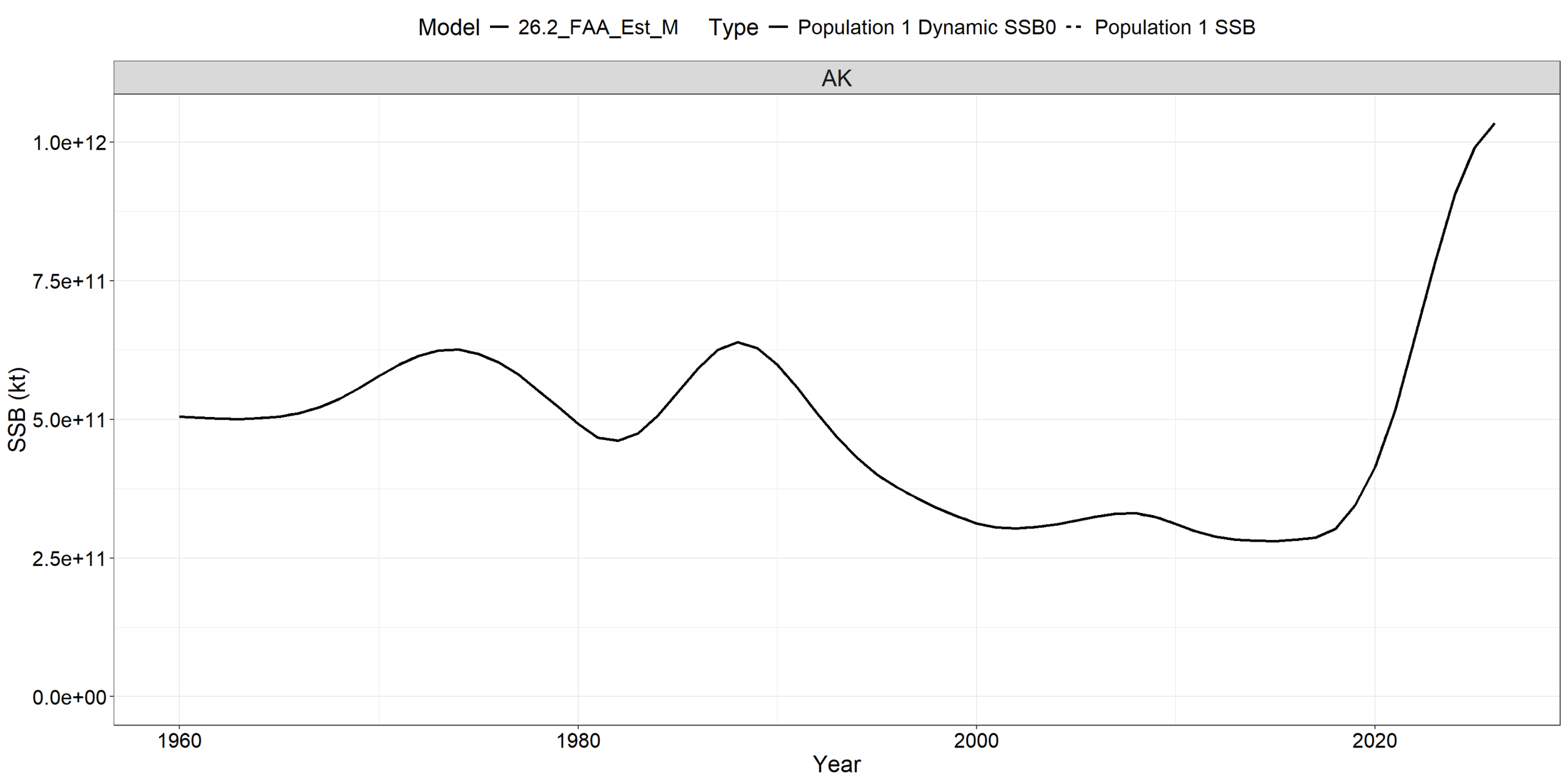

Model — 26.2_FAA_Est_M
AK
LLF F
TF F
Fishing Mortality
0e+00
1e-11
2e-11
0.0e+00
5.0e-12
1.0e-11
1.5e-11
1960
1980
2000
2020
Year

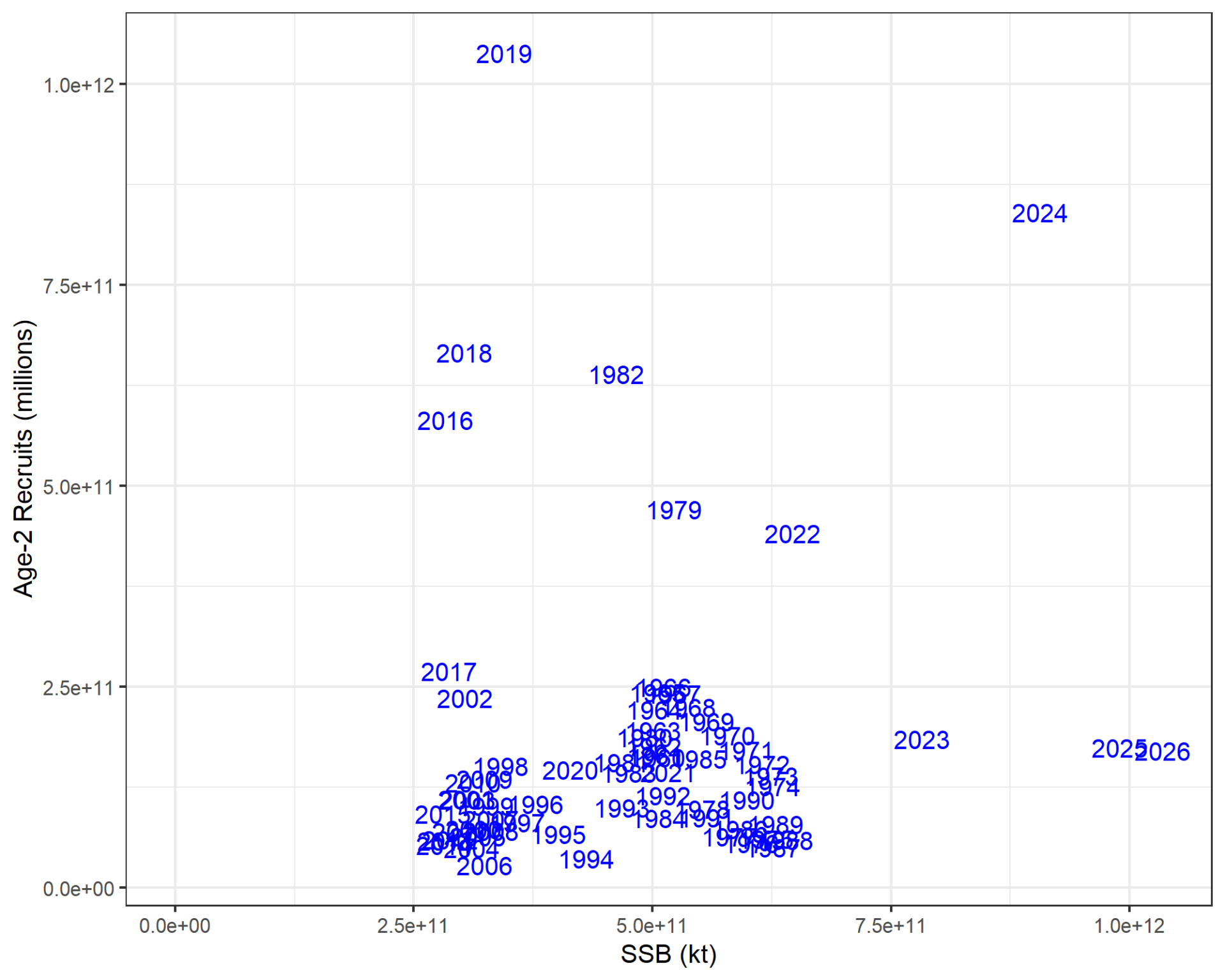

Model
26.2_FAA_Est_M
AK
Recruitment
Recruitment (Millions of Fish)
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
1960
1980
2000
2020
Year

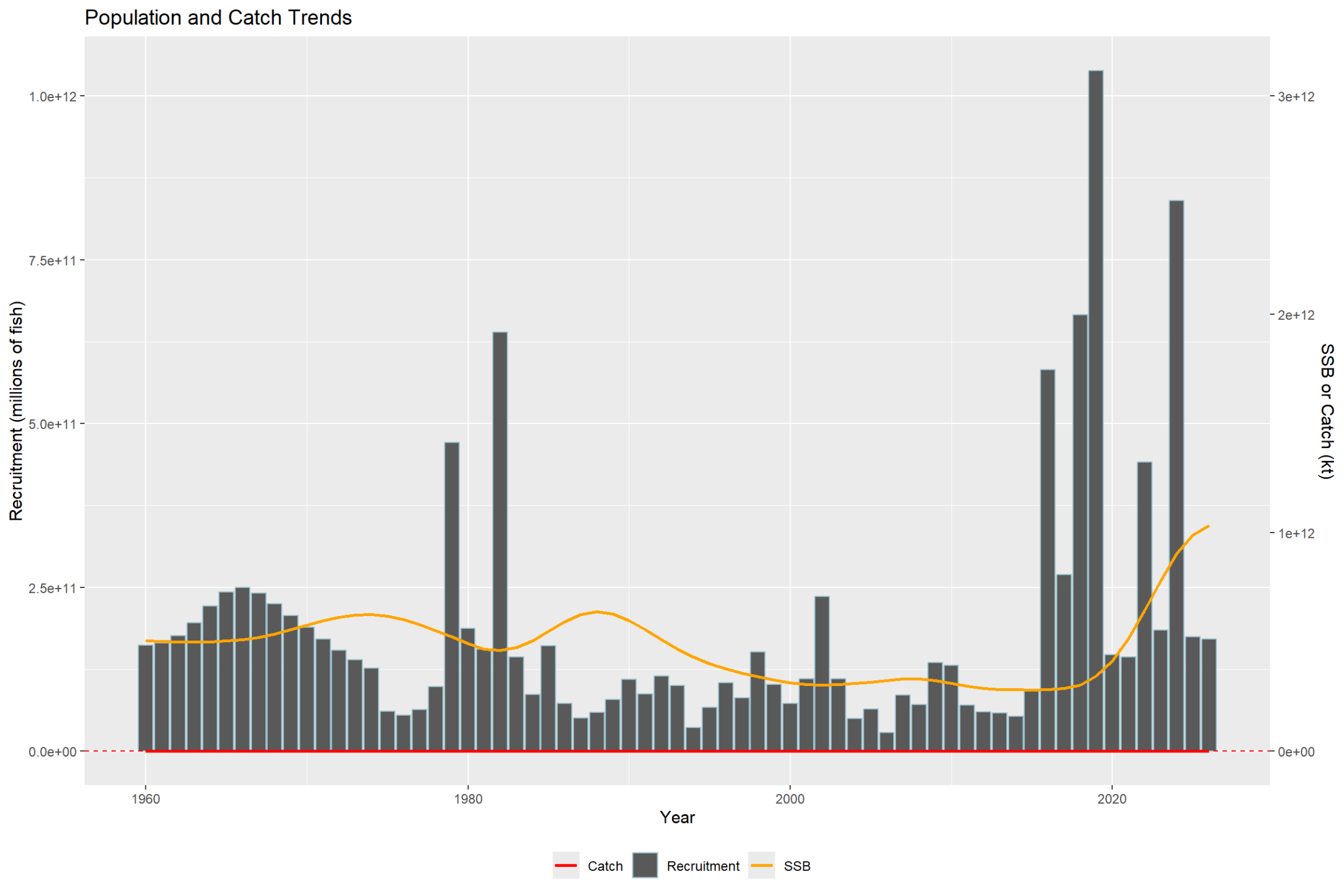

Model
26.2_FAA_Est_M
AK
SSB
SSB (kt)
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
1960
1980
2000
2020
Year


Model
26.2_FAA_Est_M
AK
Total Biom
Biomass (kt)
4e+12
3e+12
2e+12
1e+12
0e+00
1960
1980
2000
2020
Year


Model
26.2_FAA_Est_M
AK
Dynamic SSB0
Unfished SSB (kt)
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
1960
1980
2000
2020
Year


Model — 26.2_FAA_Est_M Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
SSB (kt)
1960
1980
2000
2020
Year


2019
2024
2018
1982
2016
1979
2022
2017
2002
2023
2025
2026
2020
1998
1996
1995
1994
2006
1990
1989
1974
1973
1972
1971
1970
1969
1968
1966
1992
1993
1984
1978
1985
1963
1964
1987
0.0e+00
2.5e+11
5.0e+11
7.5e+11
1.0e+12
Age-2 Recruits (millions)
SSB (kt)


Population and Catch Trends
Recruitment (millions of fish)
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
SSB or Catch (kt)
3e+12
2e+12
1e+12
0e+00
1960
1980
2000
2020
Year
Catch
Recruitment
SSB


Model — 26.2_FAA_Est_M
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA_Est_M

BS LLS

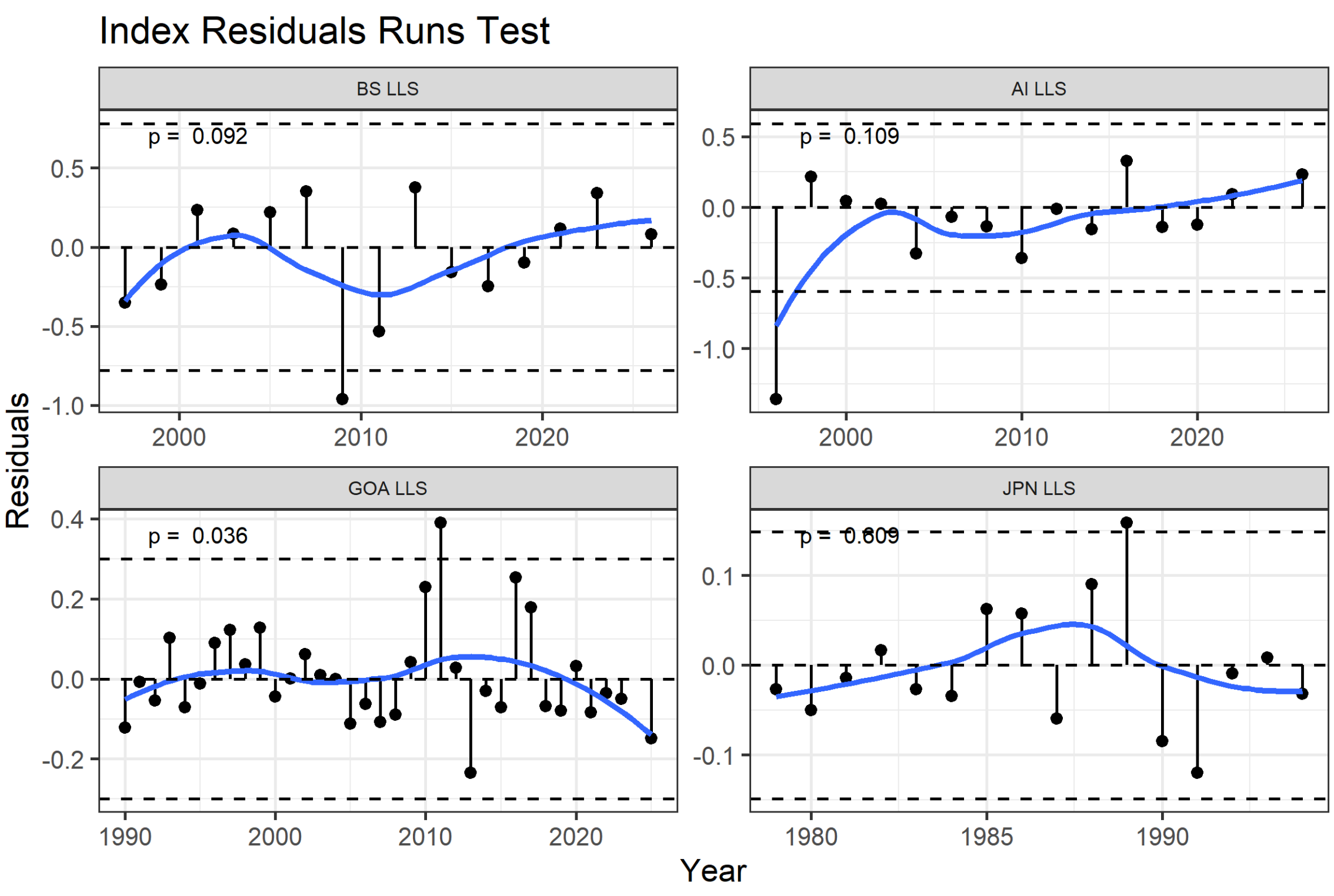
AI LLS

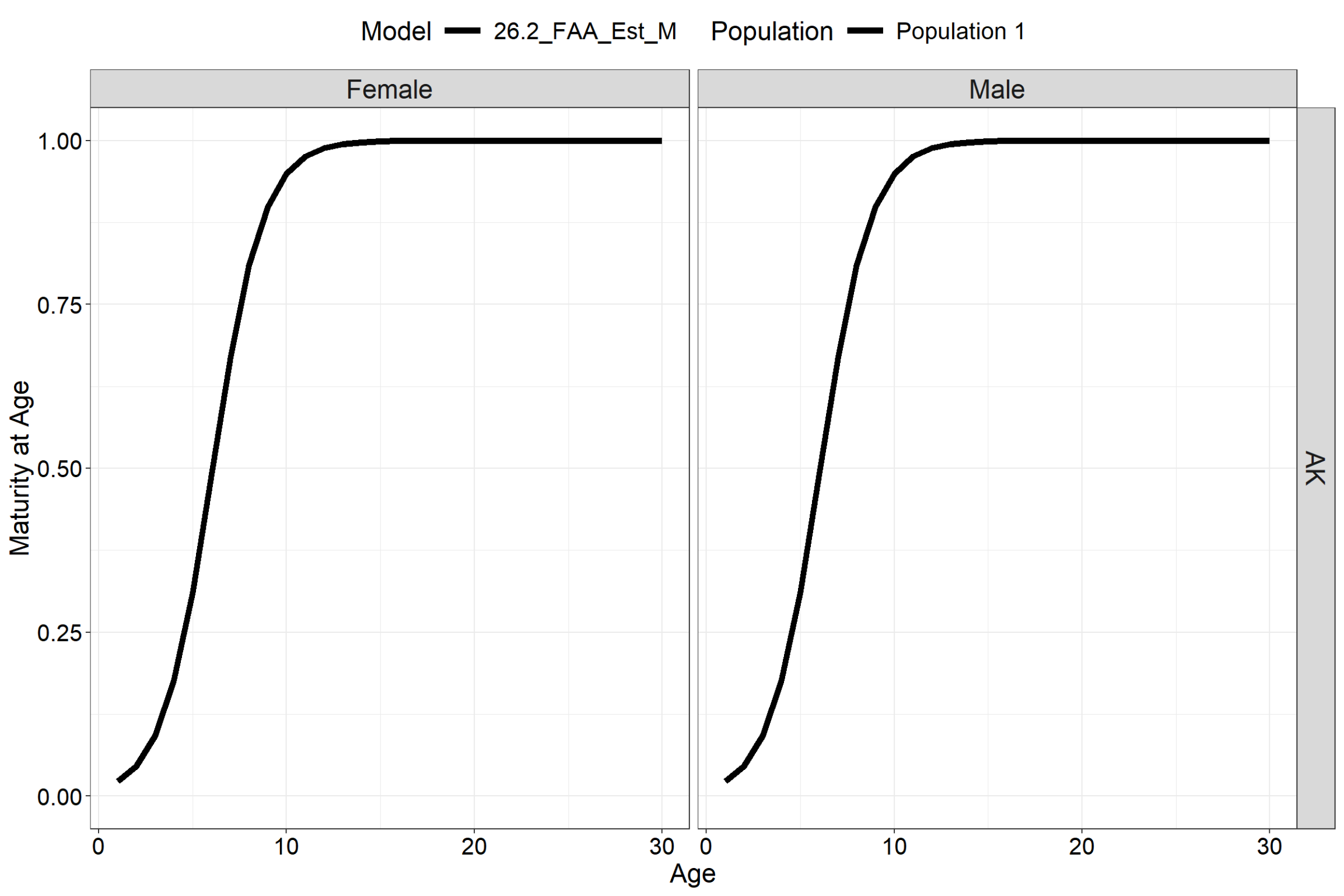
GOA LLS

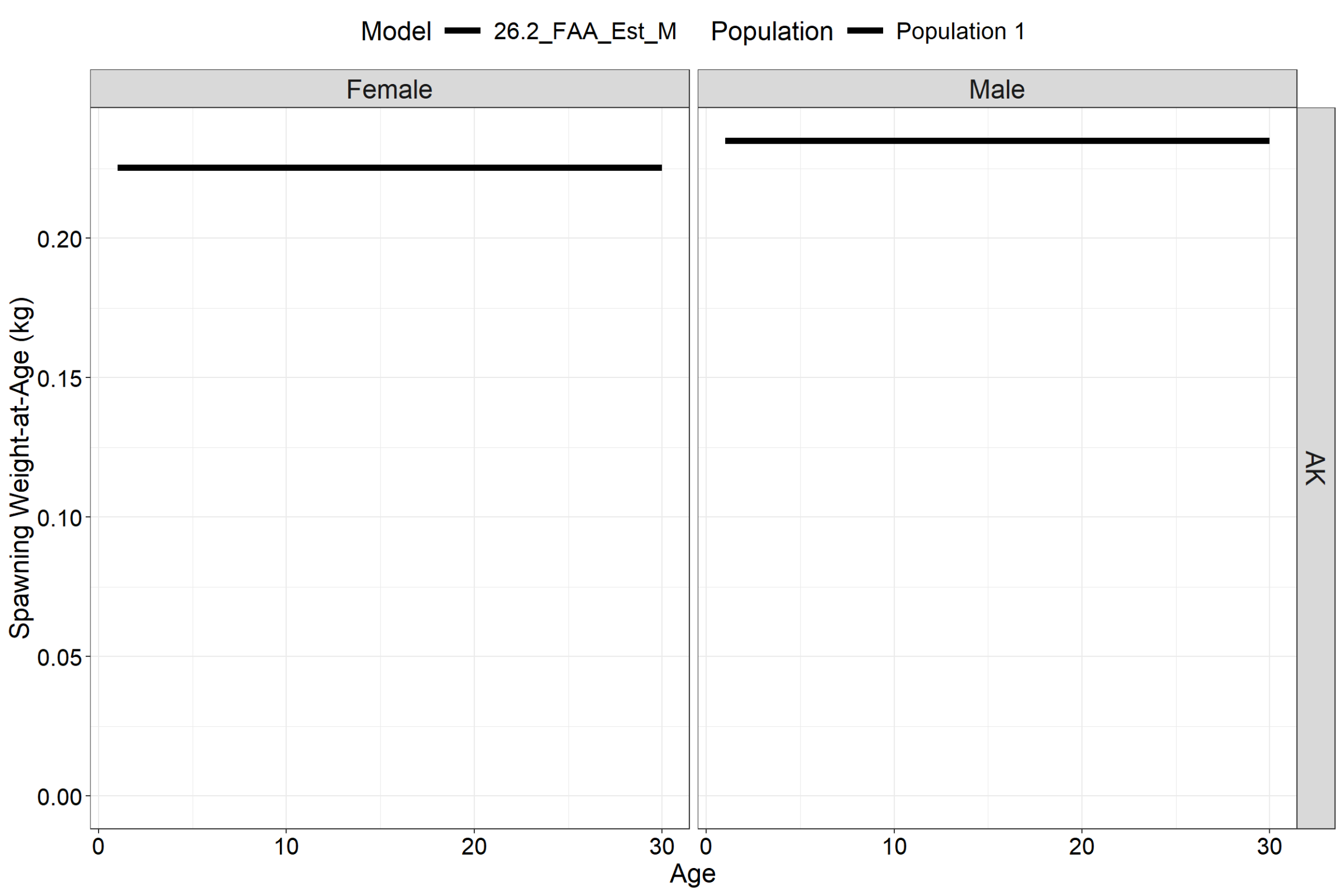
JPN LLS

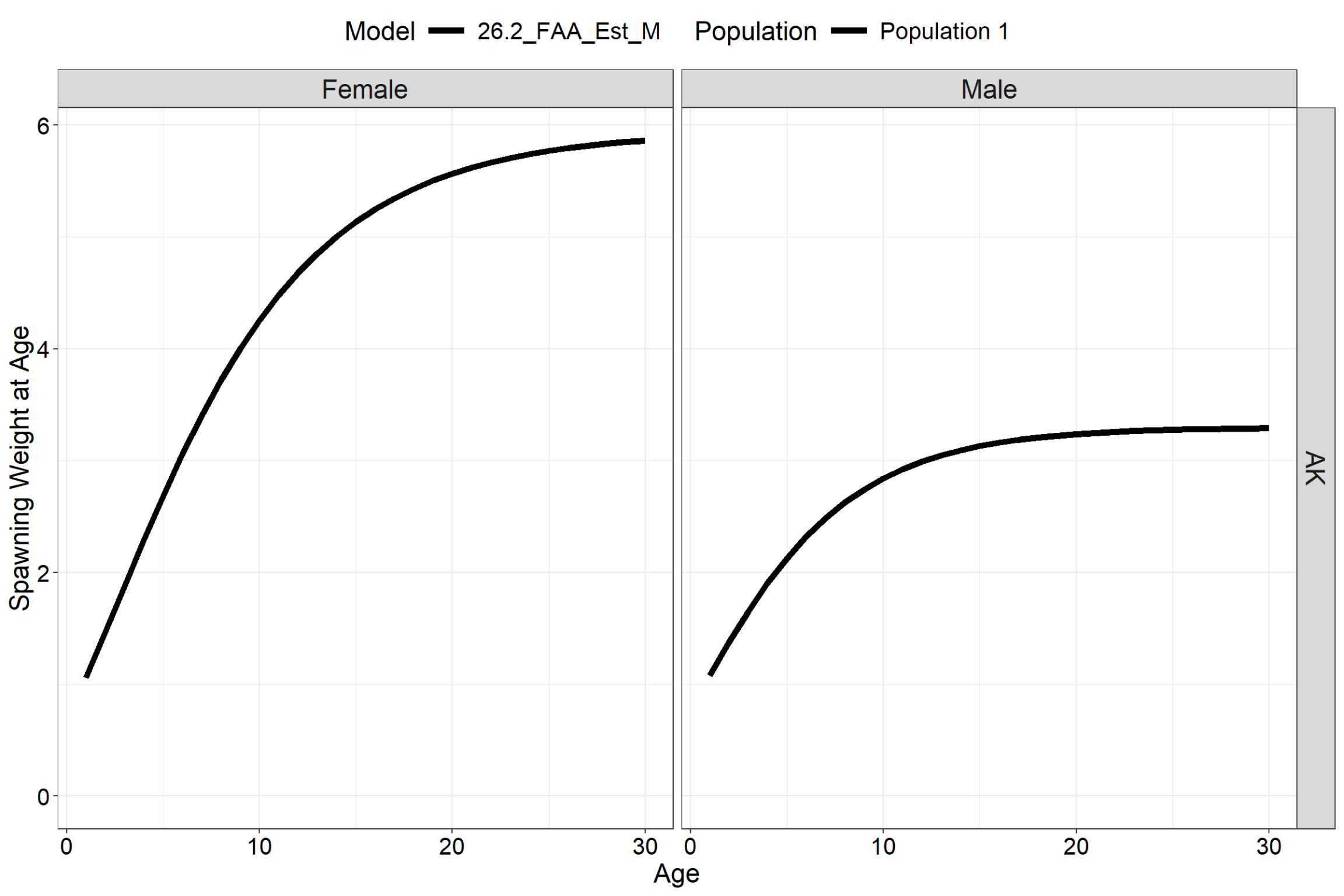
RPNs (1000s)

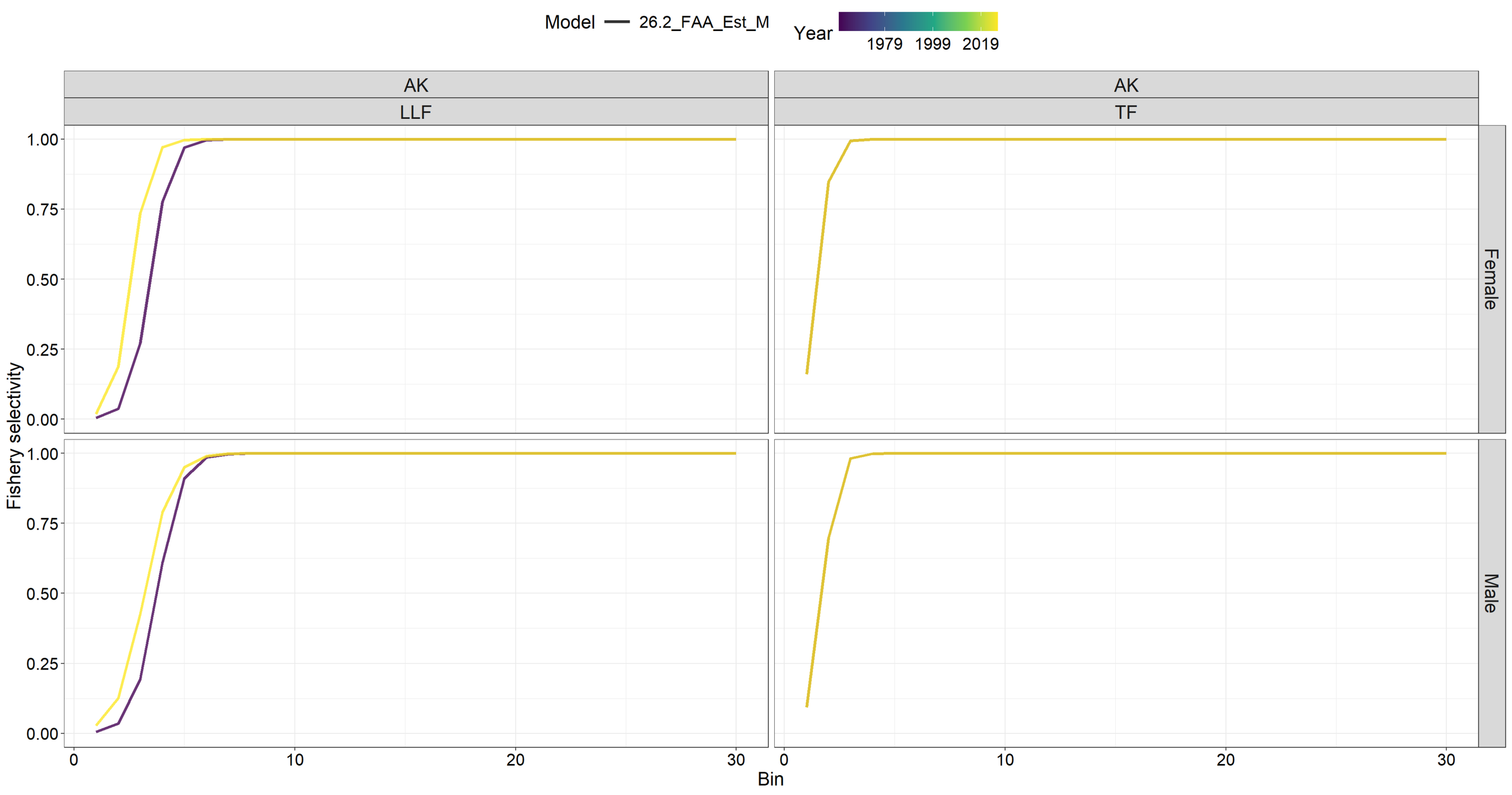
Year

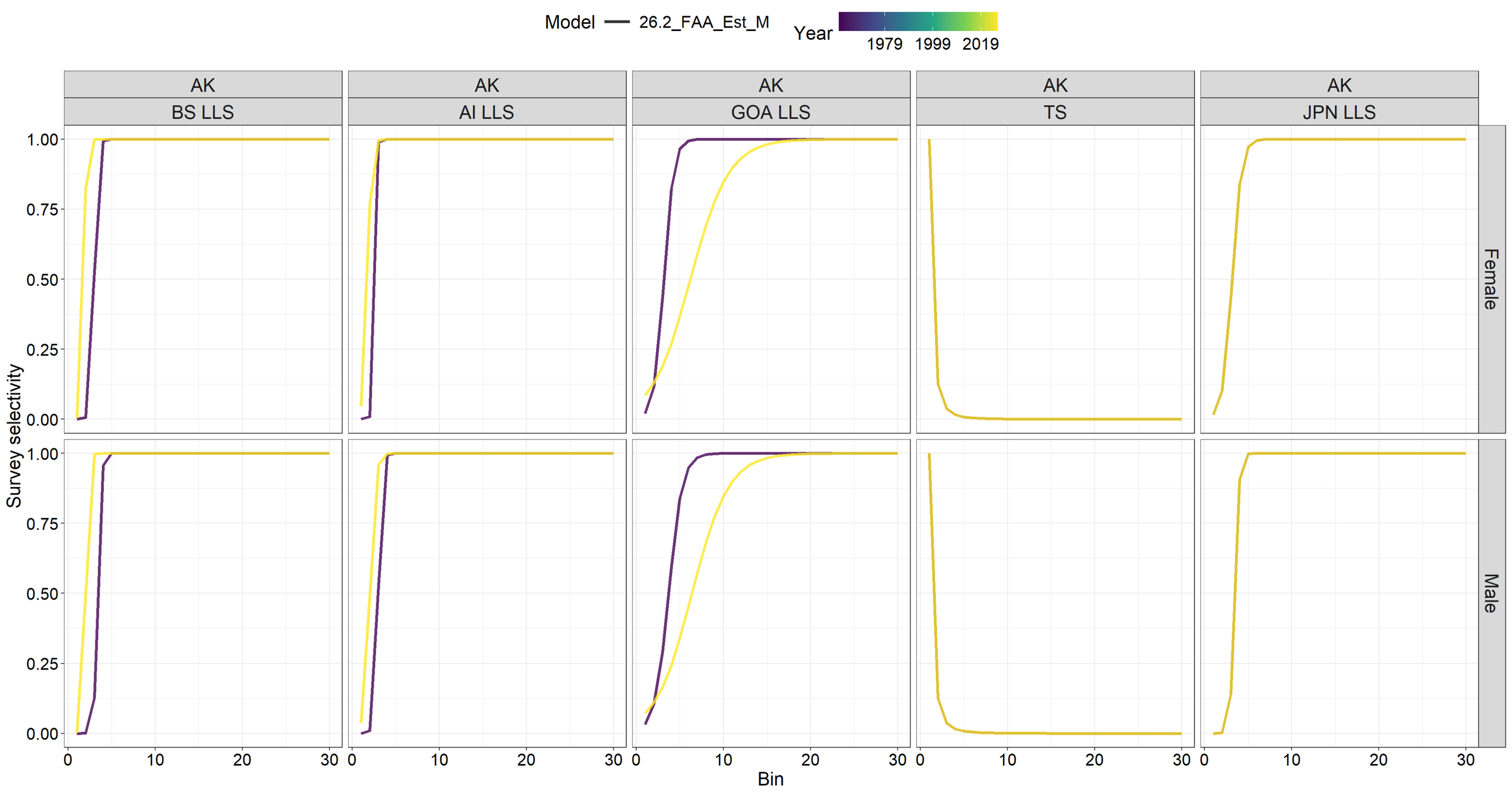

Index Residuals Runs Test
BS LLS
p = 0.092
AI LLS
p = 0.109
GOA LLS
p = 0.036
JPN LLS
p = 0.609
Residuals
Year


Model
26.2_FAA_Est_M
Population
Population 1
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age


Model
26.2_FAA_Est_M
Population
Population 1
Female
Male
AK
Spawning Weight-at-Age (kg)
0.00
0.05
0.10
0.15
0.20
0
10
20
30
Age


Model
26.2_FAA_Est_M
Population
Population 1
Female
Male
AK
Spawning Weight at Age
0
2
4
6
0
10
20
30
Age


Model
26.2_FAA_Est_M
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model 26.2_FAA_Est_M
Year
1979 1999 2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0 10 20 30
Bin

Phase Plane

1.00
0.75
0.50
0.25
0.00

$F/F_{35\%}$

0
2
4

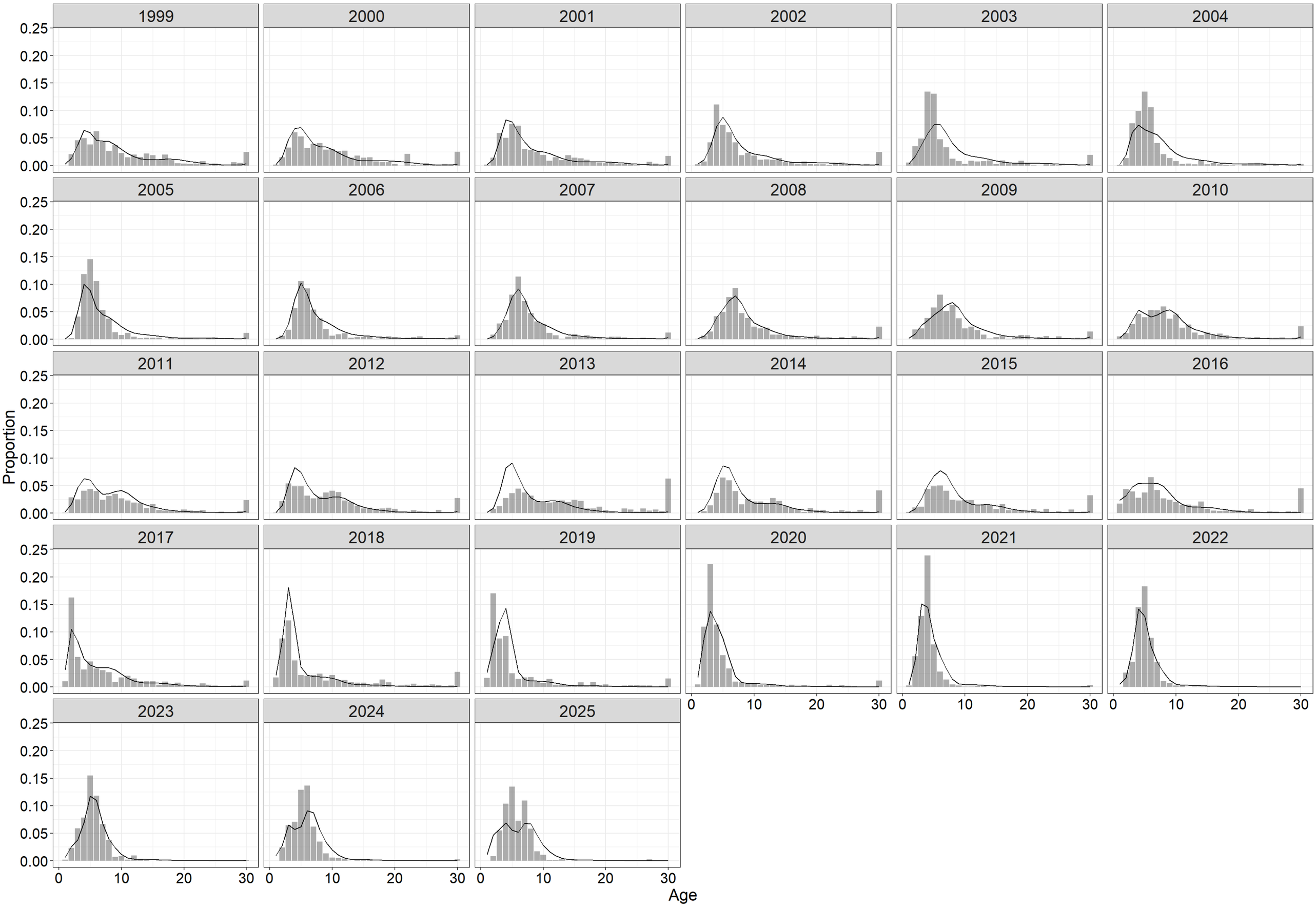
$SSB/B_{35\%}$

2010 2005 2020 1965 1985 1990
2015 2000 1995 1980 1960 1970 1975 2025 **2026**

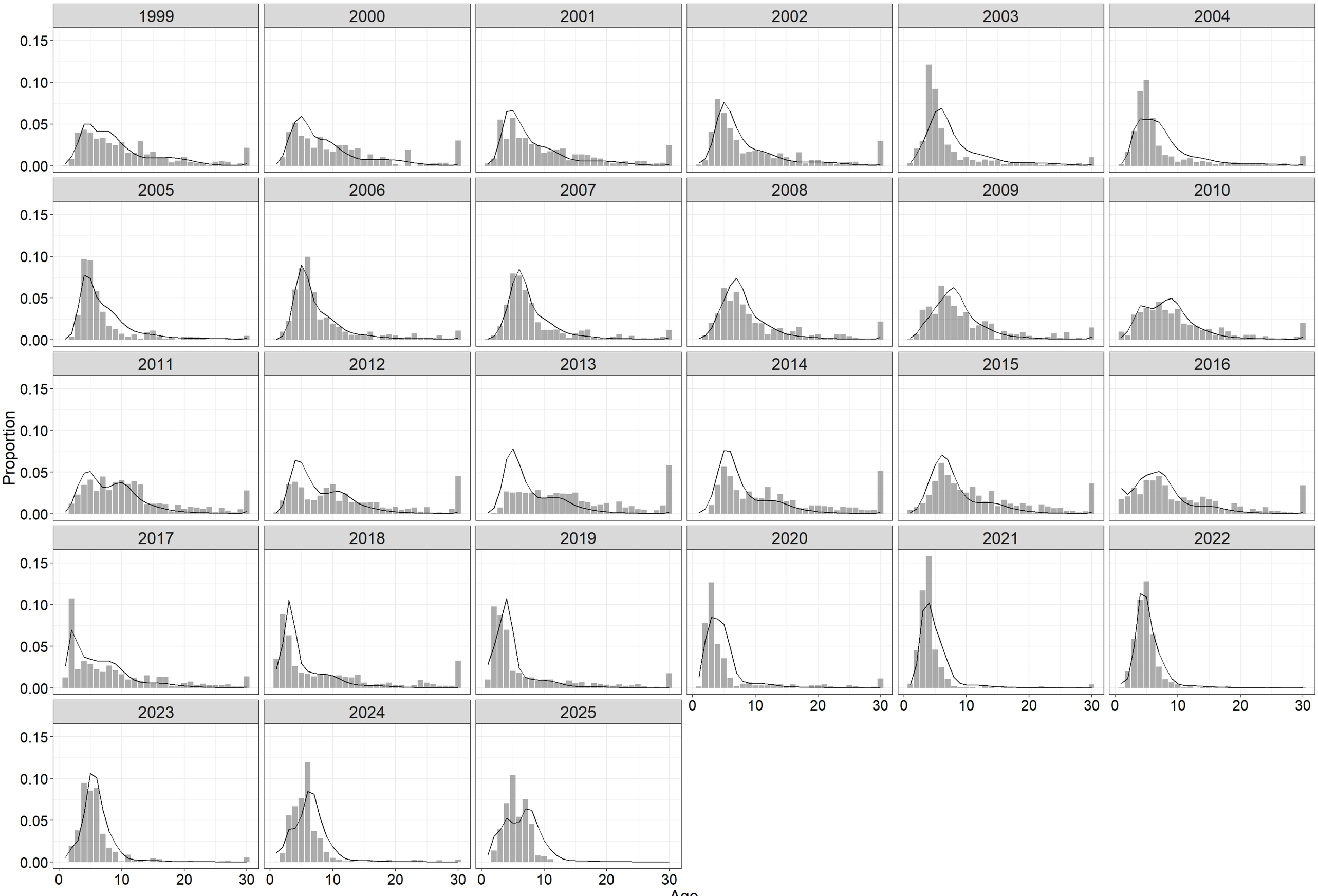
Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)

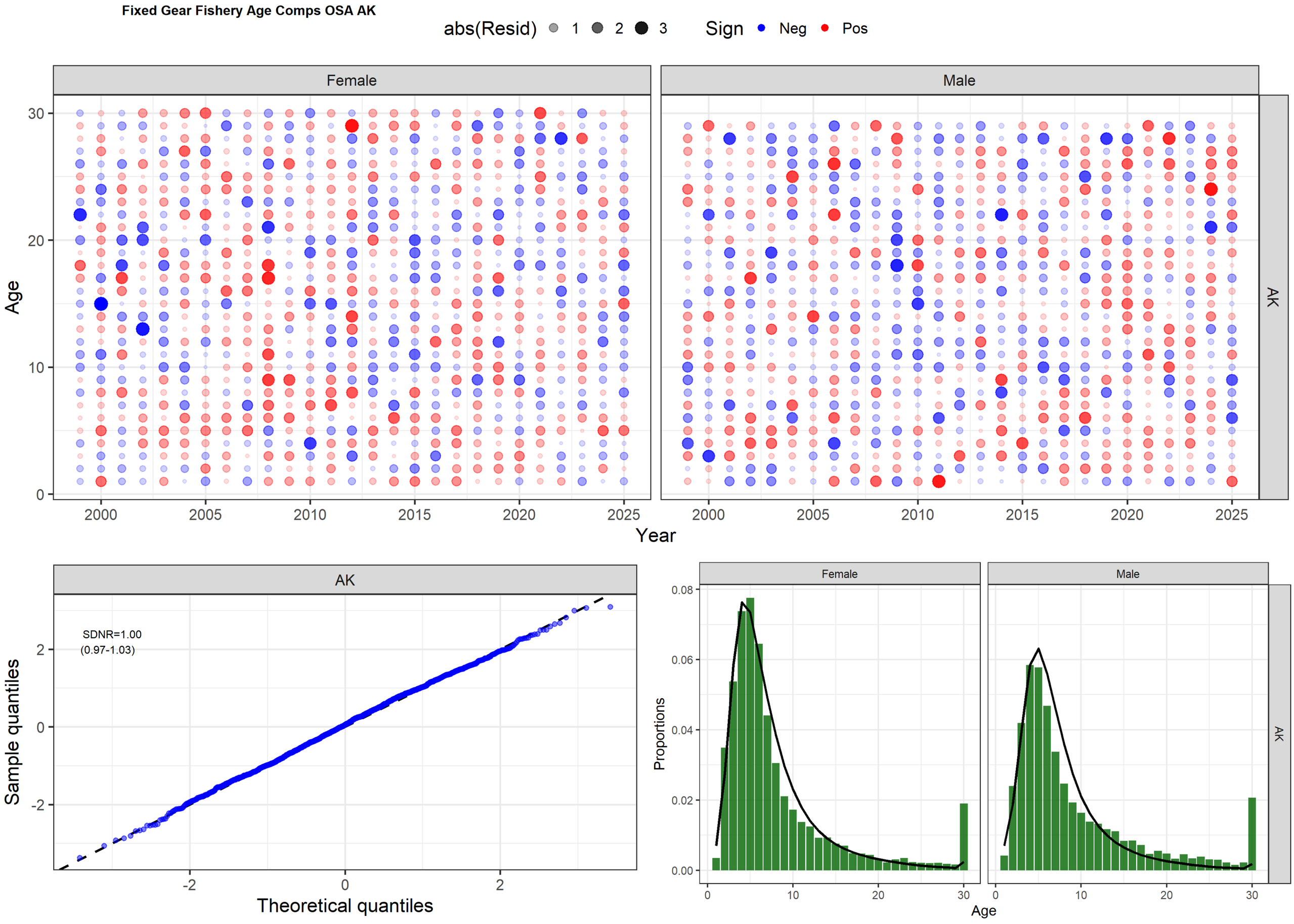
Fixed Gear Fishery Age Comps Female AK



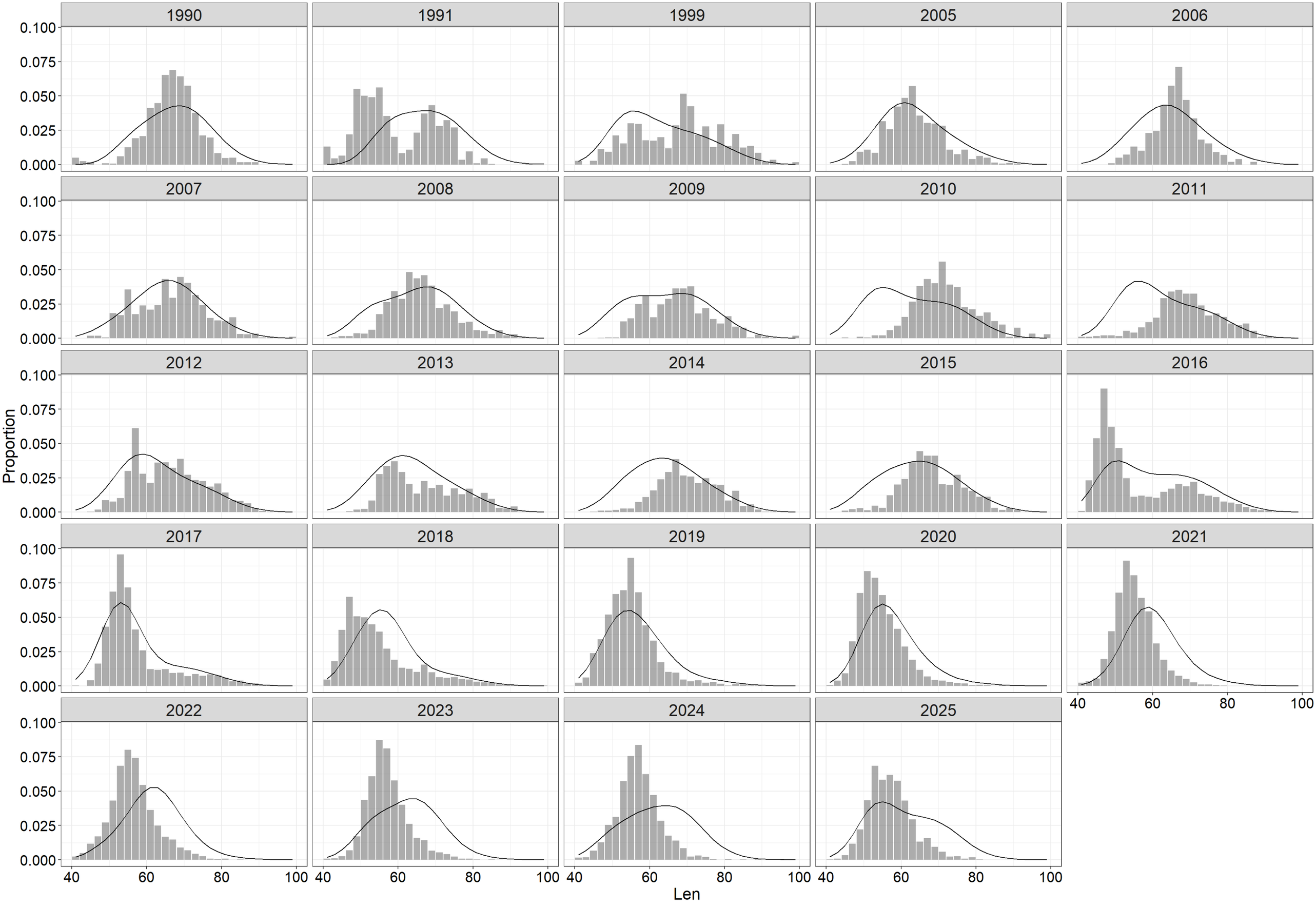
Fixed Gear Fishery Age Comps Male AK



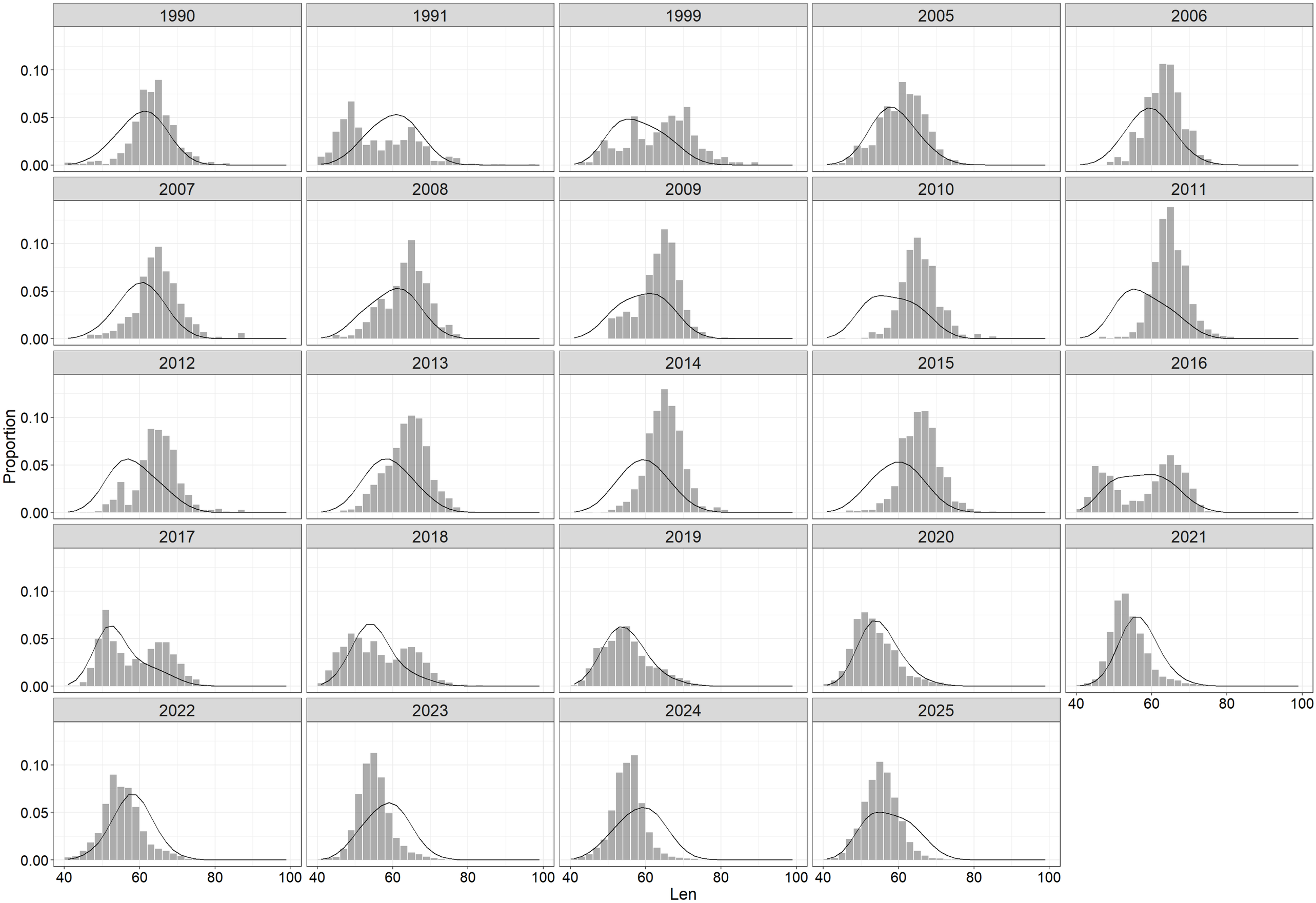
Fixed Gear Fishery Age Comps OSA AK



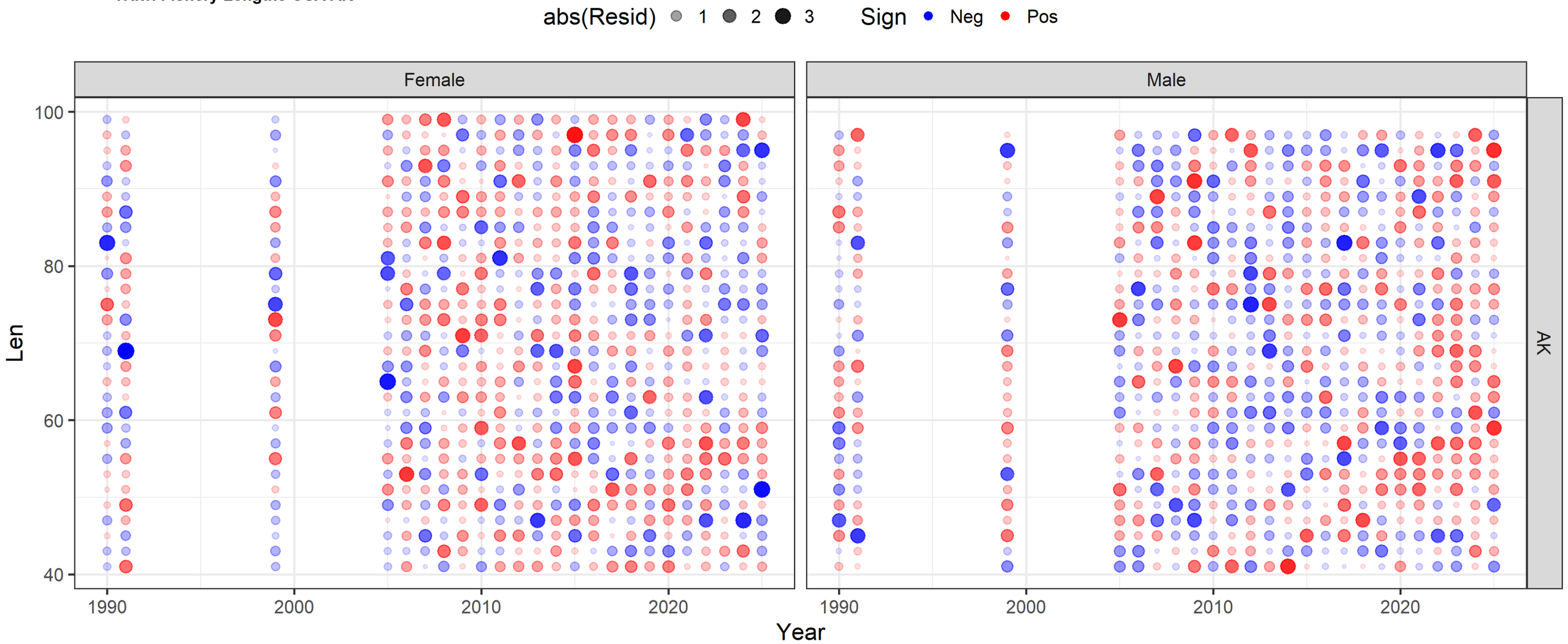
Trawl Fishery Lengths Female AK

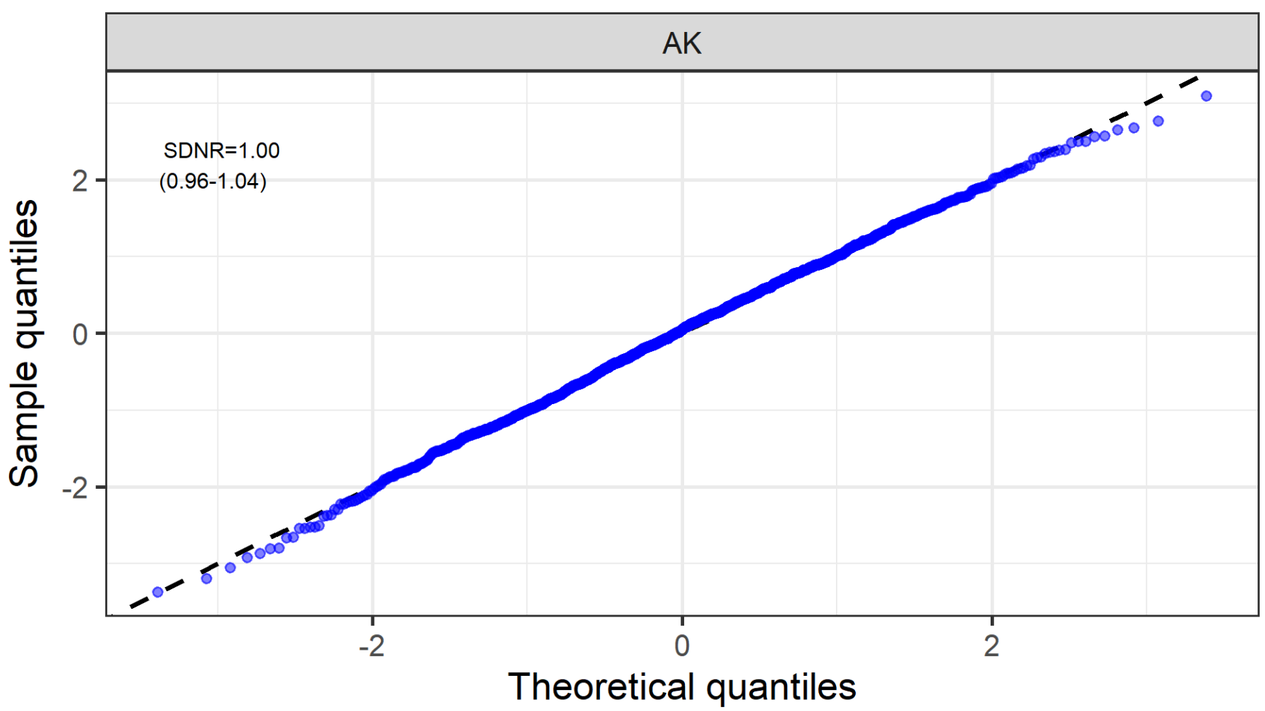


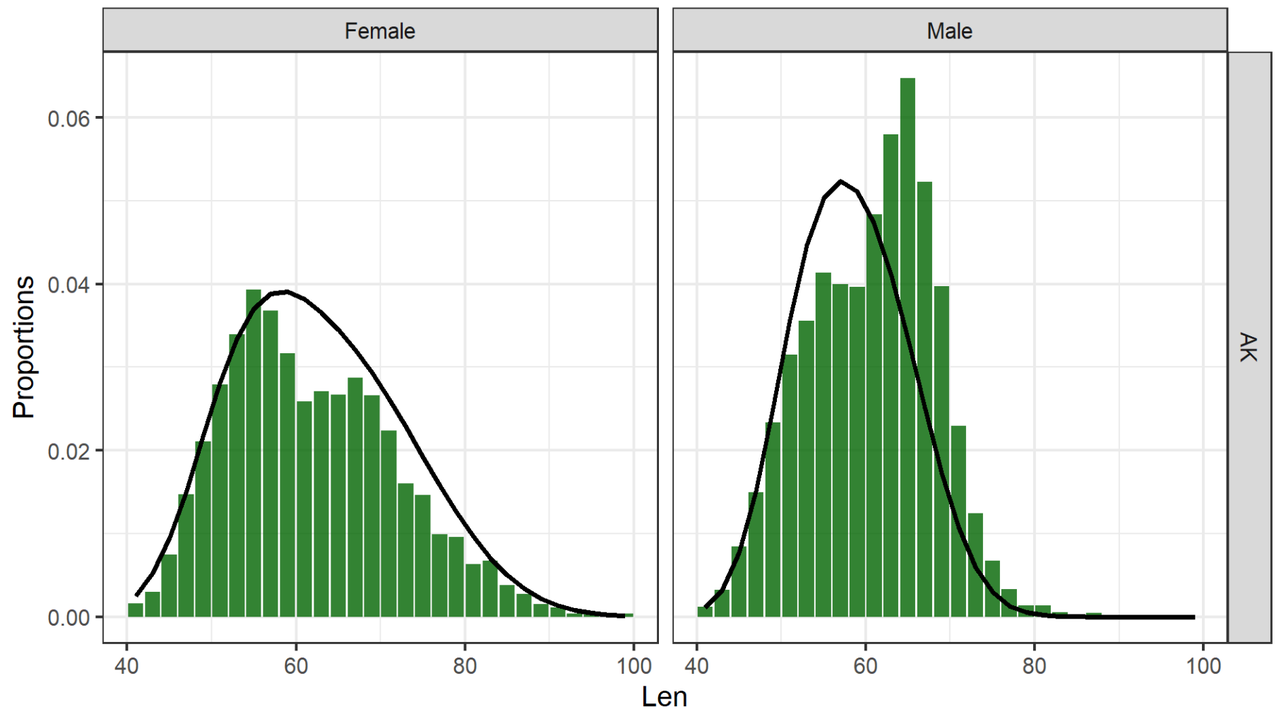
Trawl Fishery Lengths Male AK



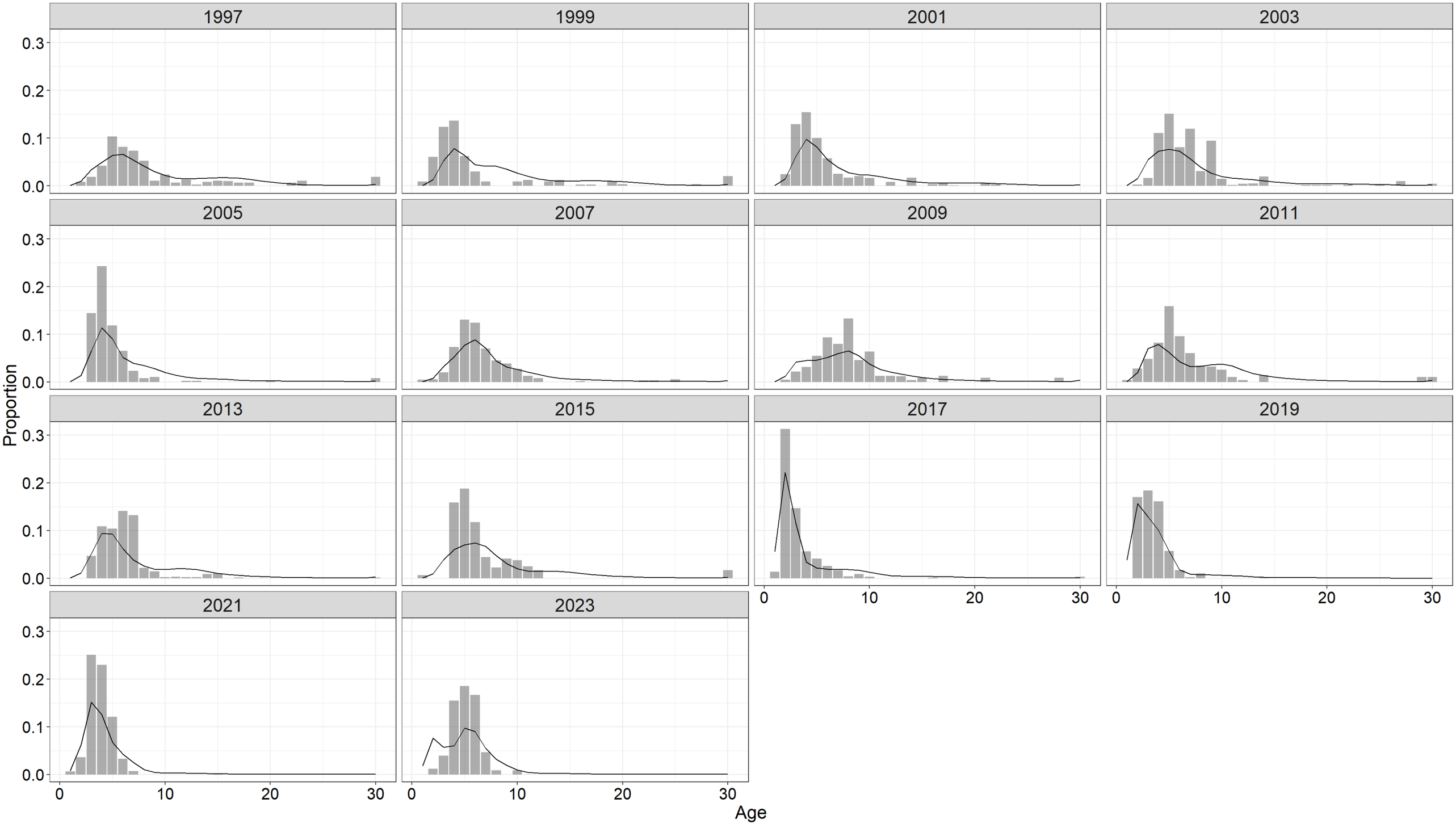
Trawl Fishery Lengths OSA AK







BS LL Survey Age Comps Female AK

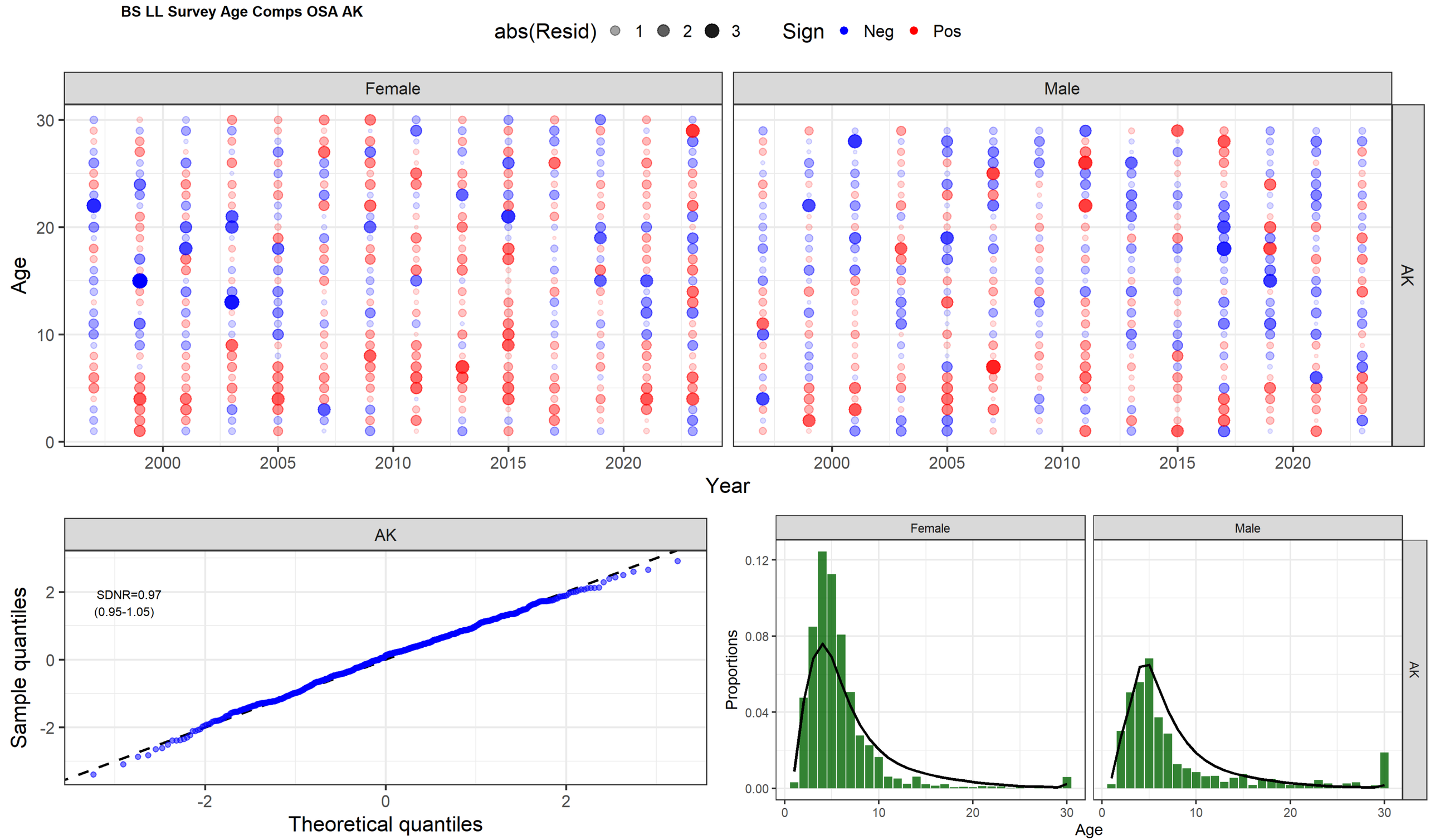


BS LL Survey Age Comps Male AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

BS LL Survey Age Comps OSA AK



AI LL Survey Age Comps Female AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.00 0.05 0.10 0.15 0.20
0 10 20 30
Age

AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.96
(0.95-1.05)

Sample quantiles

Theoretical quantiles

Female

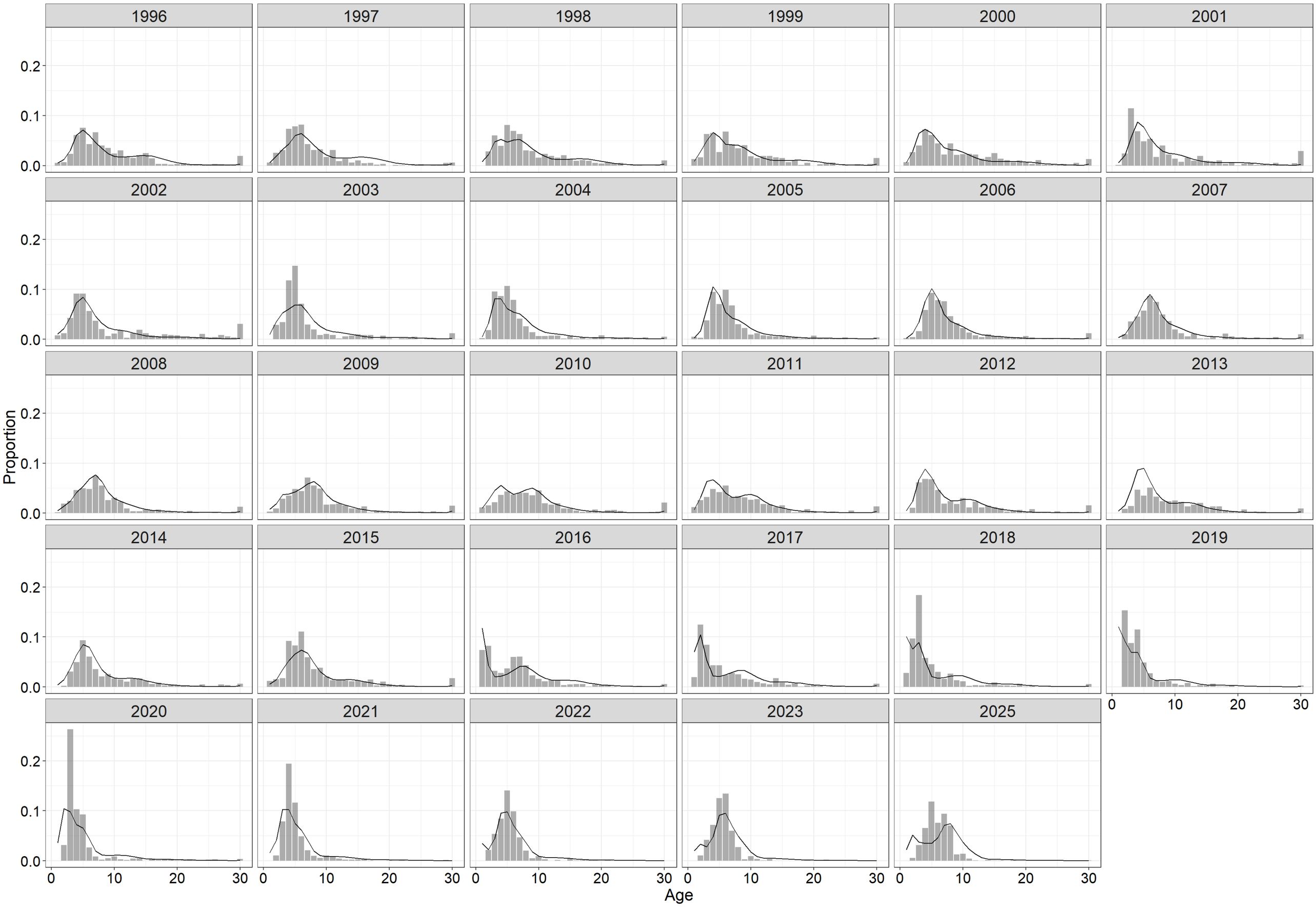
Male

AK

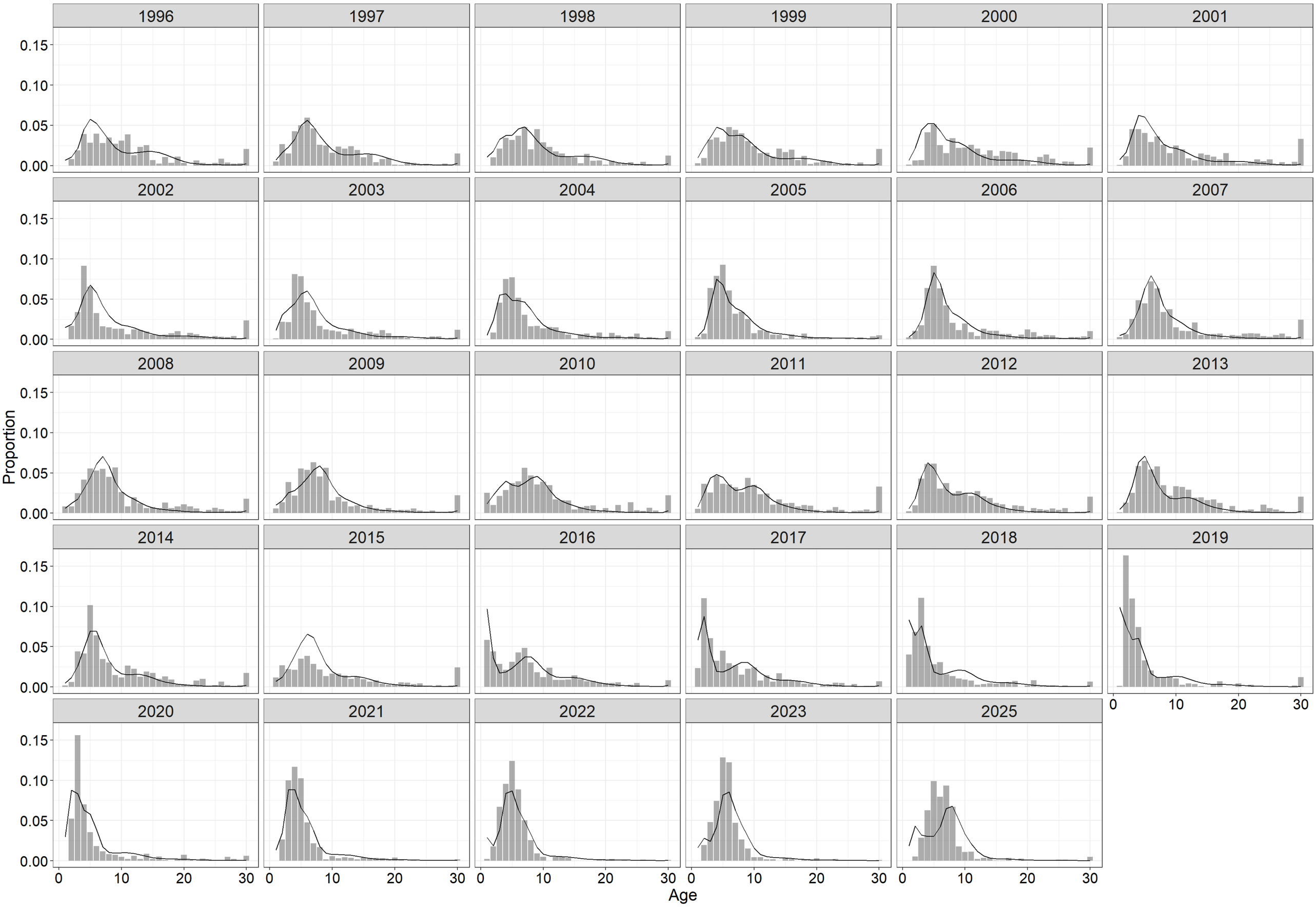
Proportions

Age

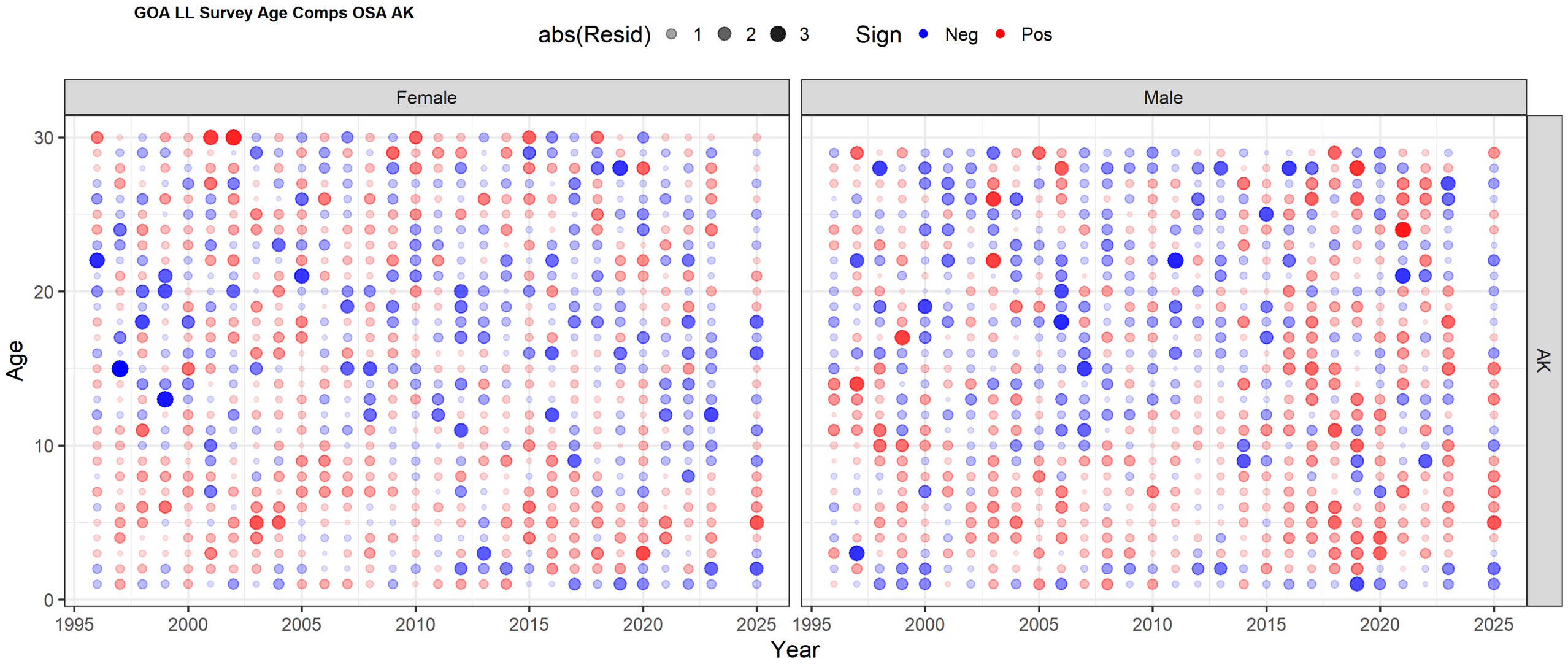
GOA LL Survey Age Comps Female AK

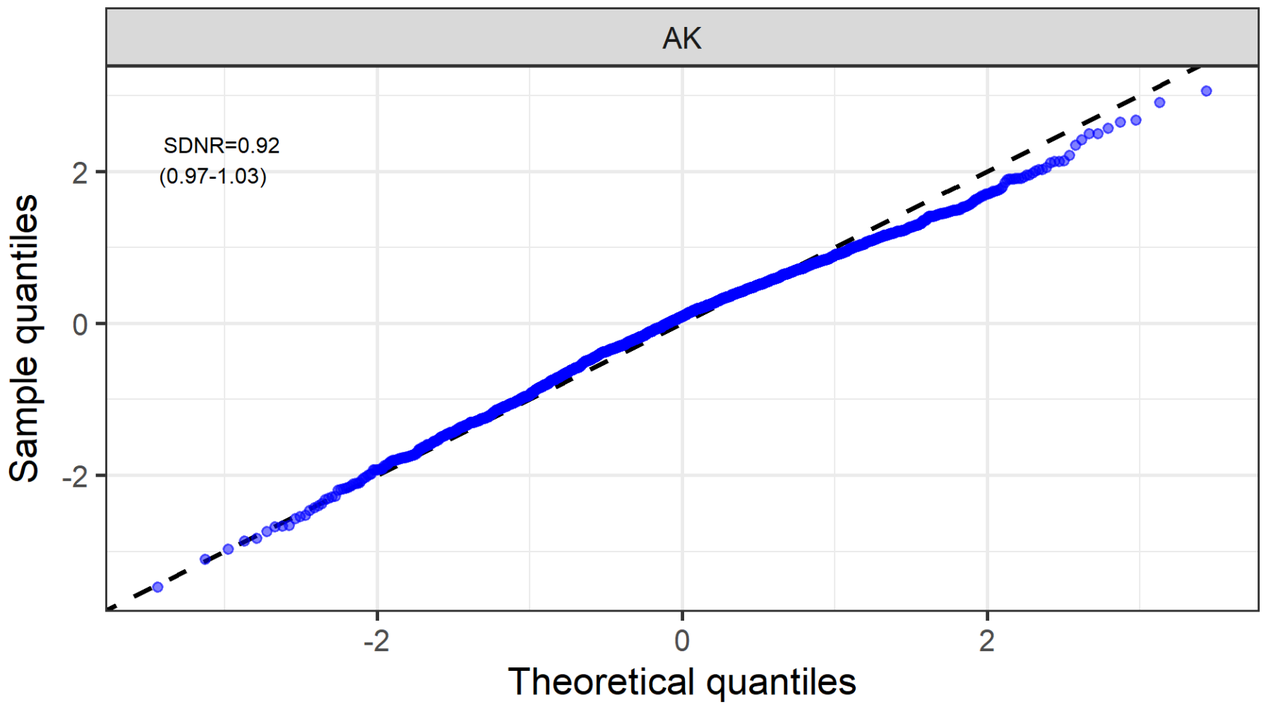


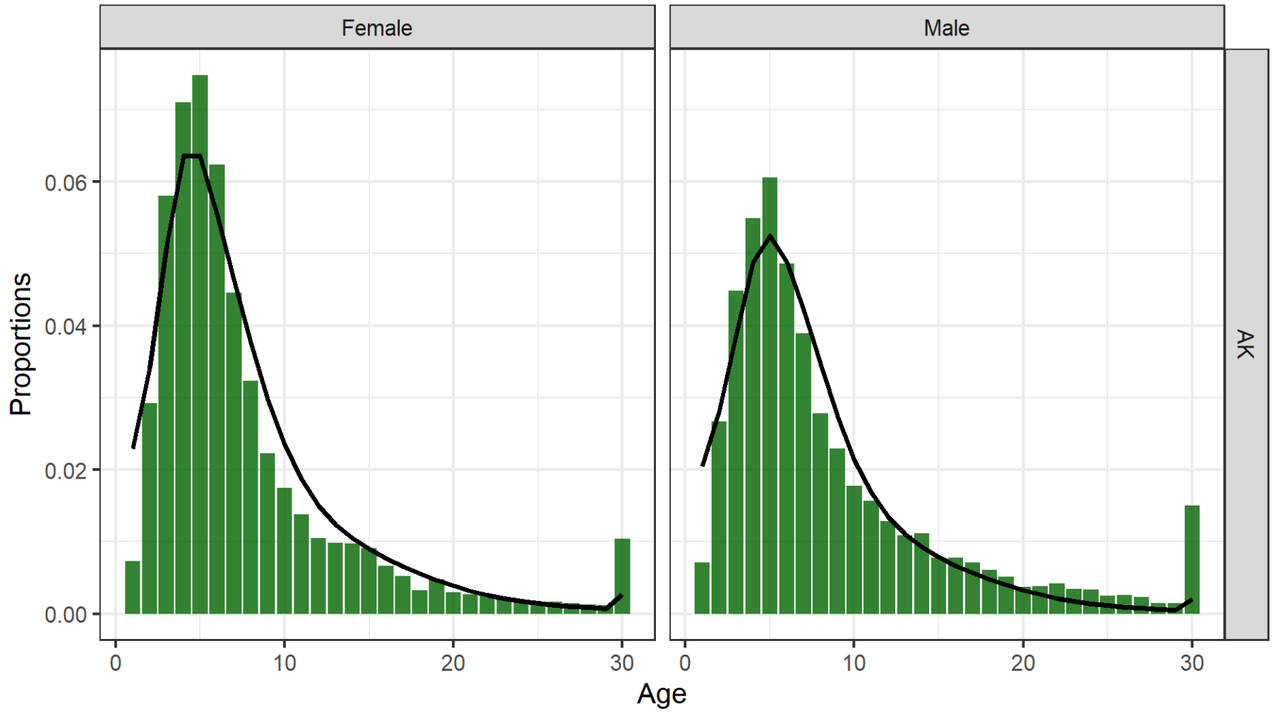
GOA LL Survey Age Comps Male AK

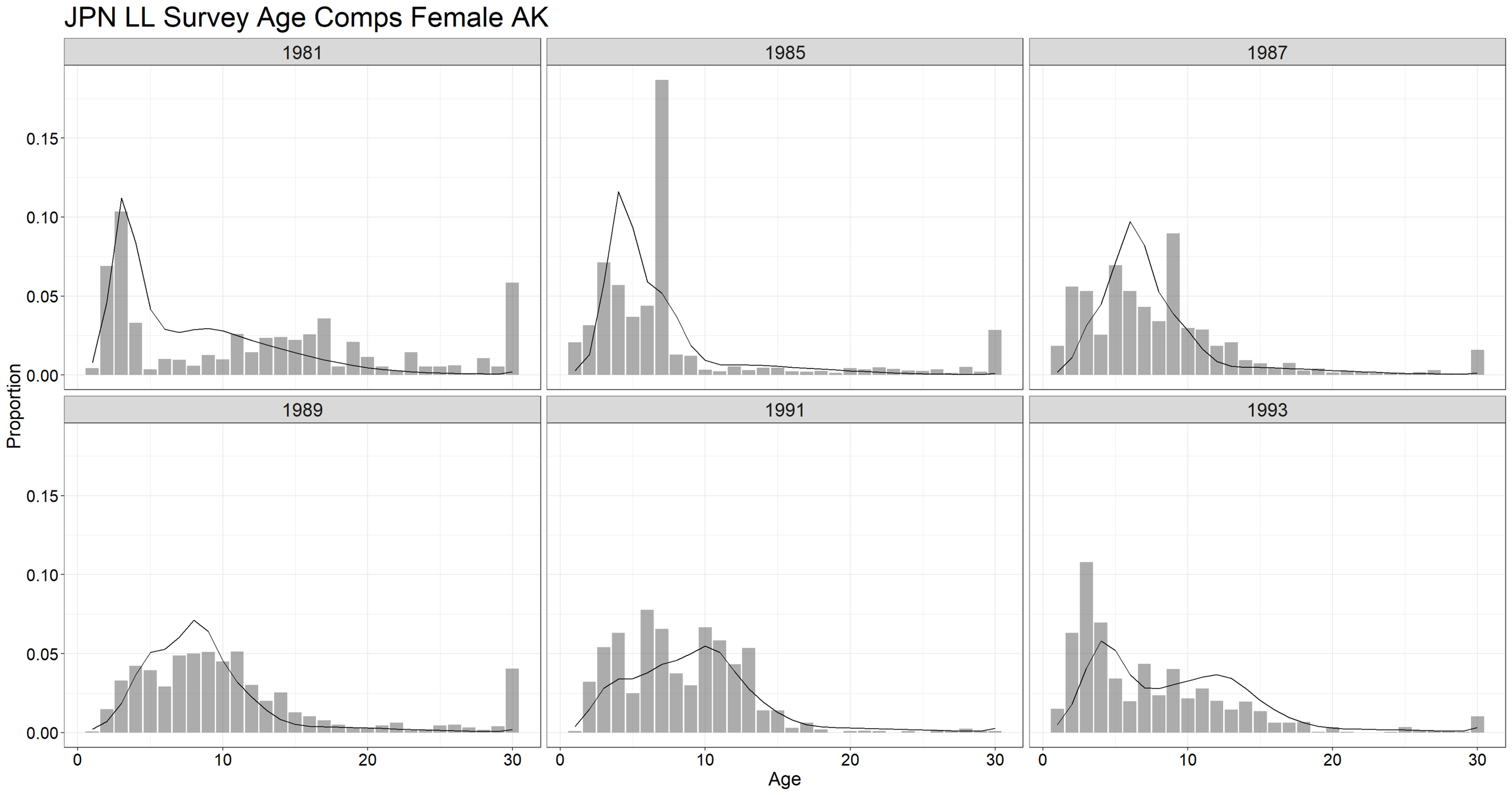


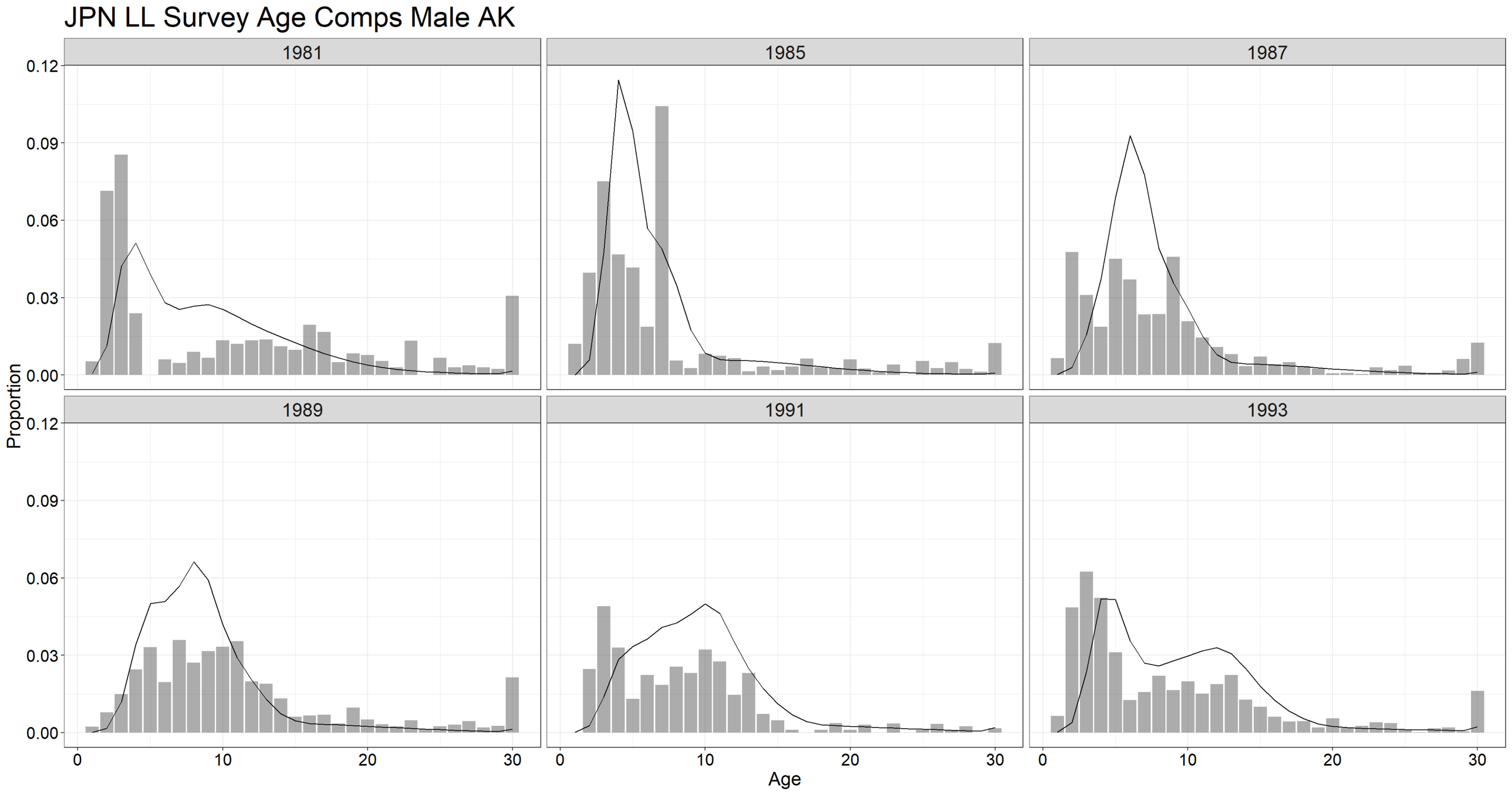
GOA LL Survey Age Comps OSA AK

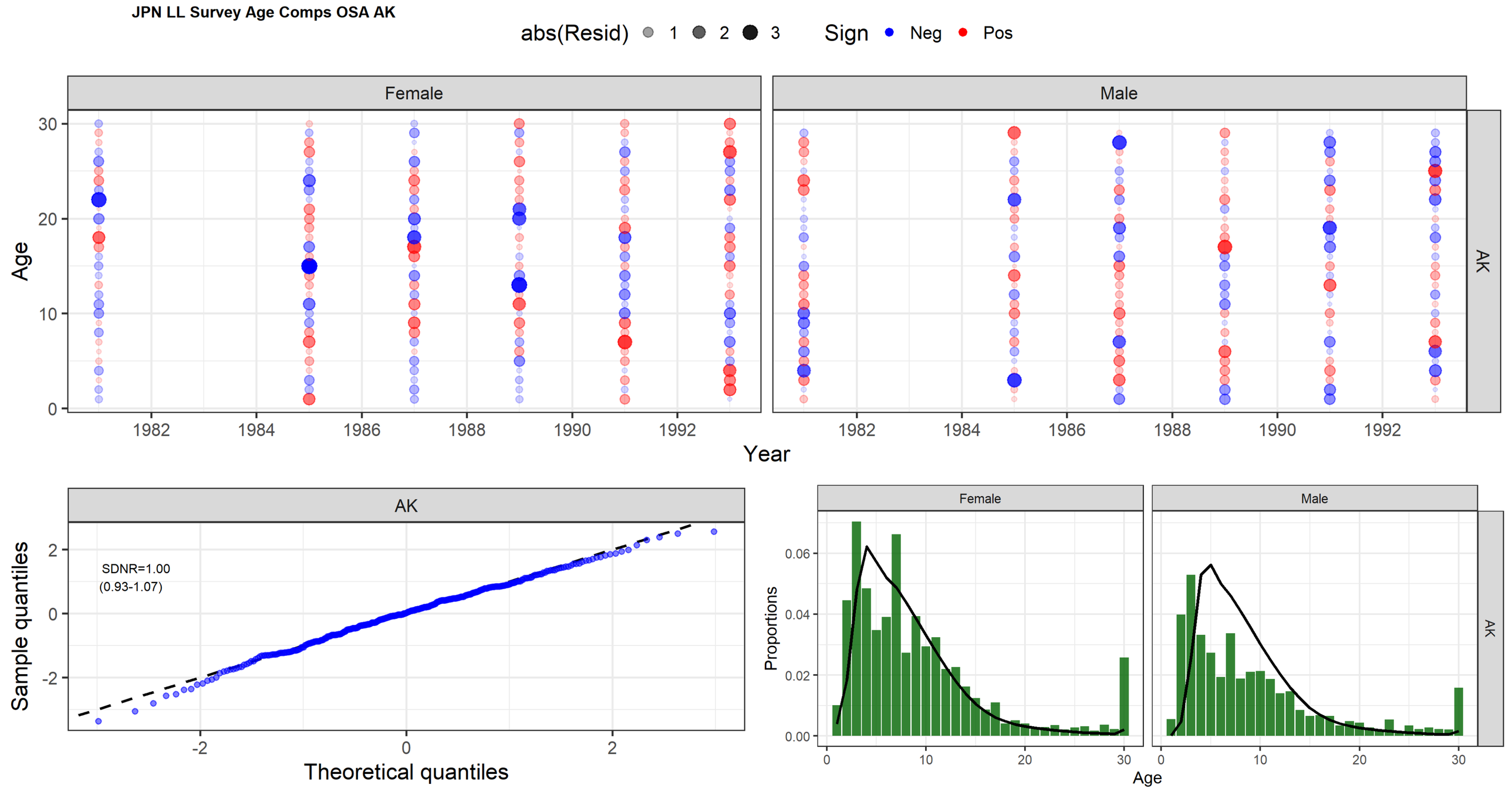




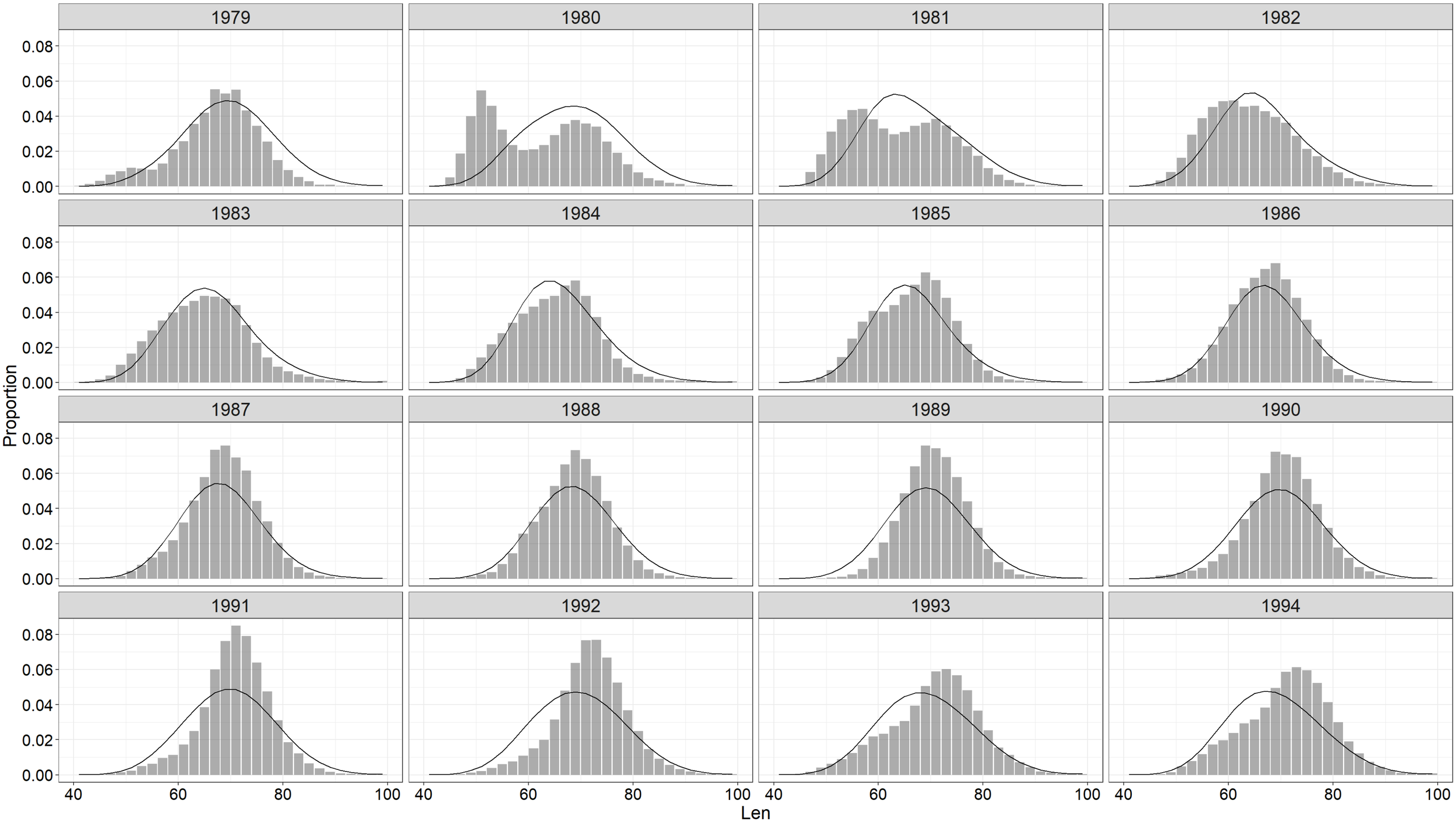



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age

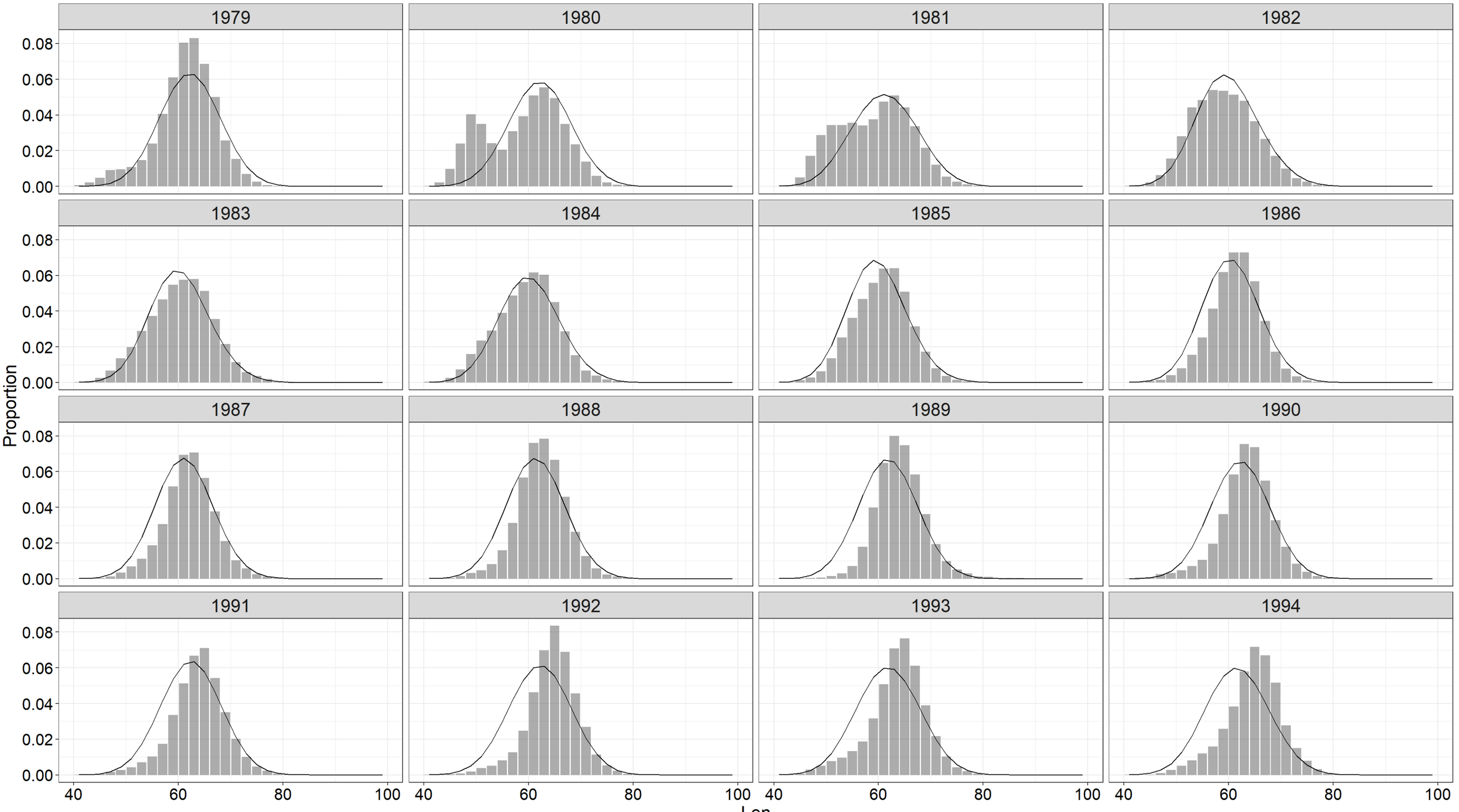

JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
Proportion
0.12
0.09
0.06
0.03
0.00
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=1.00
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions

JPN LL Survey Length Comps Female AK



JPN LL Survey Length Comps Male AK



JPN LL Survey Length Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Len

100
80
60
40

1980 1985 1990 1980 1985 1990

Year

AK

SDNR=0.90
(0.95-1.05)

Sample quantiles

2
0
-2

-2 0 2

Theoretical quantiles

Female

Male

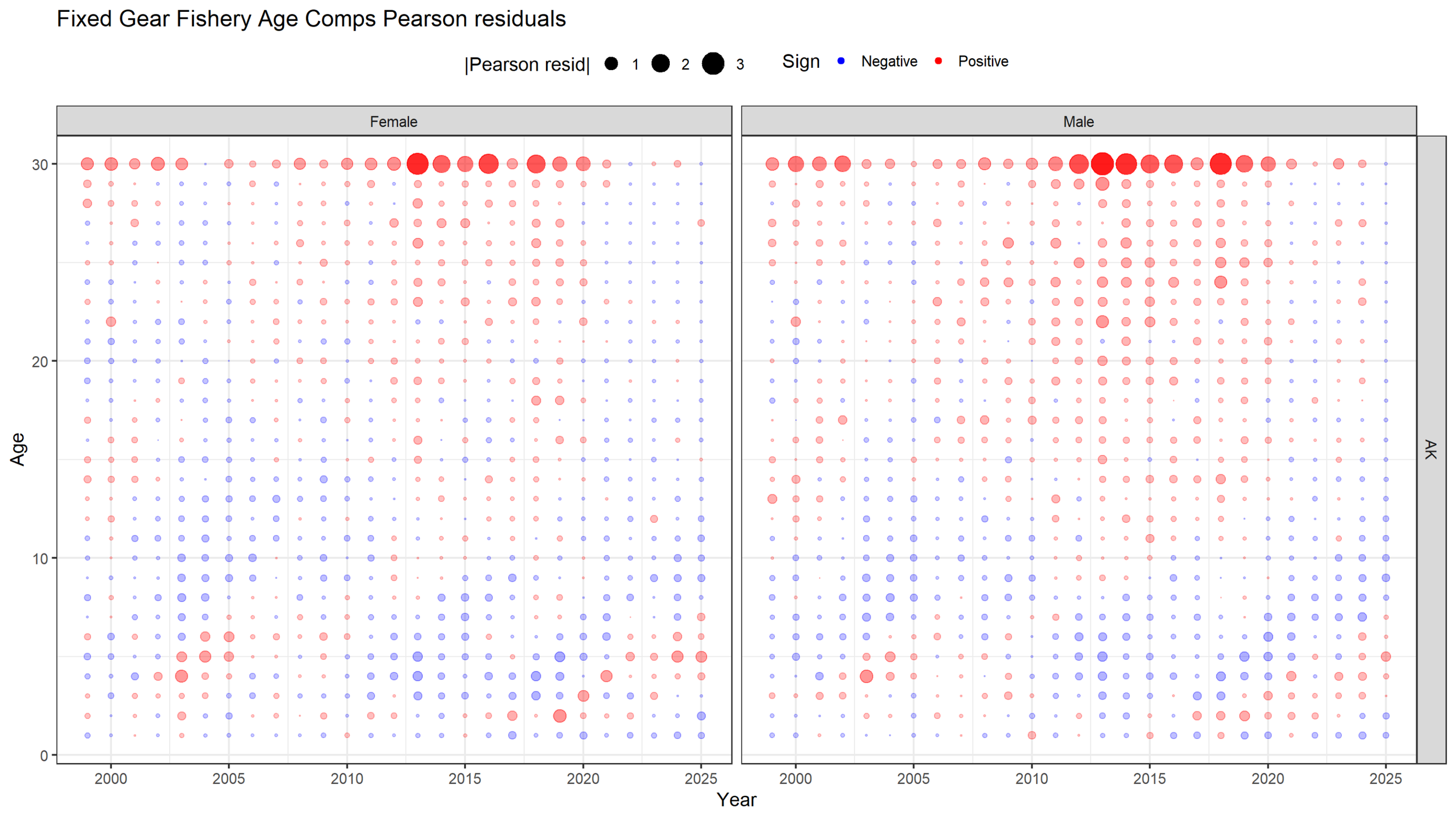
AK

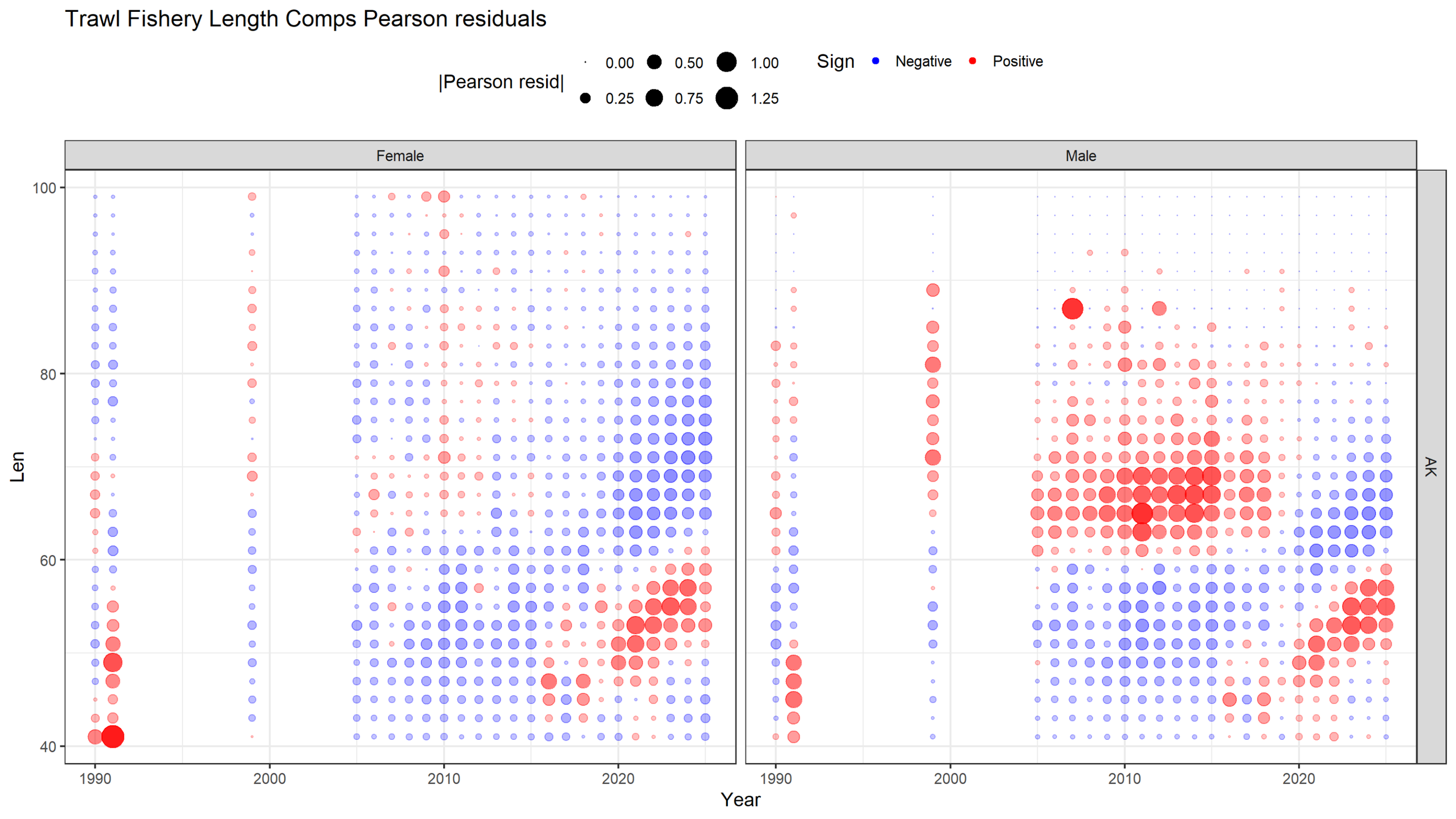
Proportions

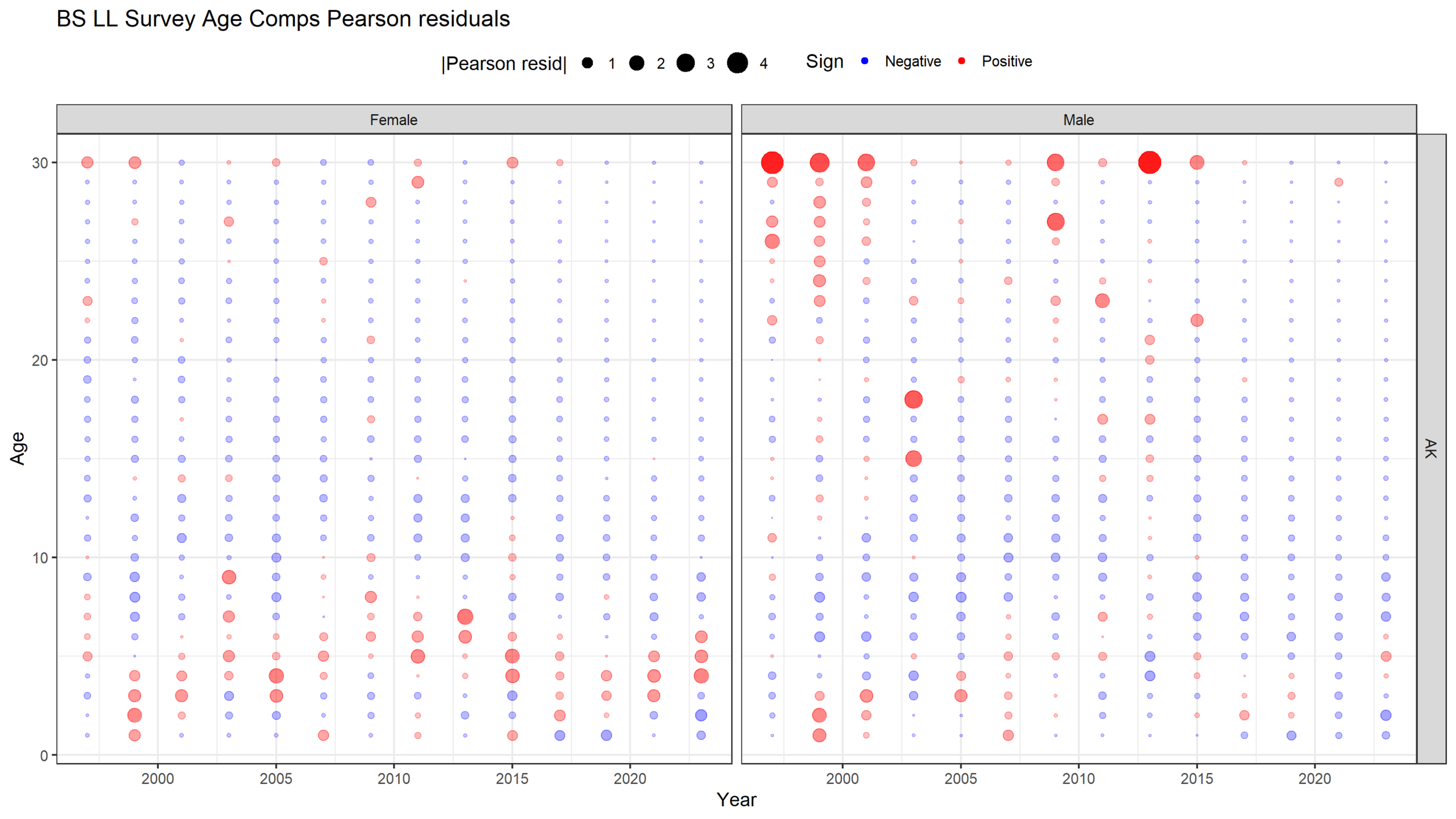
0.06
0.04
0.02
0.00

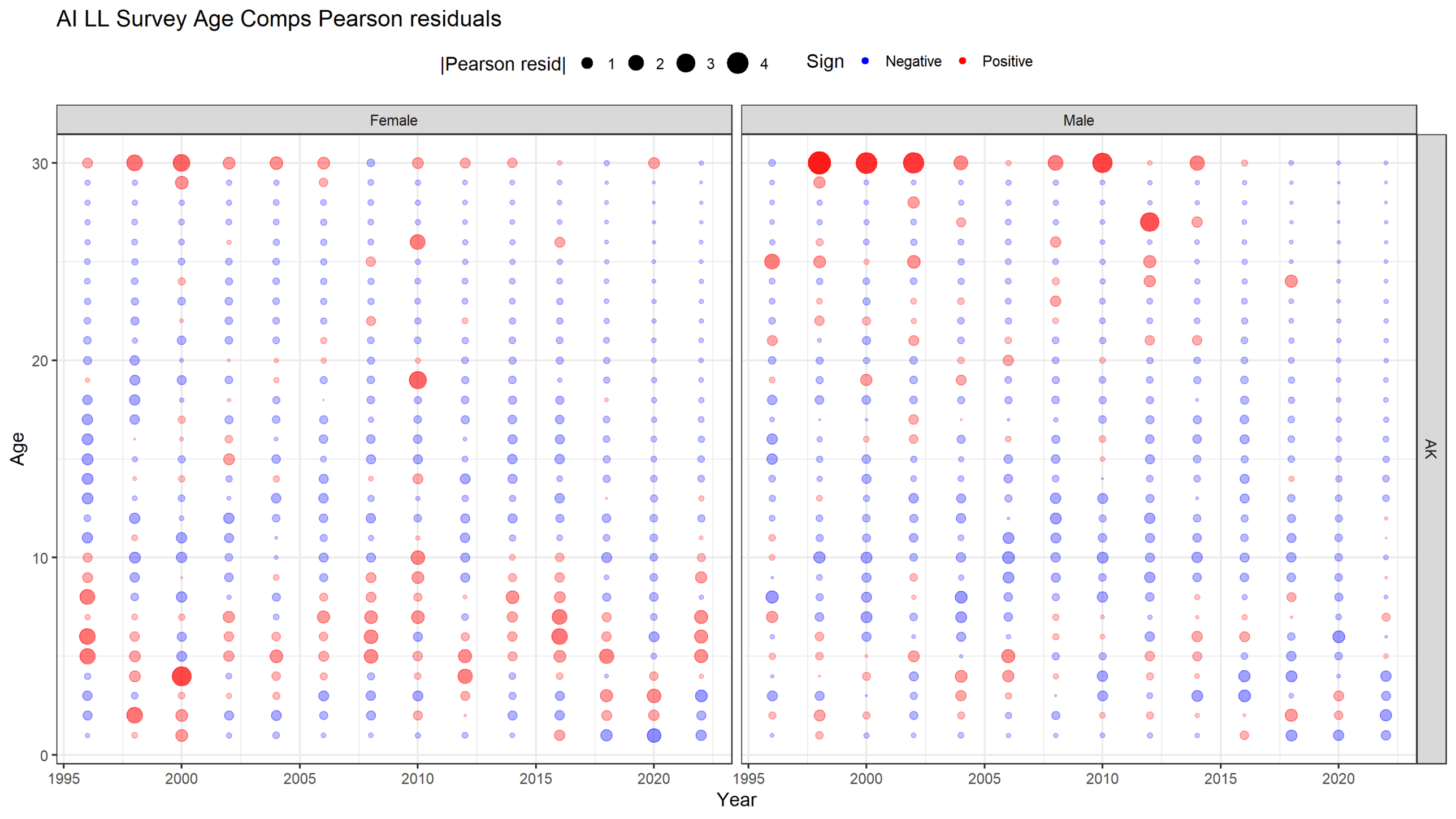
40 60 80 100 40 60 80 100

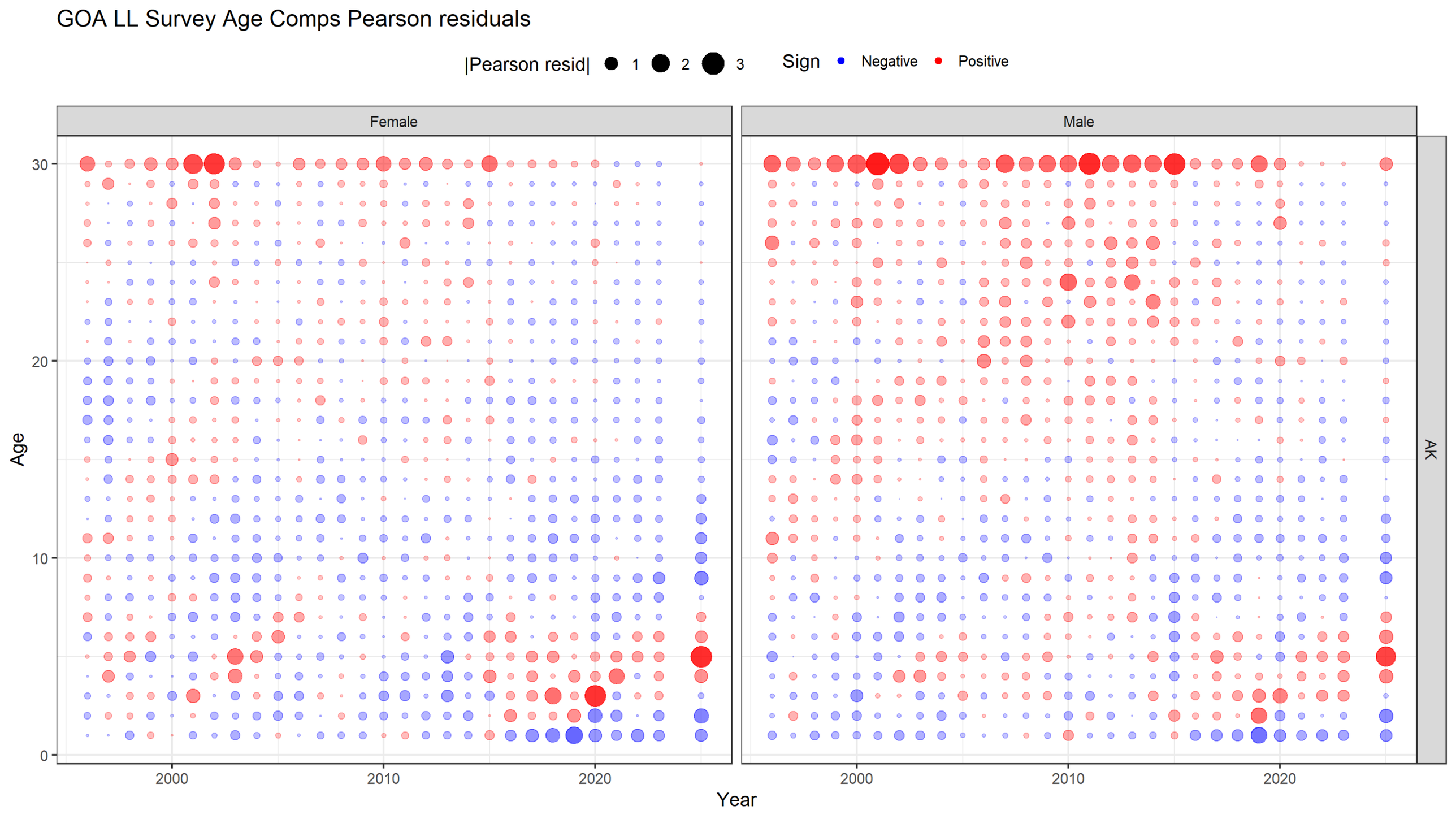
Len

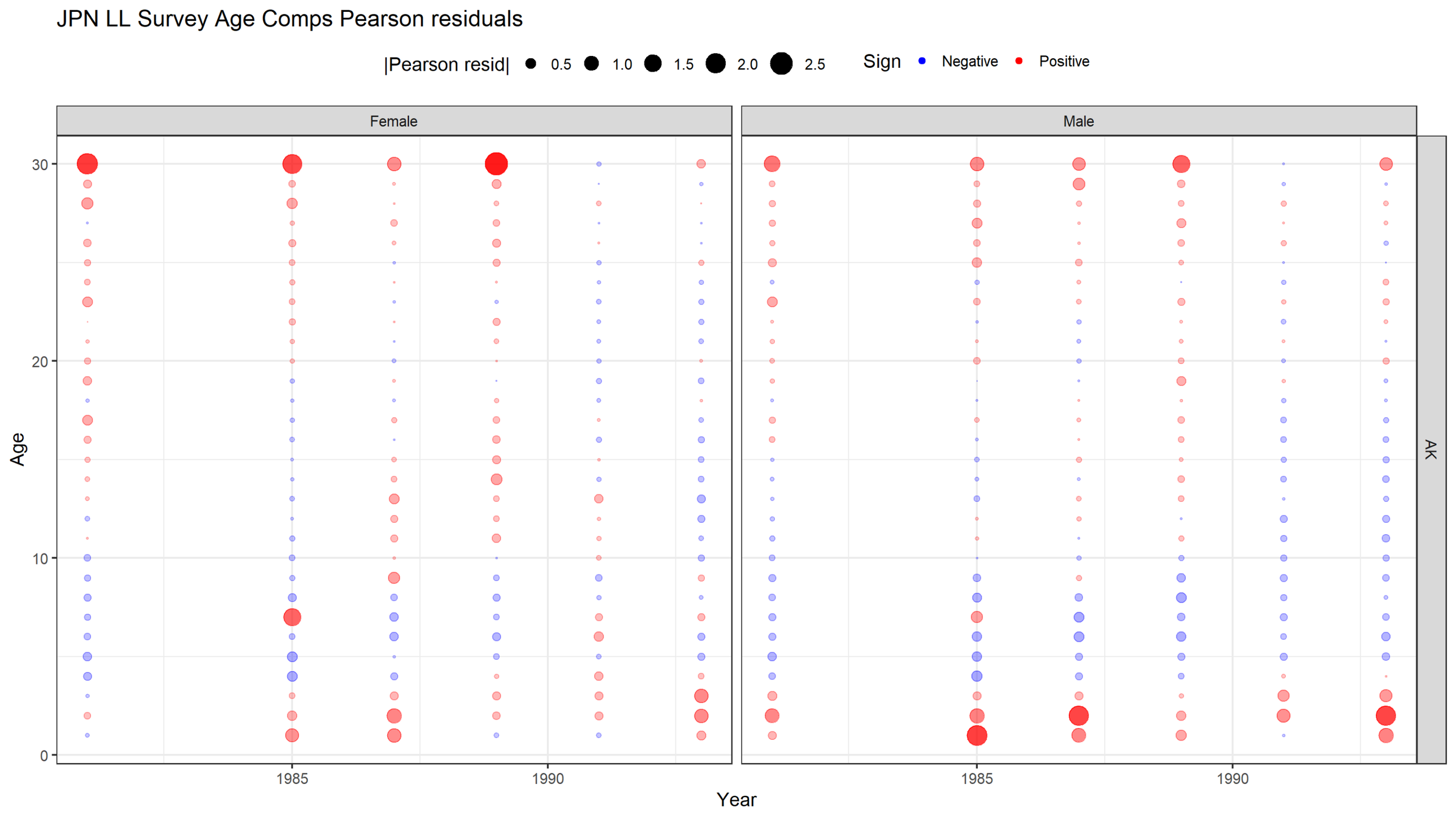

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

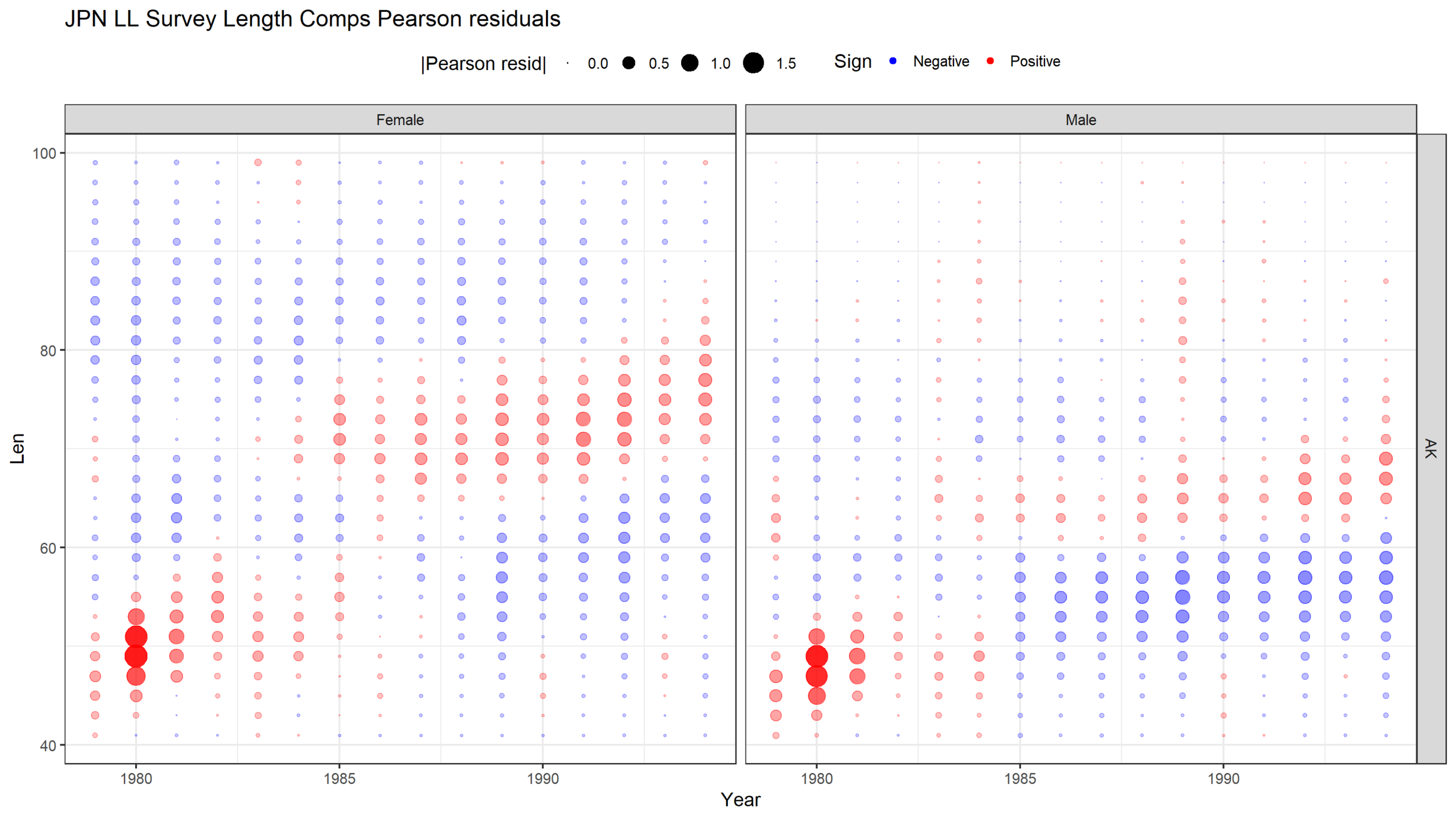

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.00
0.25
0.50
0.75
1.00
1.25
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

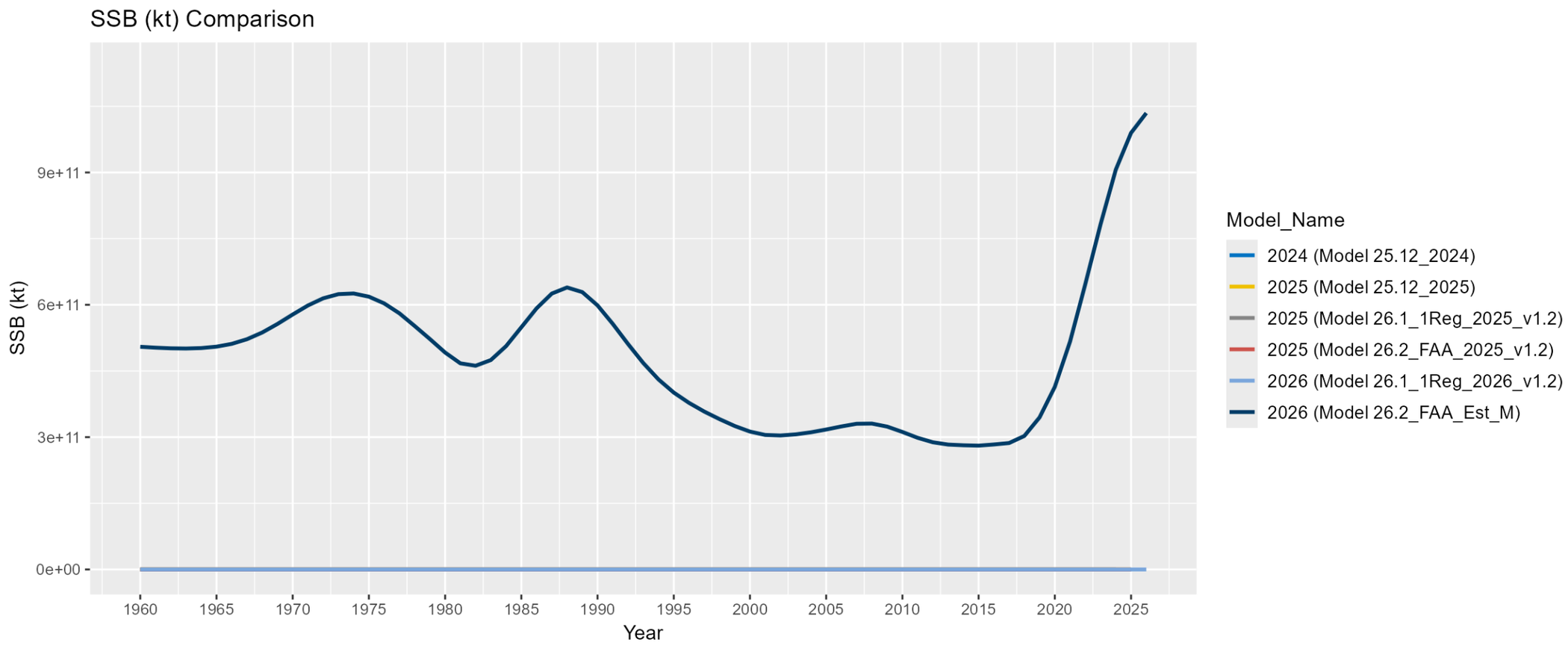

BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

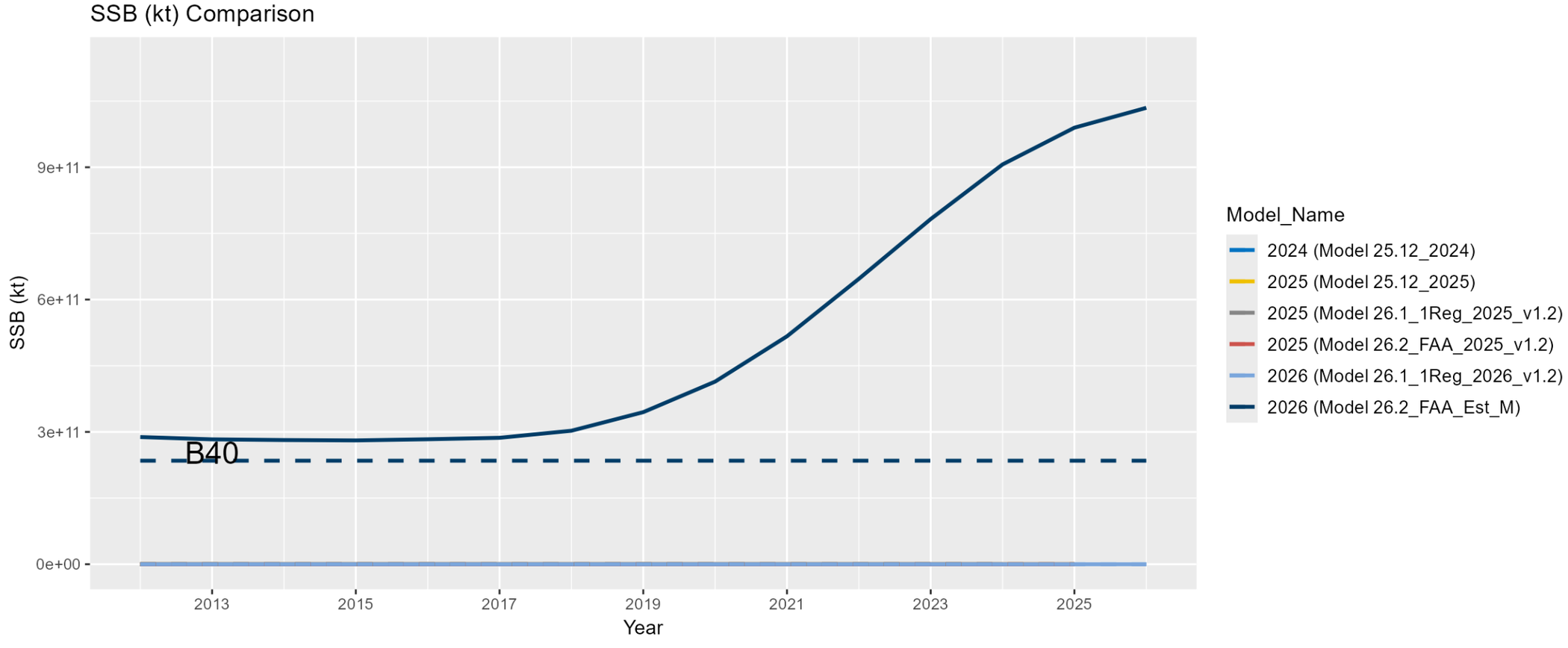

AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

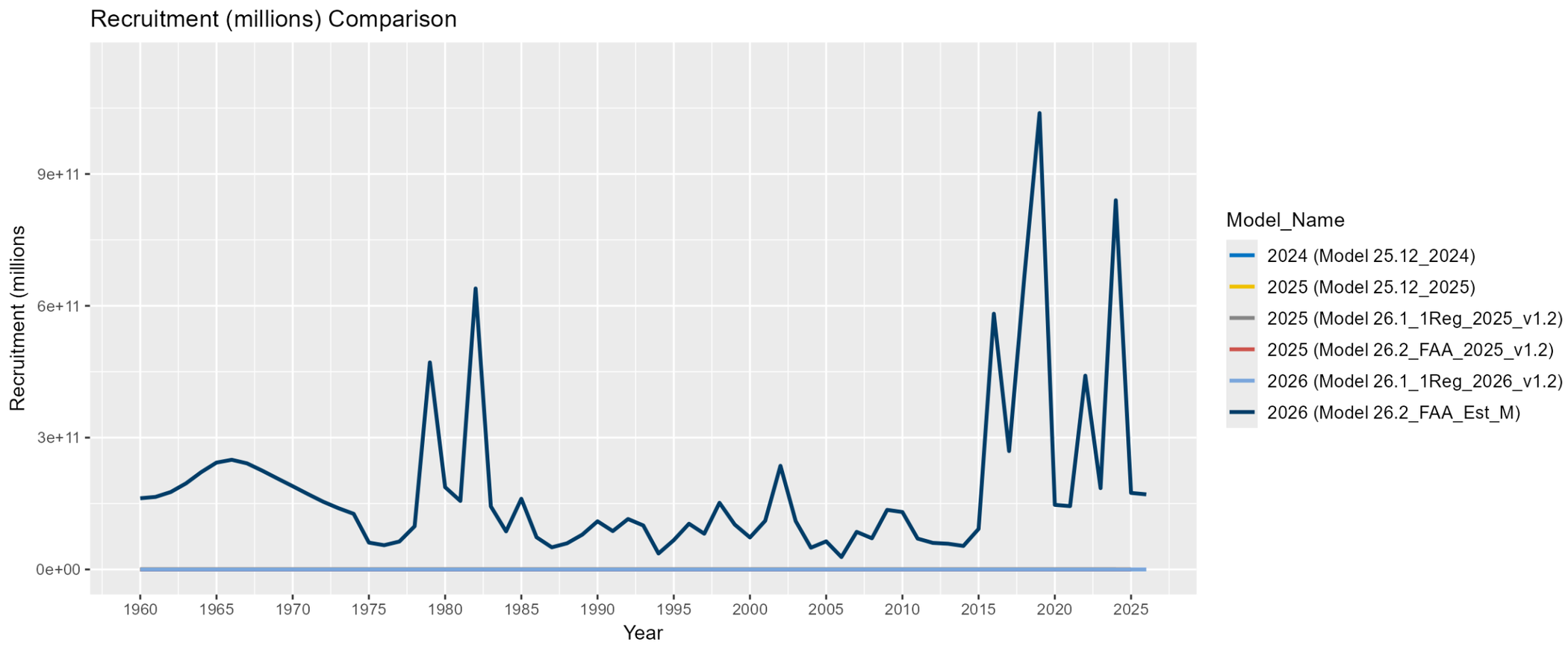

GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

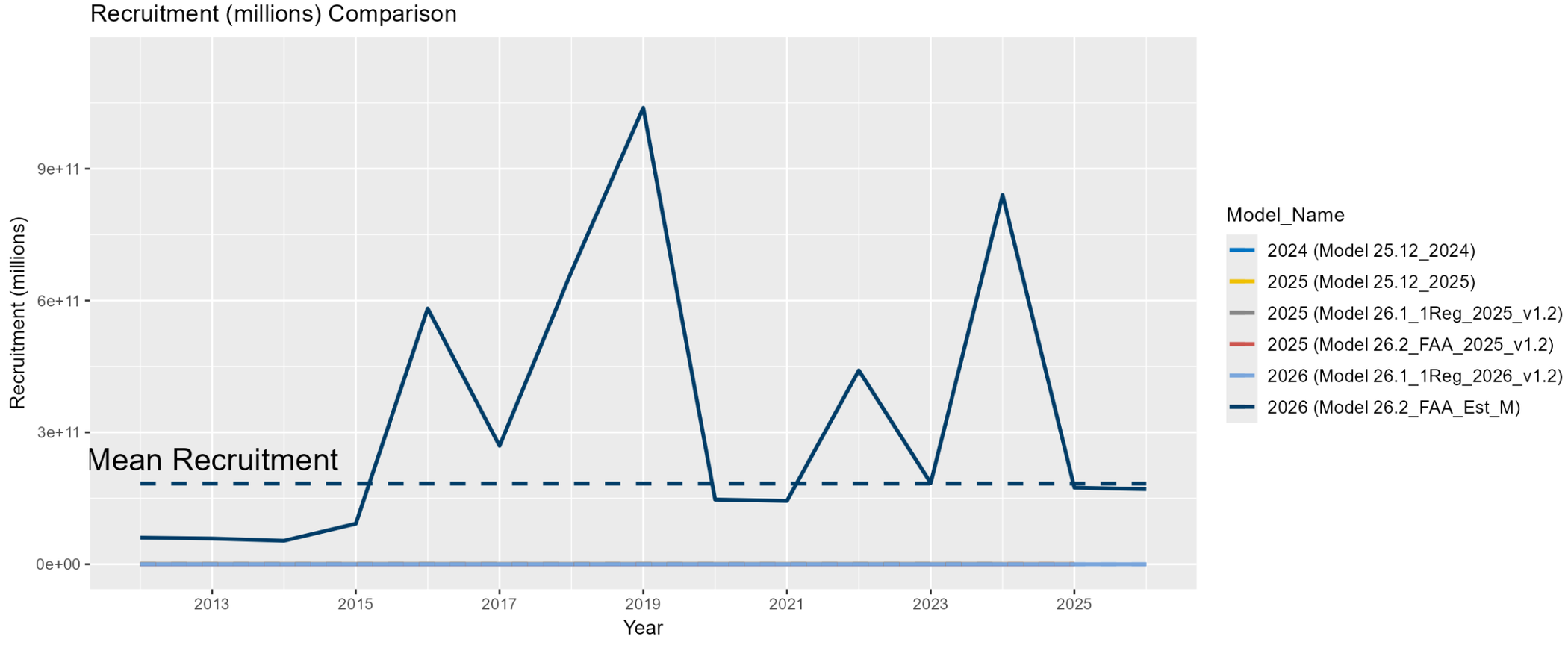

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

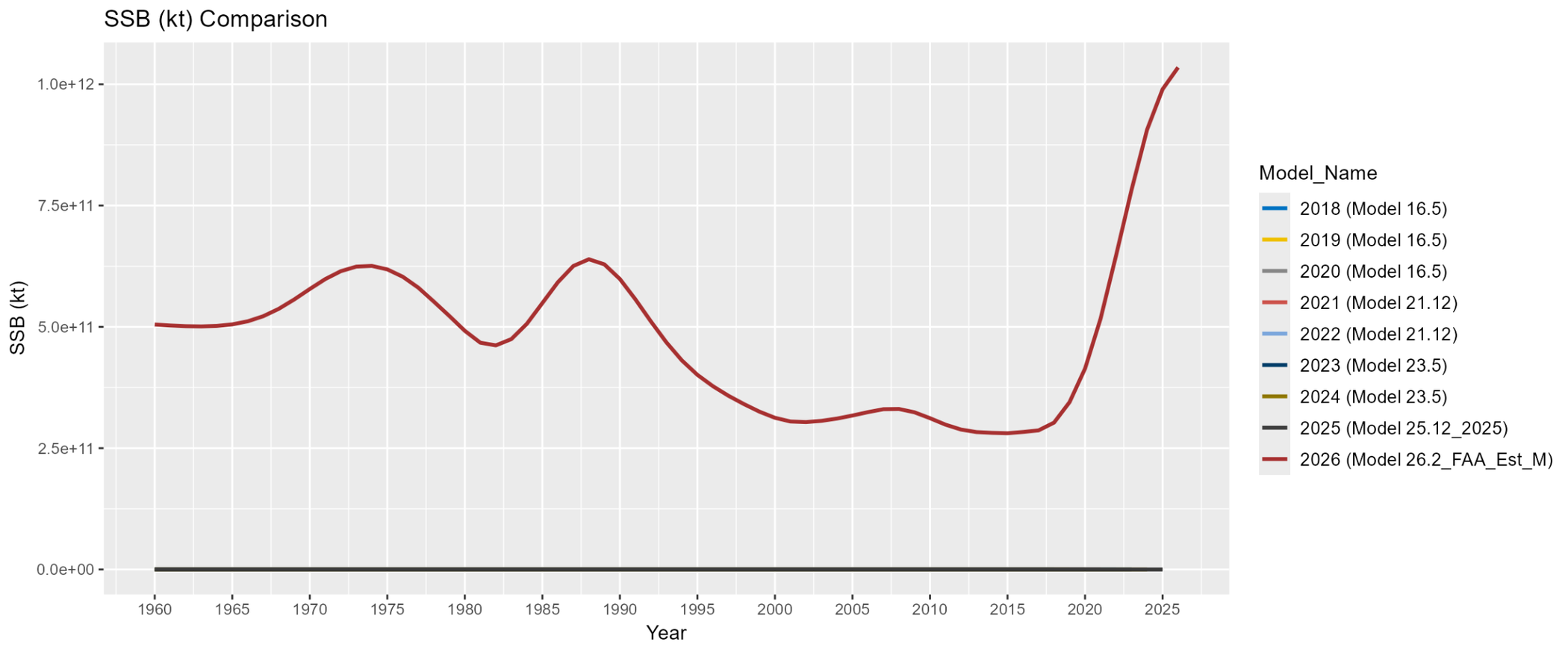

JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

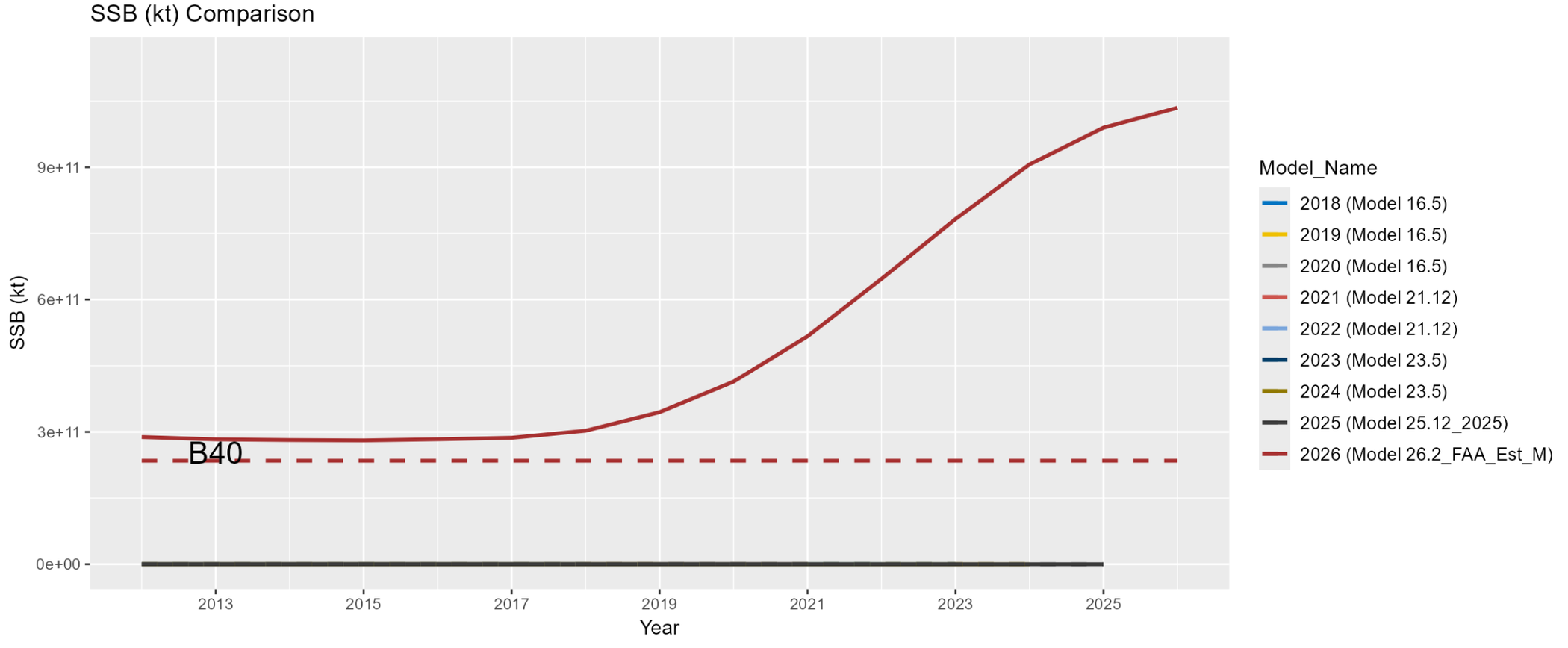

SSB (kt) Comparison
SSB (kt)
9e+11
6e+11
3e+11
0e+00
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_Est_M)

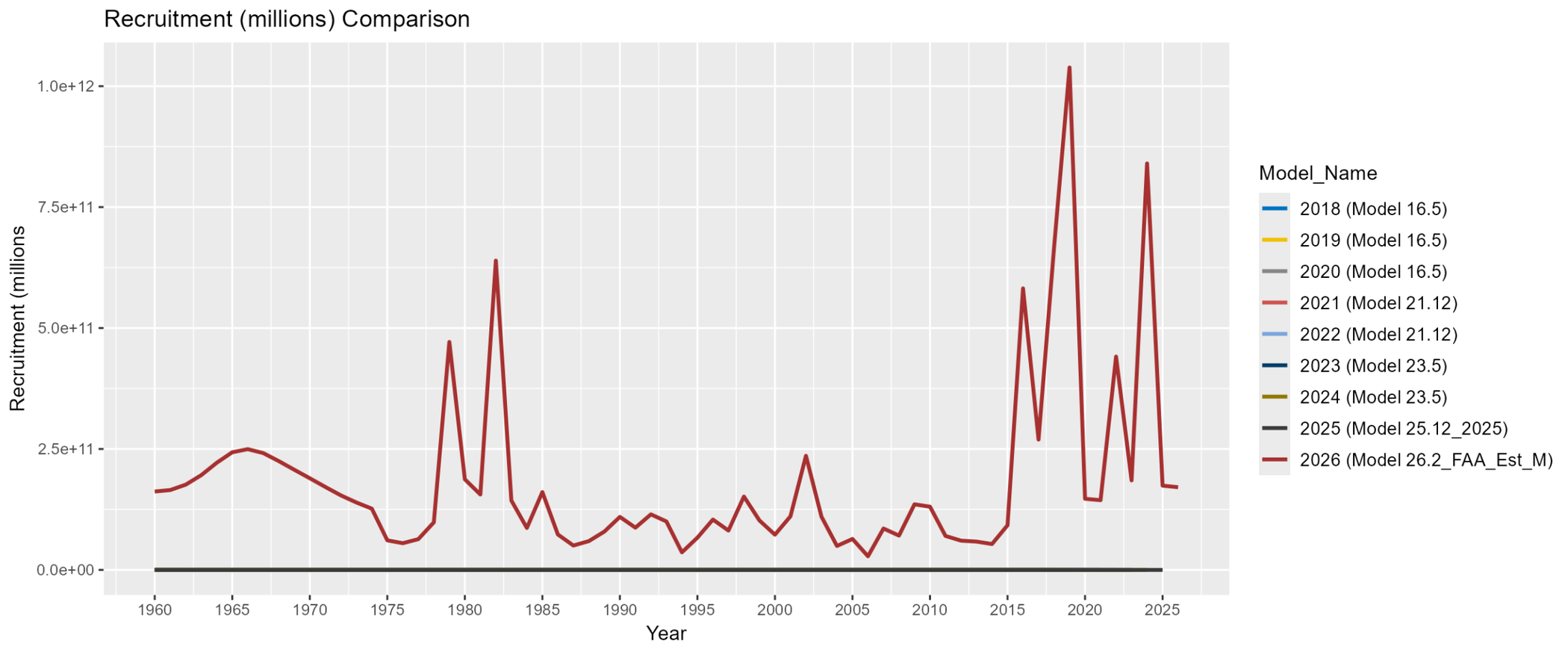

SSB (kt) Comparison
SSB (kt)
9e+11
6e+11
3e+11
0e+00
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_Est_M)

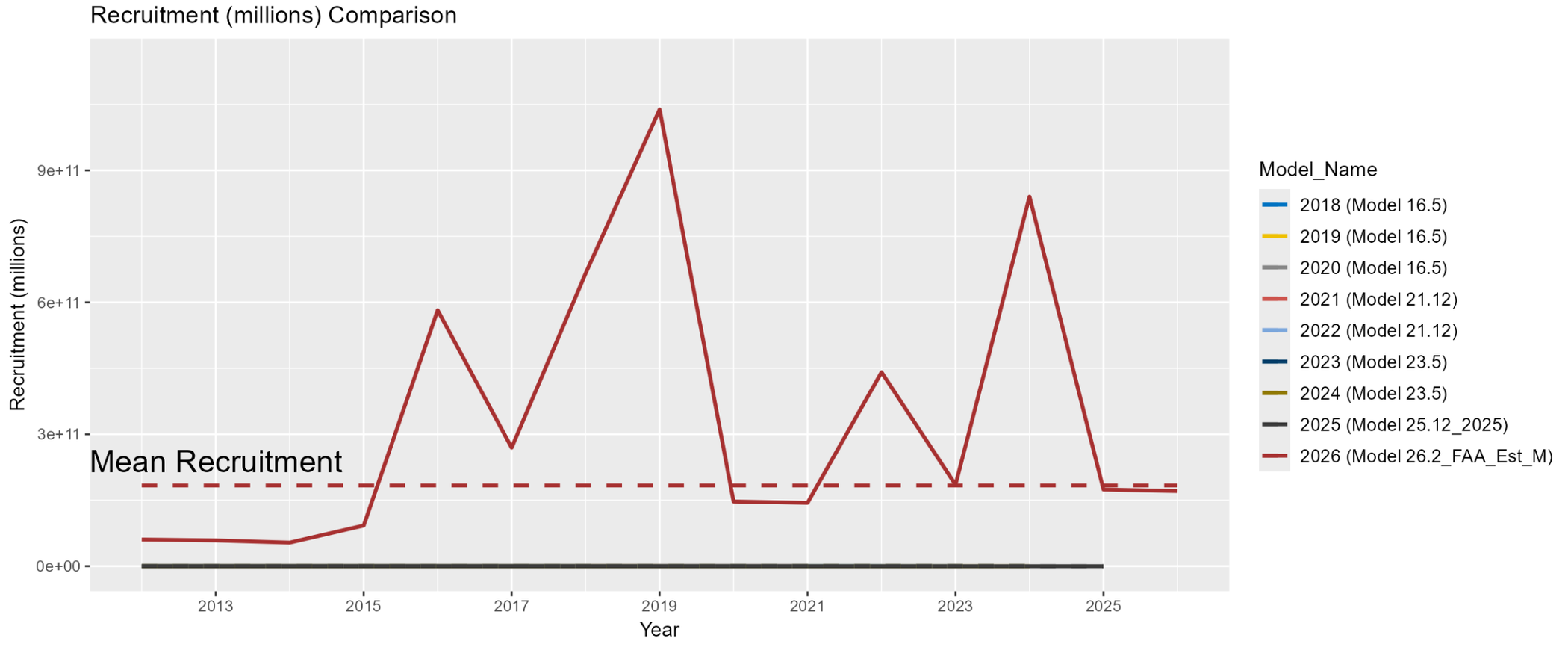

Recruitment (millions) Comparison
Recruitment (millions
9e+11
6e+11
3e+11
0e+00
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_Est_M)

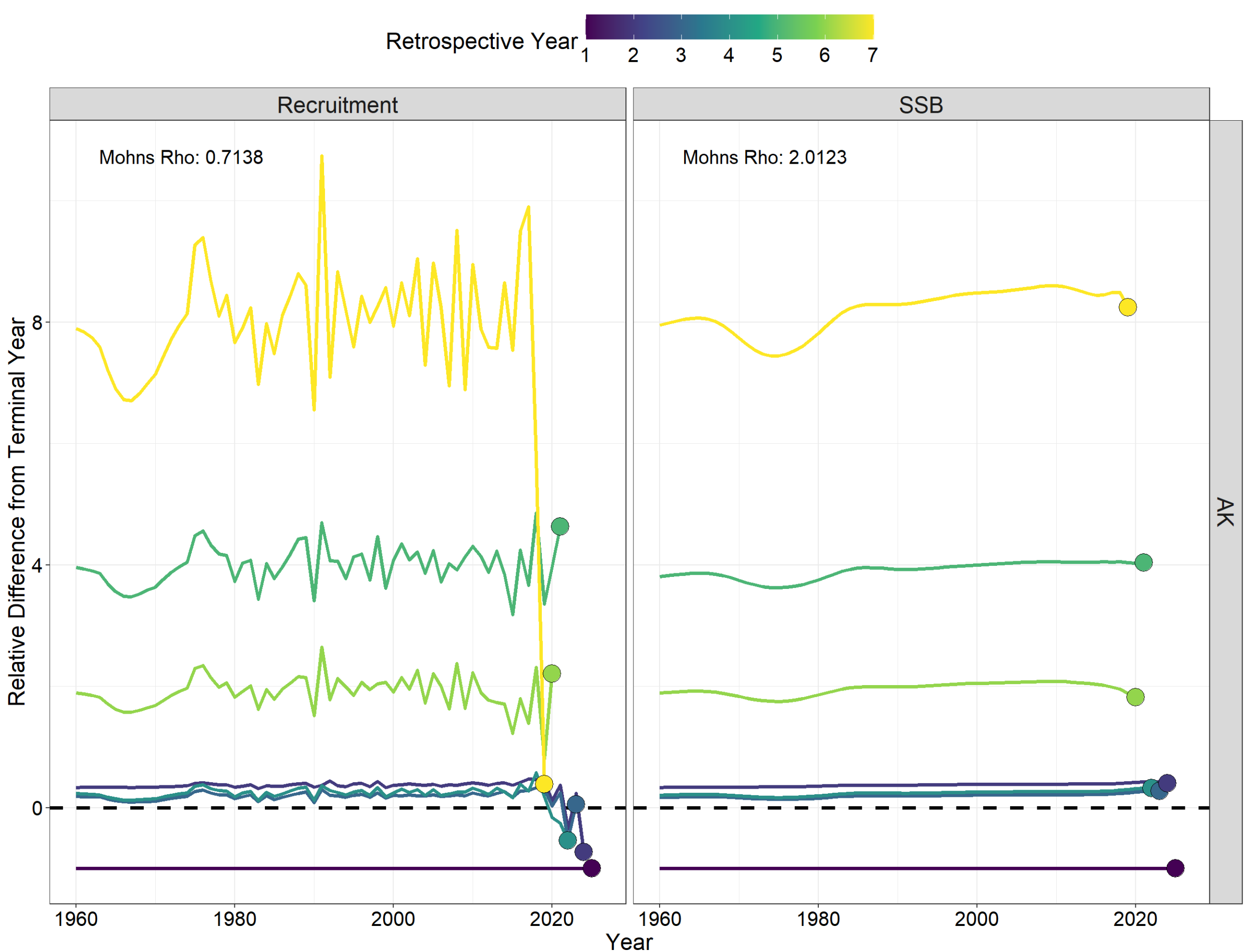

Recruitment (millions) Comparison
Recruitment (millions)
9e+11
6e+11
3e+11
0e+00
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_Est_M)


SSB (kt) Comparison
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
SSB (kt)
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_Est_M)


SSB (kt) Comparison
9e+11
6e+11
3e+11
0e+00
SSB (kt)
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_Est_M)


Recruitment (millions) Comparison
Recruitment (millions
1.0e+12
7.5e+11
5.0e+11
2.5e+11
0.0e+00
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_Est_M)


Recruitment (millions) Comparison
Recruitment (millions)
9e+11
6e+11
3e+11
0e+00
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_Est_M)


Retrospective Year
1
2
3
4
5
6
7
Recruitment
SSB
Mohns Rho: 0.7138
Mohns Rho: 2.0123
Relative Difference from Terminal Year
8
4
0
AK
1960
1980
2000
2020
Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

0e+00
1e+12
2e+12
3e+12
4e+12

1960
1980
2000
2020

Year

AK

Recruitment (millions)

3e+12
2e+12
1e+12
0e+00

0
1
2
3
4
5
6
7

Years since cohort was last estimated

Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: 0.5637

SSB

Mohns Rho: 1.1352

AK

Relative Difference from Terminal Year

3
2
1
0
-1

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

3e+12
2e+12
1e+12
0e+00

2e+12
1e+12
0e+00

1960
1980
2000
2020

Year

AK

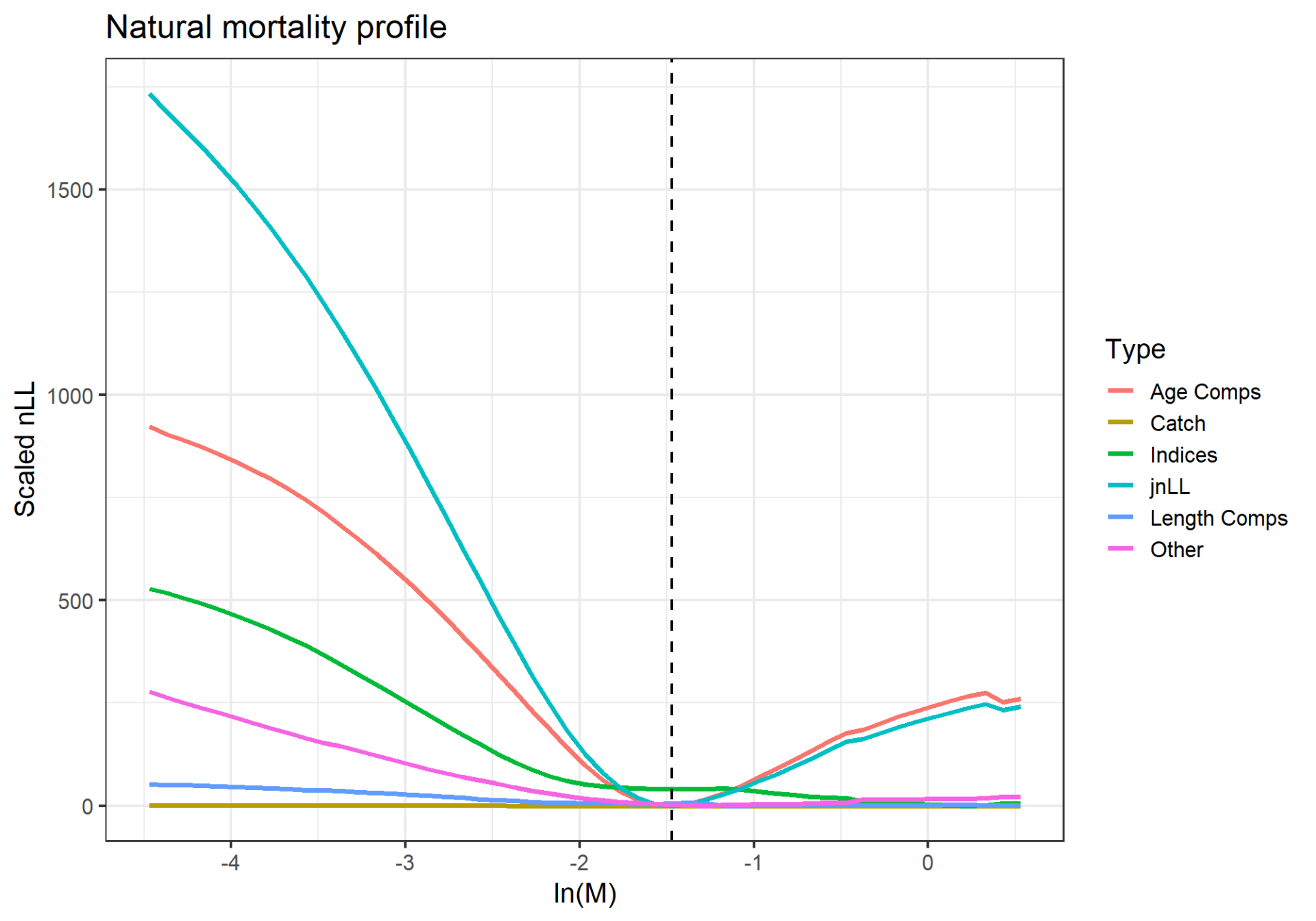
3e+12
2e+12
1e+12
0e+00

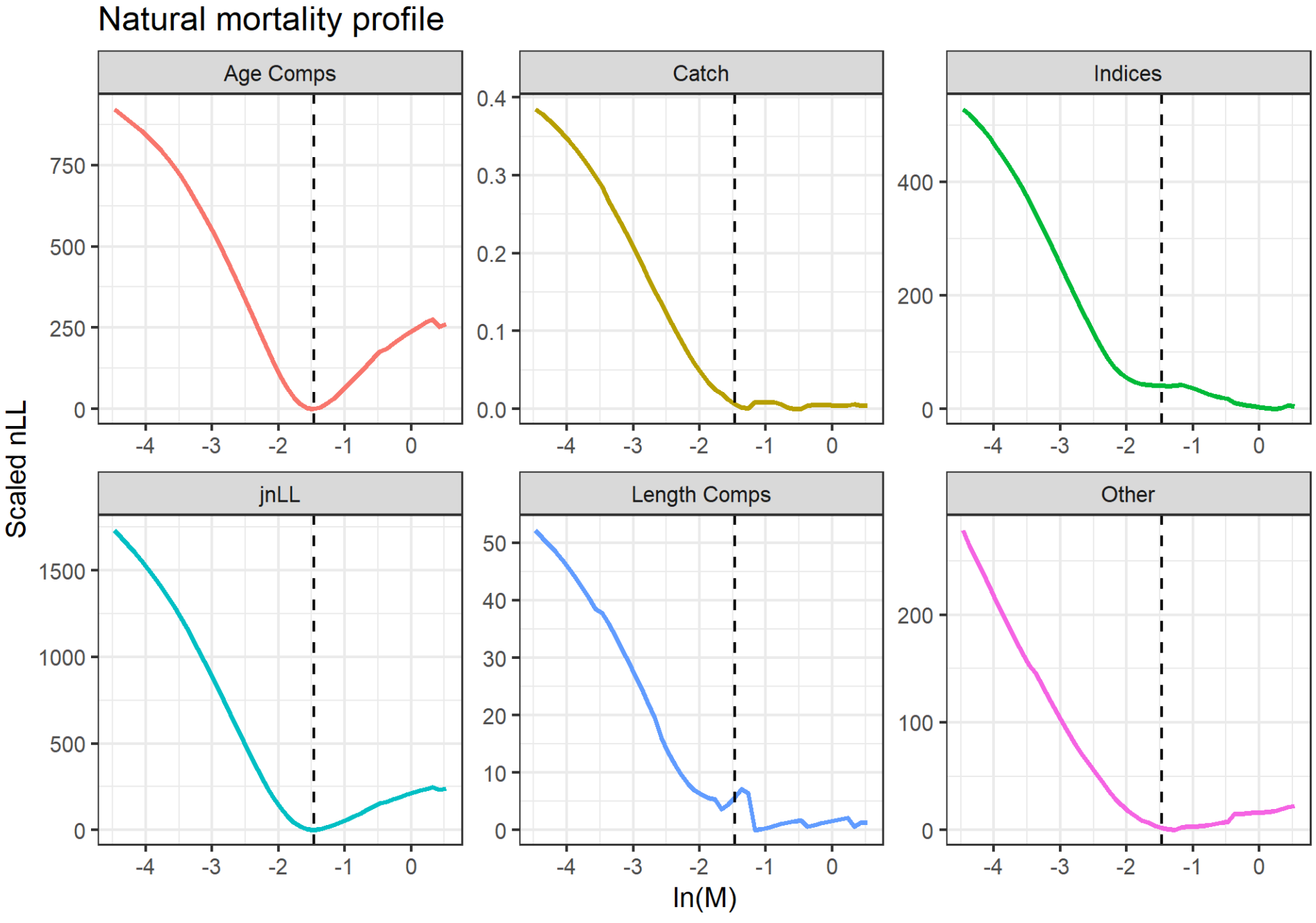
Recruitment (millions)

0
1
2
3
4
5
6
7

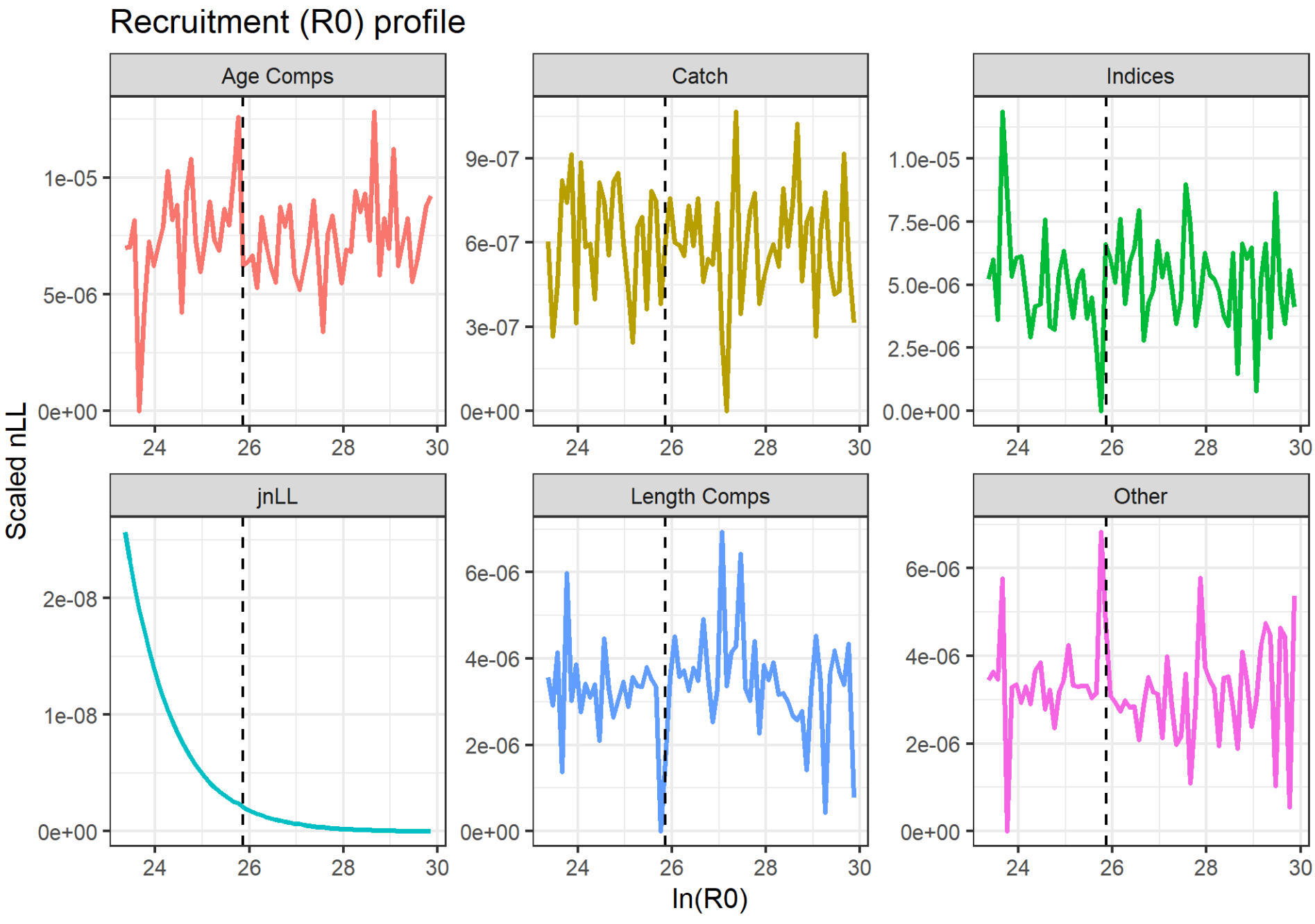
Years since cohort was last estimated

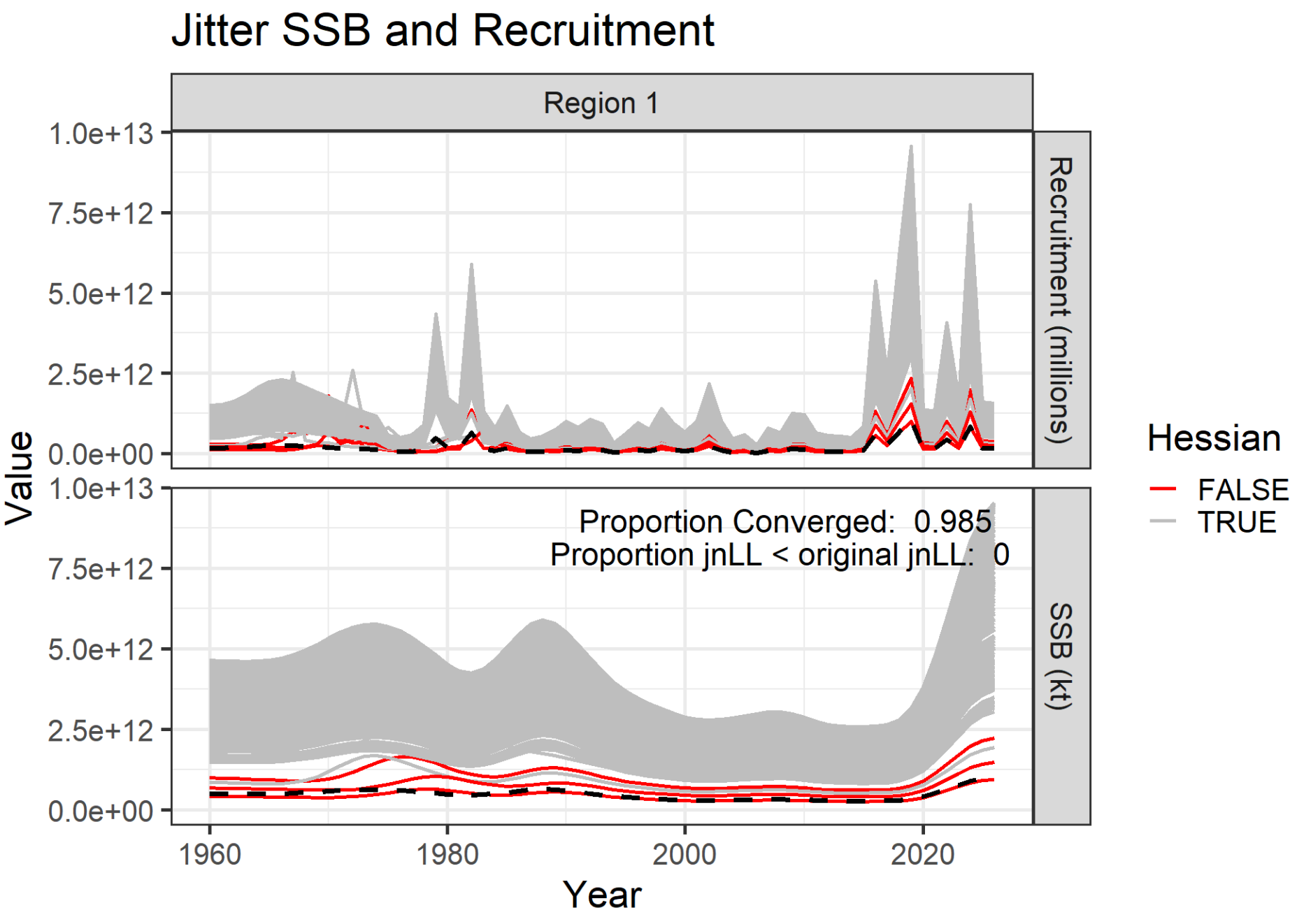
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

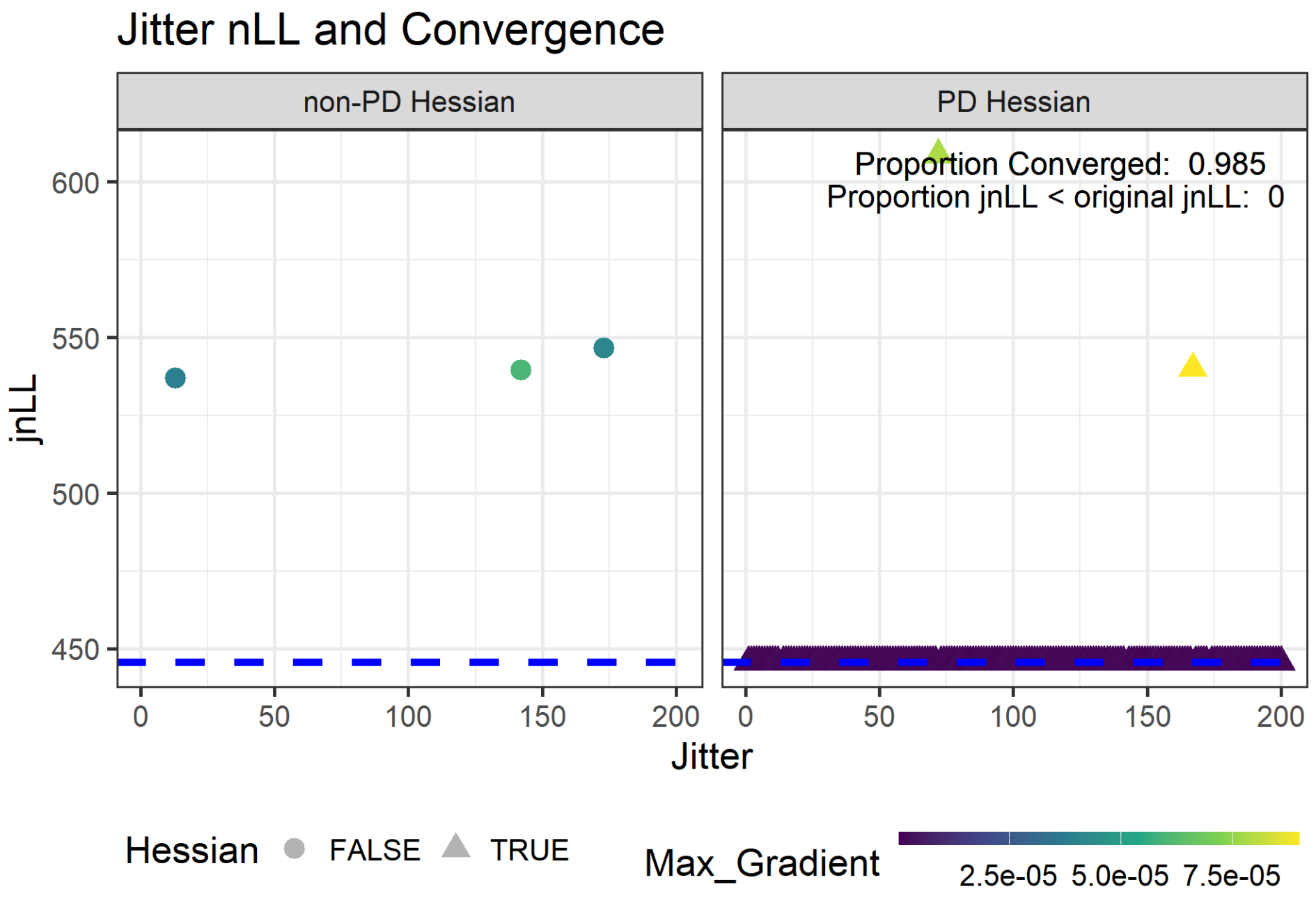

Natural mortality profile
Scaled nLL
1500
1000
500
0
-4
-3
-2
-1
0
ln(M)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

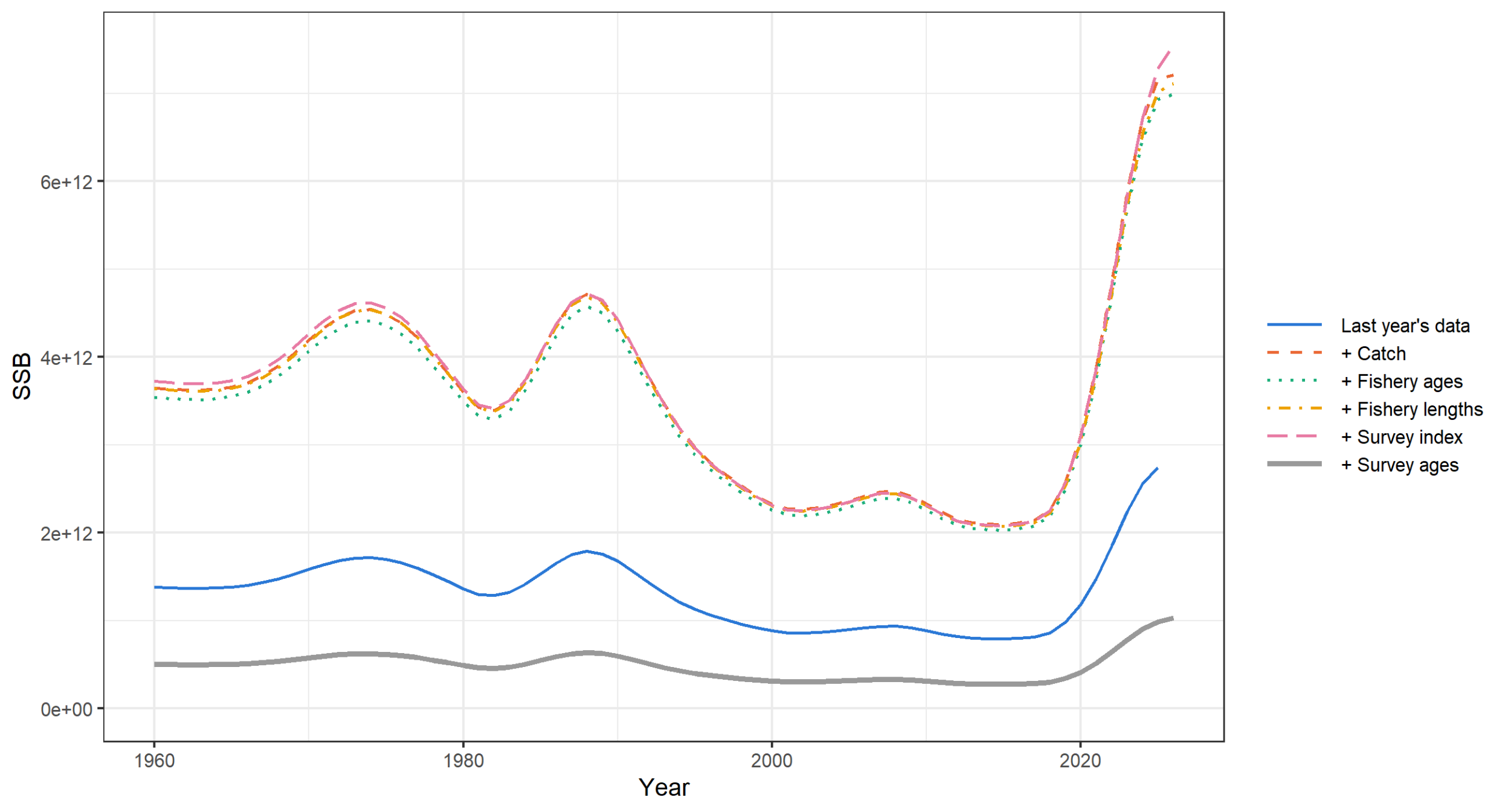

Natural mortality profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(M)

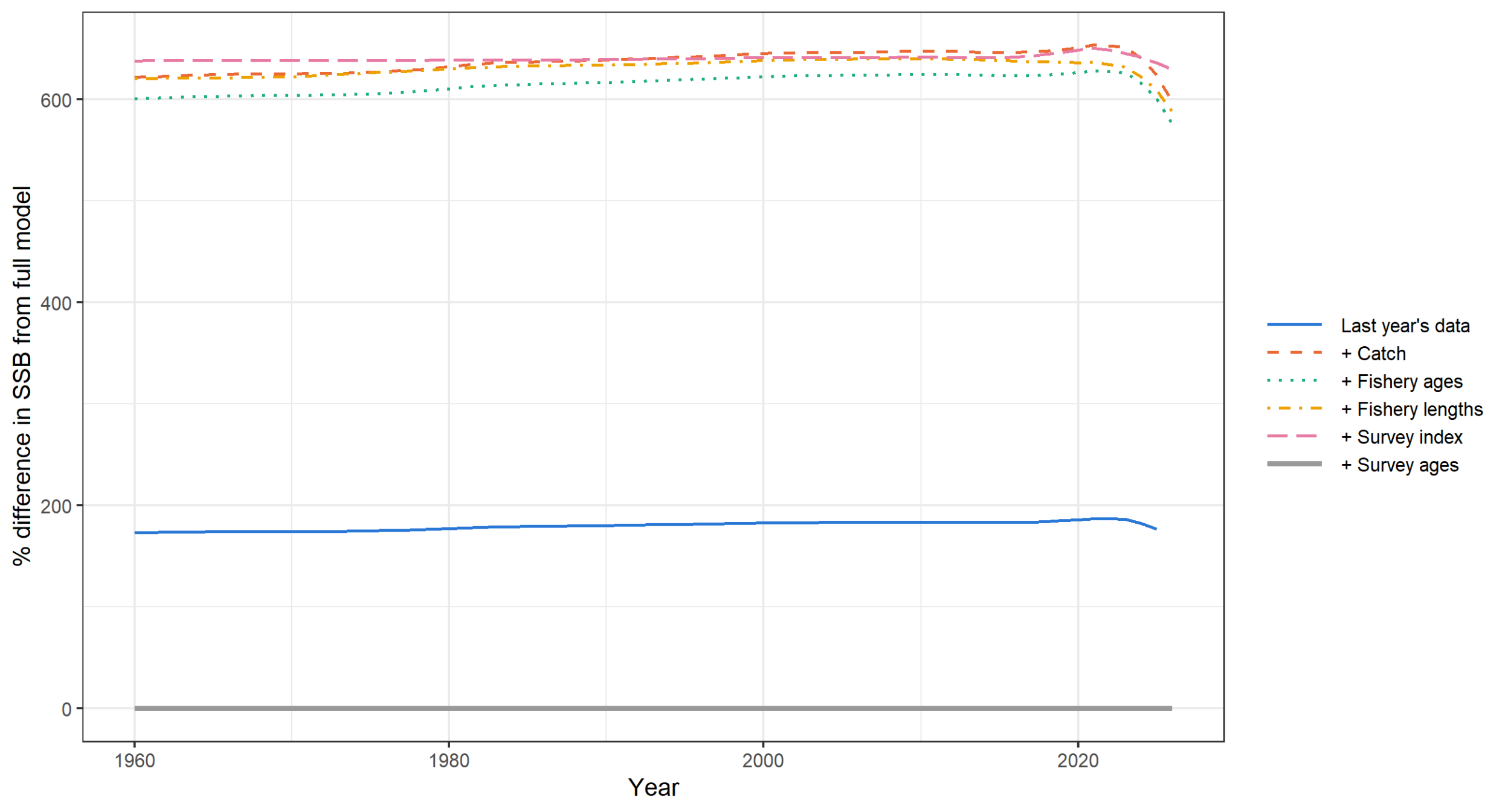

Recruitment (R0) profile
Scaled nLL
1e-05
5e-06
0e+00
24
26
28
30
ln(R0)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

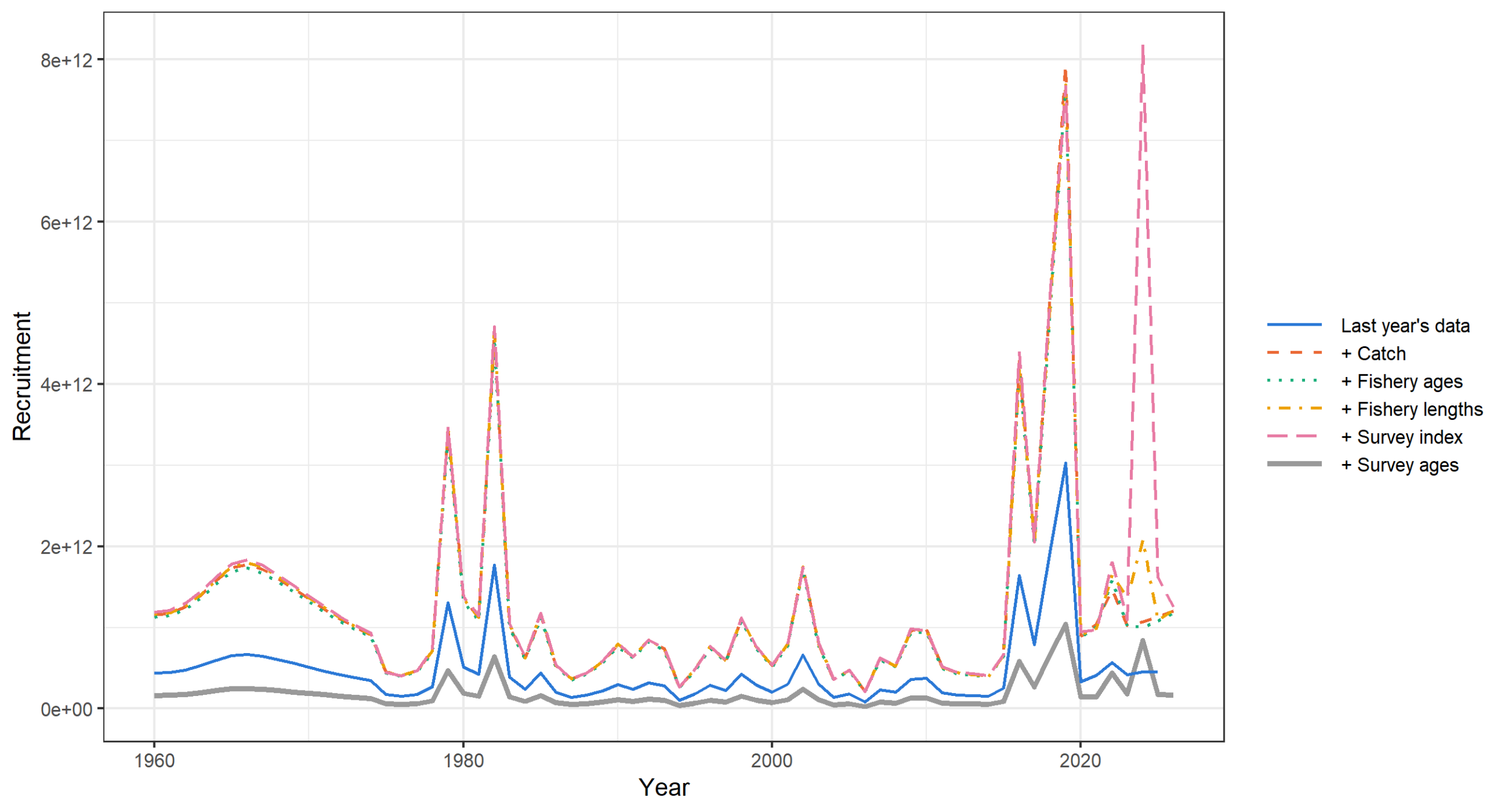

Recruitment (R0) profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(R0)
0e+00
5e-06
1e-05
3e-07
6e-07
9e-07
2.5e-06
5.0e-06
7.5e-06
1.0e-05
1e-08
2e-08
2e-06
4e-06
6e-06
24
26
28
30

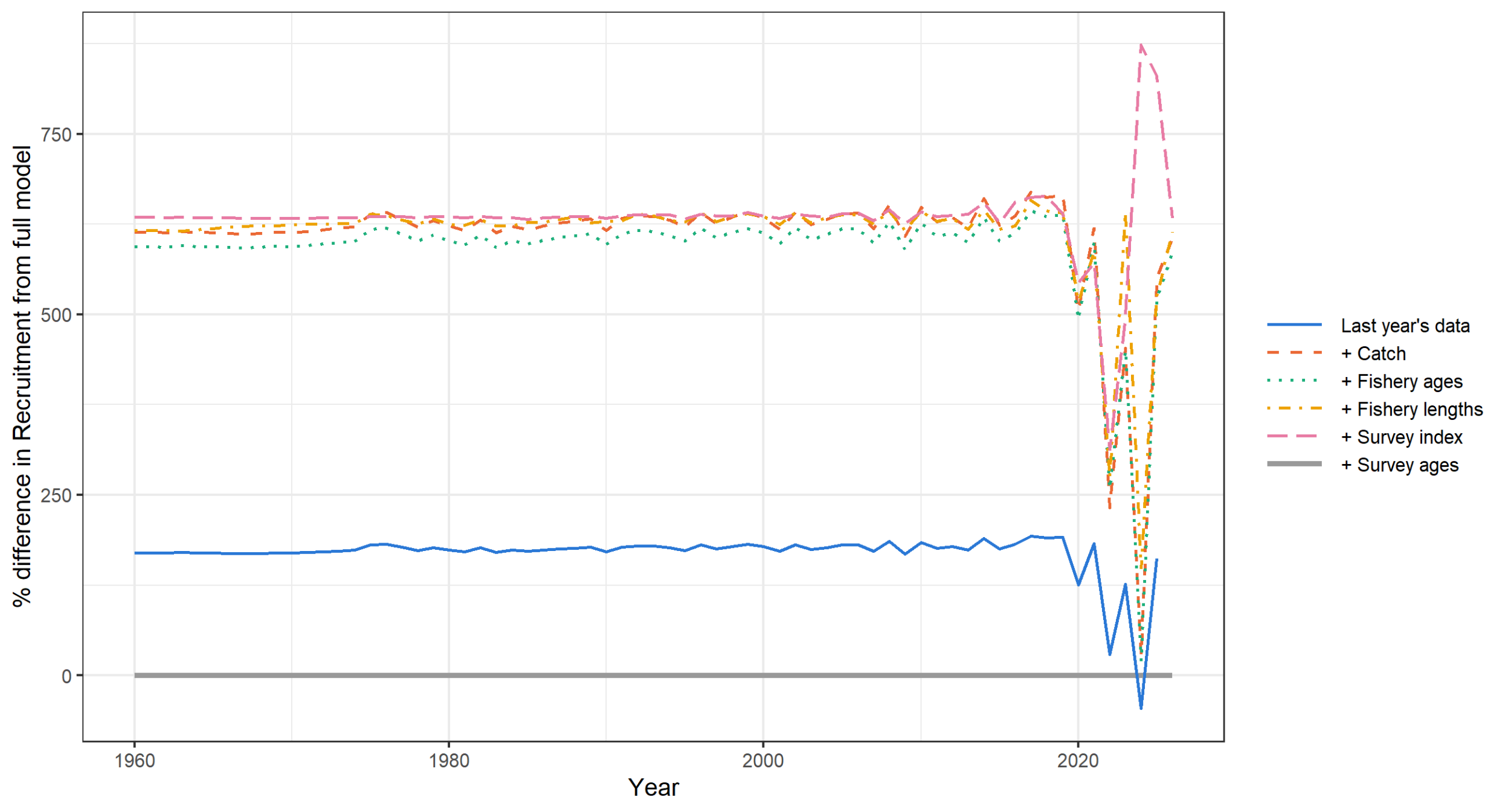

Jitter SSB and Recruitment
Region 1
Recruitment (millions)
SSB (kt)
Proportion Converged: 0.985
Proportion jnLL < original jnLL: 0
Value
Year
1.0e+13
7.5e+12
5.0e+12
2.5e+12
0.0e+00
1960
1980
2000
2020
Hessian
FALSE
TRUE

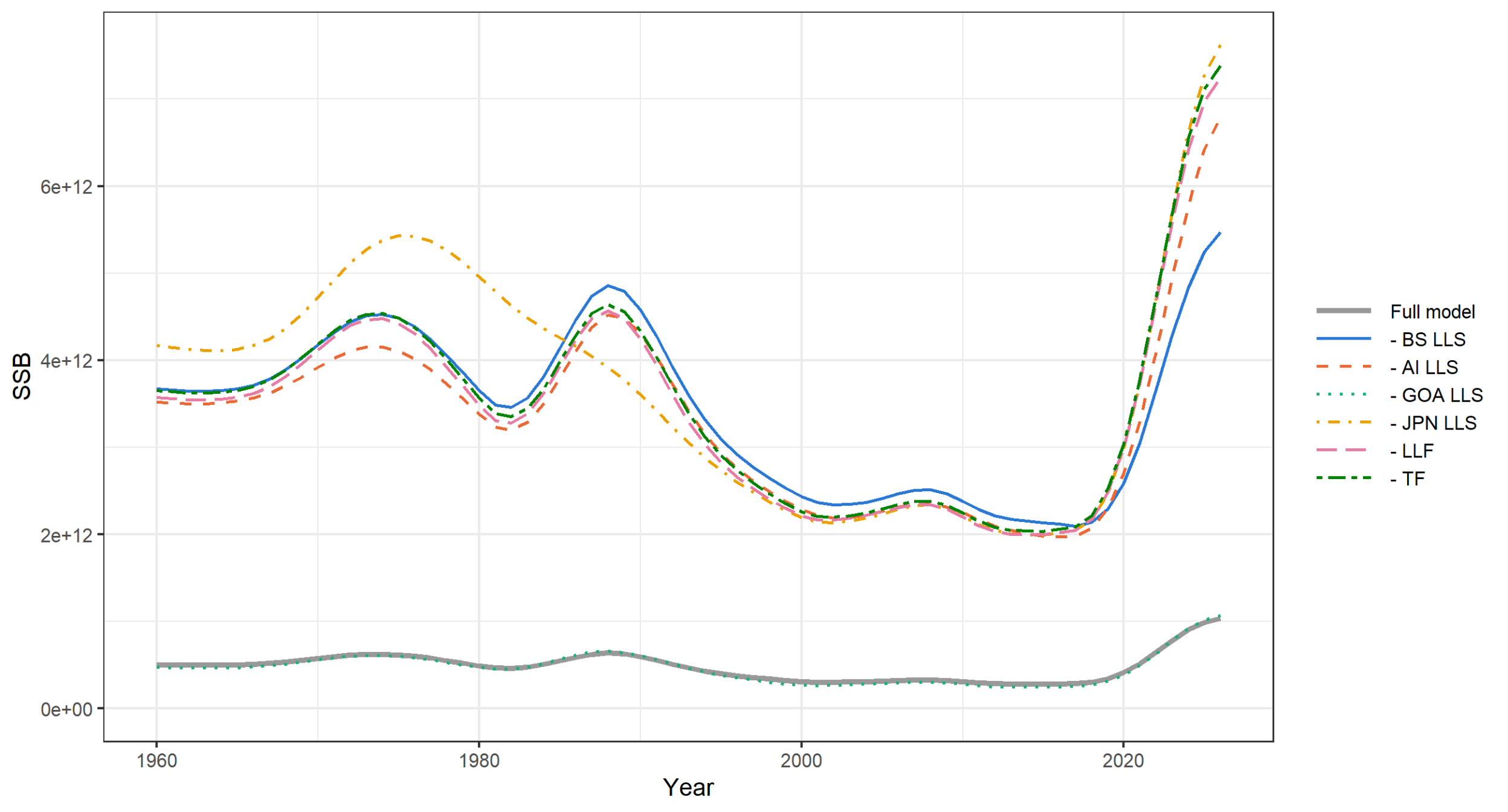

Jitter nLL and Convergence
non-PD Hessian
PD Hessian
Proportion Converged: 0.985
Proportion jnLL < original jnLL: 0
jnLL
600
550
500
450
0
50
100
150
200
Jitter
Hessian
FALSE
TRUE
Max_Gradient
2.5e-05
5.0e-05
7.5e-05

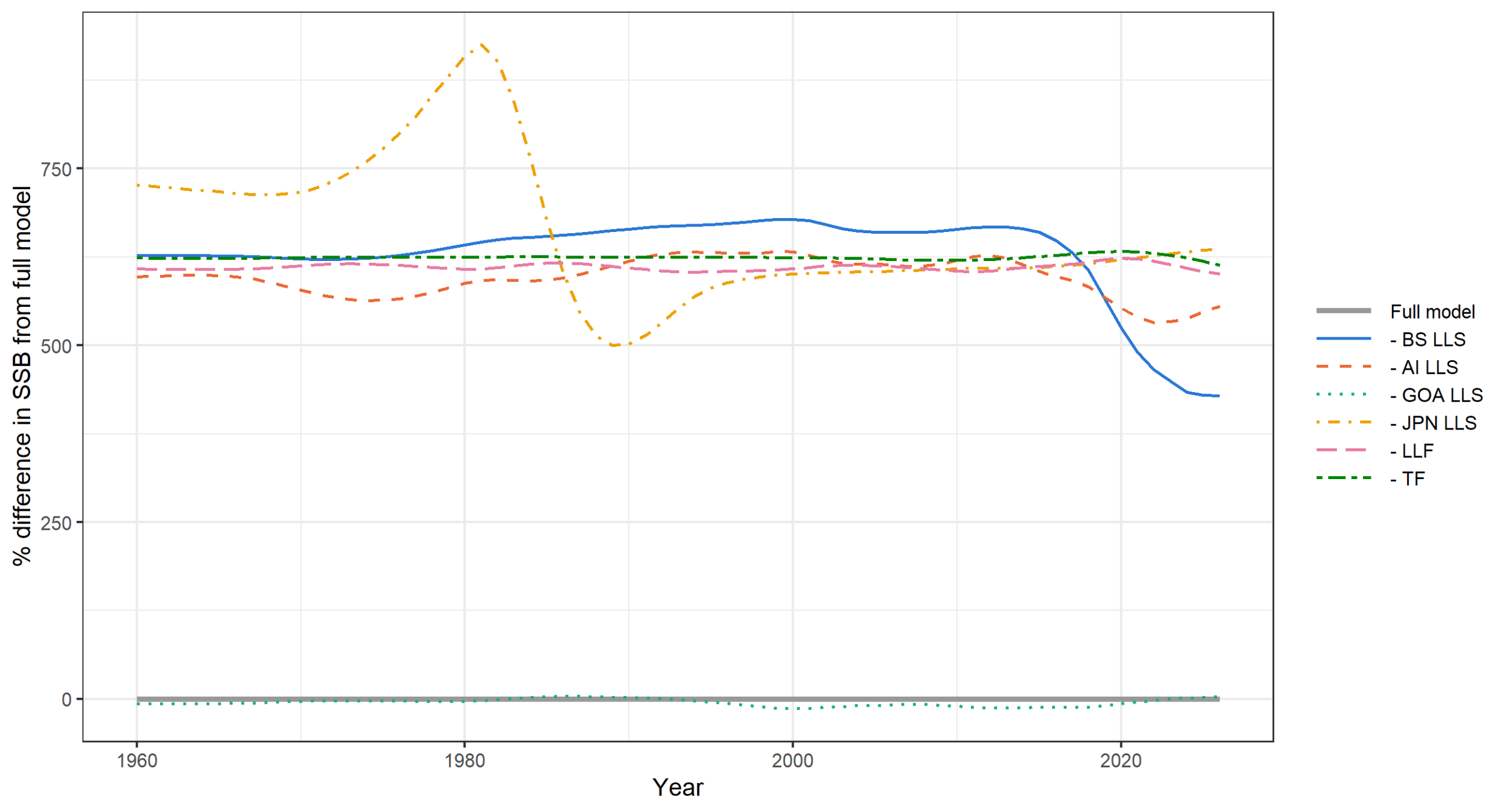

SSB
6e+12
4e+12
2e+12
0e+00
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

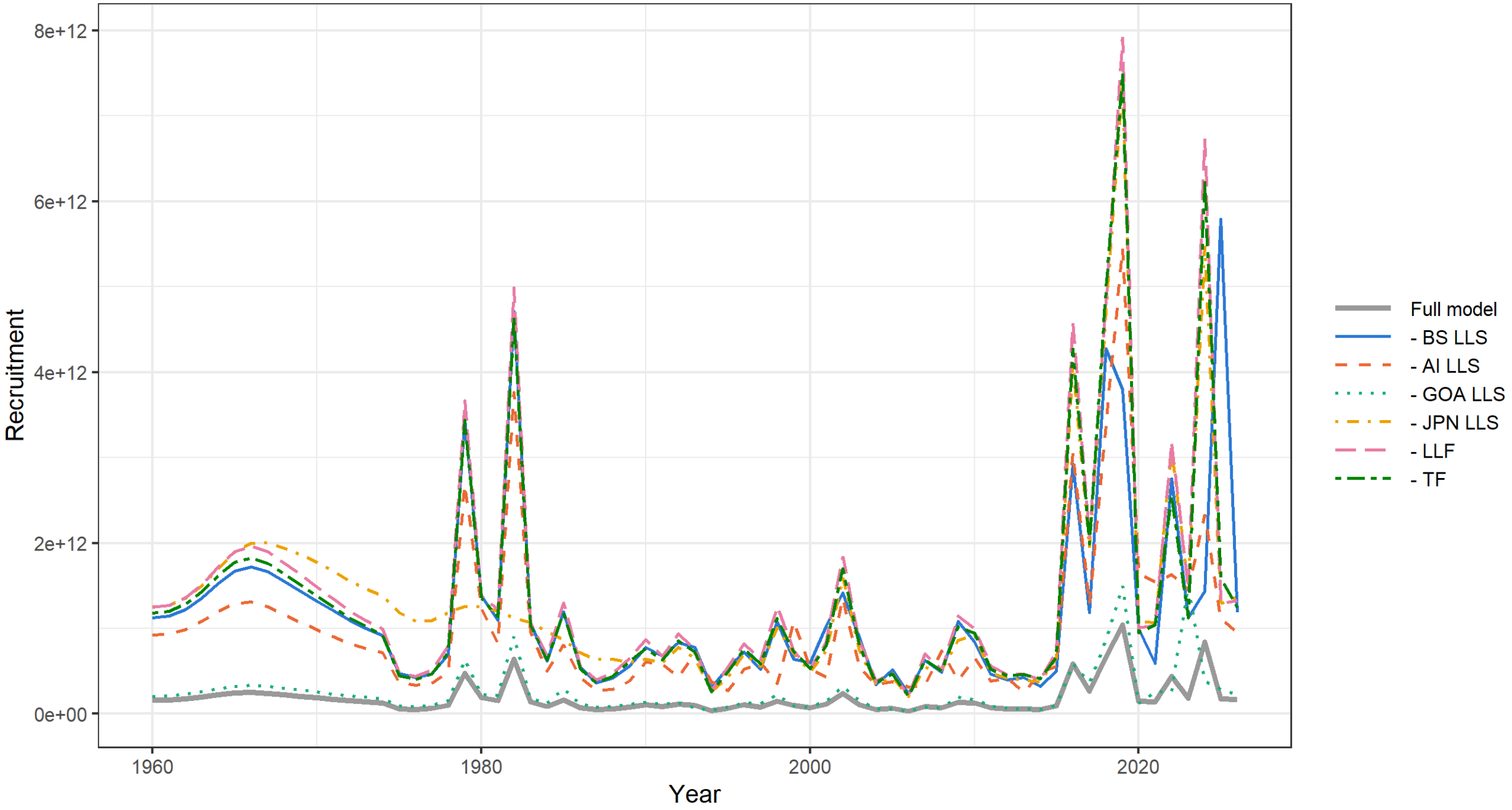

% difference in SSB from full model
600
400
200
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

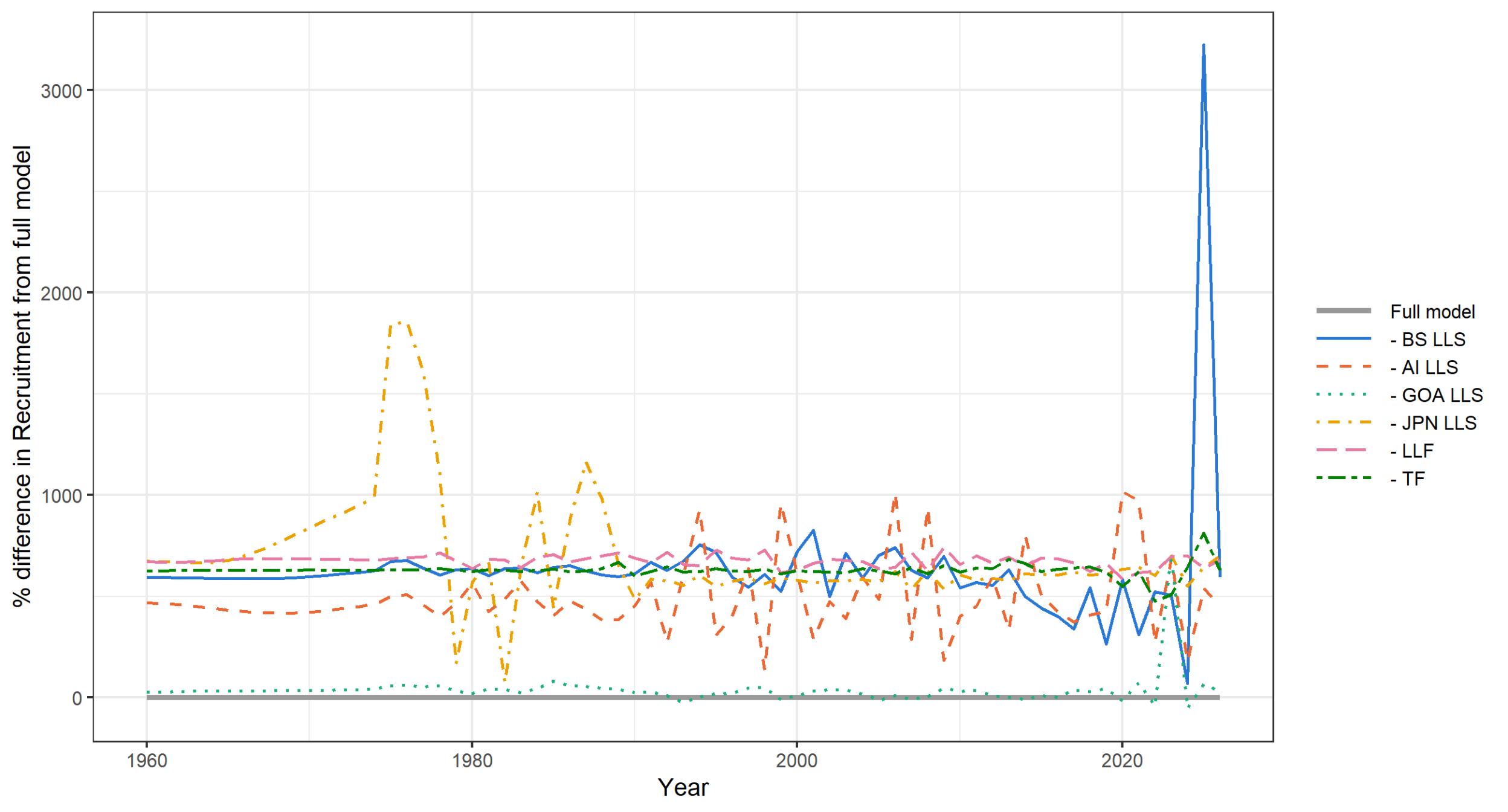

Recruitment
8e+12
6e+12
4e+12
2e+12
0e+00
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in Recruitment from full model
750
500
250
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


SSB
6e+12
4e+12
2e+12
0e+00
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


% difference in SSB from full model
750
500
250
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


8e+12
6e+12
4e+12
2e+12
0e+00
Recruitment
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


% difference in Recruitment from full model
3000
2000
1000
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.62	34.07	21.09
AI LL Survey Ages	NA	1.74	35.57	62.00
GOA LL Survey Ages	NA	0.38	67.17	25.57
JPN LL Survey Ages	NA	0.08	83.33	6.35
JPN LL Survey Lengths	NA	0.37	54.62	20.00
Fixed Gear Fishery Ages	NA	0.05	142.85	7.12
Trawl Fishery Lengths	NA	0.11	47.62	5.05

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Re
6.2_FAA_Est_M	1	1.034699e+12	1.034699e+12	9.558112e-12	367417465368	234702500156	0.14792	4.40855	1	0

Phase Plane Assuming B40 from ~2018

1.00
0.75
0.50
0.25
0.00

$F/F_{35\%}$

0
2
4
6

$SSB/B_{35\%}$

2010 2005 2020 1965 1985 1990
2015 2000 1995 1980 1960 1970 1975 2025 **2026**

Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)